

Sixteen Refined Theories of Longitudinal Vibration of Rods: A Review, From Lord Rayleigh to Modernity

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Dedicated to the 80th Birth Anniversary of Prof. J.N. Reddy

Abstract

In this review paper, we discuss sixteen available refined theories, to the best of our knowledge, on longitudinal vibration of rods, starting nearly 150 years ago with Rayleigh-Love theory. We herewith try to remain as faithful as possible to the derivations of the original developers of the theories at hand. Finally, the solutions provided by all the above theories are compared with the exact solution of Pochhammer's and Chree's derivation from the three-dimensional equations of the theory of elasticity.

1. Introduction

Understanding the longitudinal vibration of rods is fundamental to the dynamics of elastic bodies and wave propagation in continuous media. Over the past century and a half, multiple theoretical approaches have been developed to model such vibrations with increasing accuracy by incorporating more complex physical effects. This paper presents a comprehensive review of sixteen refined theories developed for modeling longitudinal vibrations of rods, each progressively

building on its predecessor by accounting for phenomena such as radial inertia, radial shear, axial shear, and multimodal displacement fields.

There are several established methods for deriving the equations of motion for dynamic systems. These include Newton's motion laws, the equivalent system parameters (or energy) method, and Hamilton's principle. While Newton's laws are based upon clear physical intuition, they become increasingly difficult to apply in complex systems with coupled motions or multiple interacting effects. Hamilton's principle, on the other hand, provides a general and systematic variational framework and is especially useful for deriving equations of motion for systems involving multiple degrees of freedom or field variables. The energy method lies between the two in terms of complexity, offering both physical insight and analytical tractability [1].

The derivation of the classical theory of longitudinal wave propagation in rods is based on Newton's laws. It assumes that vibrations arise from the net force acting on an infinitesimal rod element and considers only axial motion. This basic approach leads to the well-known one-dimensional wave equation. According to Achenbach [2]: *"The longitudinal vibrations of rods and beams represent one of the classic problems of the theory of wave propagation in elastic solids. The propagation characteristics depend sensitively on the physical assumptions embodied in the mathematical model."* A major refinement to this classical theory came from John W. Strutt (Lord Rayleigh) [3], who introduced the concept of radial inertia resulting from lateral contraction due to Poisson's effect. Lord Rayleigh used the equivalent energy method to compare the kinetic energy (or effective mass) of rods vibrating with and without the radial inertia effect. However, he assumed that the potential energy (or effective stiffness) remained constant in both cases. As a result, the final expression for the ratio of time periods was simply the ratio of the kinetic energies or, equivalently, the effective masses, without accounting for changes in potential energy. The

resulting Rayleigh–Love and Bishop formulations, along with their predicted natural frequencies of longitudinal vibration, have been presented and discussed in detail in standard vibration texts by J. S. Rao [4] and S. S. Rao [5].

Interestingly, in the first edition of his monograph, Augustus E. H. Love [6] questioned this assumption and pointed out that the potential energy due to radial contraction should also be considered, as it is of the same order as kinetic energy in the radial direction. He remarked: *“The terms of the kinetic energy arising from the inertia of the lateral motion are of the order neglected, the terms of the potential energy arising from the lateral contraction are of the same order as those arising from the longitudinal extension.”* Because of this observation, Love initially omitted radial inertia from his theory of rod vibrations. However, he included it a decade later in the second edition of his monograph [7], where he derived the governing equations using Hamilton’s principle. Still, as Grigoljuk and Selezov [8] (in a free translation from the Russian) observed, *“Love’s theory offers little improvement over the classical one, as it represents a single-mode approximation of the equations of elasticity theory. A significant improvement was achieved only after shear deformations were taken into account and a two-mode approximation was developed.”* Notably, Rayleigh had derived the ratio of time periods with and without radial inertia but never proceeded to the full equations of motion. Had Rayleigh and Love included both the kinetic and potential energies due to radial motion, they would have anticipated Bishop’s theory [9] by more than 50 years. Richard E. D. Bishop incorporated both radial inertia and radial shear into the model using Newton’s laws. However, with the exception of Bishop’s work, most subsequent theories adopted Hamilton’s principle as the preferred method for formulating the equations of motion.

According to Graff [10]: *“The equations of motion for elastic media are most efficiently formulated by variational principles such as Hamilton’s principle, which automatically yield the*

necessary boundary conditions and incorporate all relevant energy contributions.” Hamilton’s principle requires expressing both kinetic and potential energies in terms of displacement fields. The kinetic energy expression is usually straightforward. In contrast, expressing potential energy requires a more indirect approach: it begins as the internal energy due to deformation (the negative of the internal work done), formulated as the product of stress and strain. Stress is related to strain via Hooke’s law, and strain is connected to displacement. Finally, minimizing the difference between the kinetic and the potential energies yields the governing equations and the associated boundary conditions.

The theories reviewed in this paper differ mainly in terms of the physical phenomena they capture and the complexity of their mathematical formulations. Classical theory considers only axial motion of the rod and assumes the cross-section remains planar. The Rayleigh-Love theory includes radial motion due to Poisson’s effect but still retains the planar cross-section assumption. Bishop’s theory builds upon Rayleigh-Love by adding radial shear effect, which further refines the model’s accuracy. The Mindlin-Herrmann theory [11] represents the first of the multimodal theories, treating axial and radial displacements as combinations of distinct mode shapes. It uses a single mode for each and applies Hamilton’s principle for the derivation. Because it incorporates a wider range of physical effects, it is considered more general than Bishop’s theory and should match the latter with a simplifying assumption discussed in section 2.3 of this paper. As noted by Grigoljuk and Selezov [8] (in a free translation from the Russian): *“The two-mode approximation by R.D. Mindlin and G. Herrmann describes the dynamics of the rod significantly better than single-mode approximations. Their model accounts for both radial and axial motions, resulting in a hyperbolic system that agrees well with the exact solutions of elasticity theory.”* These four theories assume that the rod’s cross-section remains plane during vibration and are referred to as

plane cross-sectional theories. The next group of theories removes this assumption and allows for non-planar cross-sectional deformation. The Zachmanoglou–Volterra theory [12] provides a three-mode model: two modes for each of axial and radial displacements are introduced at first, and then one mode is removed by satisfying the zero radial stress condition at the rod’s outer surface. This theory, together with the Rayleigh–Love and Bishop formulations, was also discussed in the influential 1965 monograph [13] by Volterra and Zachmanoglou. The Mindlin–McNiven theory [14] is also a three-mode theory, with two modes for axial displacement, including axial shear deformation, and one for radial displacement. Doyle’s theory [15] also uses a three-mode model and constructs the displacement fields using Legendre polynomials. The four-mode theory by Anderson [16] builds on this multimodal framework by including an additional degree of freedom to enhance predictive accuracy. More recently, Arkadiusz Żak and Marek Krawczuk [17] proposed higher-order two-, three-, and four-mode theories, extending the multimodal approach to capture additional kinematic effects. These higher-order formulations are also reviewed in a 2024 monograph [18] Żak. In a subsequent contribution, Michael Shatalov and co-authors [19] developed three- and four-term Rayleigh–Love and Rayleigh–Bishop type unimodal theories, as well as three- and four-mode theories, by representing axial and lateral displacements as power series expansions.

All multimodal theories strive to approximate Pochhammer’s and Chree’s exact solution, which describes wave propagation in an infinitely long, elastic cylinder. In that exact solution, displacement fields must satisfy boundary conditions on the external surface of the rod. However, real-world rods are of finite length, where end boundary conditions must also be satisfied, which in turn emerge naturally from Hamilton’s principle. The multimodal theories achieve increasingly accurate approximations by using a higher number of mode shapes, often built from Jacobi,

Legendre, or general power series polynomials. Specifically, Mindlin-Herrmann theory uses Jacobi and Legendre polynomials (which are equivalent in the first order); Zachmanoglou-Volterra theory employs a general power series; Mindlin-McNiven and Anderson theories utilize Jacobi polynomials; and Doyle's theory resorts to the Legendre polynomials. According to Carrera et al. [20], *"Moreover, the refined theories are very important for wave processes, oscillation problems, static and dynamic stability, especially when high-order oscillations and stability are concerned."*

In addition to the foundational theories reviewed above, a substantial body of literature has further examined longitudinal rod vibrations from both theoretical and applied perspectives. Kynch and Green [21] applied second-order perturbation theory to the exact elasticity equations to determine dispersion relations for non-circular cylinders. Classical and refined theories, including Rayleigh-Love, Mindlin-Herrmann, Bishop, and Volterra-Zachmanoglou formulations, were also reviewed by Abramson et al. [22] in the context of stress-wave propagation. Several authors have focused on higher-order and power-series-based formulations. Boström [23] employed power-series expansions to derive a four-mode theory and compared its dispersion characteristics and displacement fields with the Rayleigh-Love model and the lowest branch of the Pochhammer-Chree solution. Corrections to Anderson's four-mode theory were later provided by Brizard et al. [24]. More recently, Xie et al. [25] developed displacement expansions for one-through five-mode theories and introduced strain correction factors to better match the cut-off frequencies of the exact Pochhammer-Chree solution.

Beyond uniform geometries, longitudinal vibration characteristics of rods with spatially varying properties have also received considerable attention. Natural frequencies of nonuniform bars were systematically investigated by J. S. Rao [4], while vibrations of bars with variable cross-

sections were comprehensively treated by Leissa and Qatu [26], who provided analytical and numerical results applicable to tapered and stepped geometries.

Beyond one-dimensional modeling, several studies have addressed three-dimensional effects. Leissa and So developed three-dimensional analytical methods to compare vibration frequencies of rods and beams, later extending this approach to fully free elastic cylinders [27], [28]. Zhao and Gary [29] derived a three-dimensional analytical solution for longitudinal wave propagation in infinite viscoelastic cylindrical bars, with applications to experimental characterization techniques. Additional applications include the analysis of longitudinal vibrations in tall buildings and high-rise structures [30], thick tapered rods and beams [31], and attenuation relations for longitudinal and transverse waves in isotropic solids [32]. More recently, Meng et al. [33] proposed a three-dimensional wave-based model for longitudinal vibration of large-diameter floating pipe piles embedded in viscoelastic media. The importance of longitudinal vibration effects has also been highlighted in biofluid applications, where Yue and Mester [34] demonstrated that the longitudinal component of vibration can significantly increase wall shear stress in large vessels.

The propagation characteristics associated with the classical, Rayleigh–Love, Mindlin–Herrmann, and Doyle three-mode theories have been analyzed in detail by Krawczuk et al. [35] and Mei [36]. Several studies by Fedotov and co-authors have considered longitudinal vibrations of stepped and nonuniform rods, bars with lumped masses and elastic restraints, and conical rods analyzed within the framework of Bishop’s theory [37], [38], [39]. Orthogonality relations associated with longitudinal vibration modes were also established in this context [40]. A later contribution by Fedotov et al. [41] proposed a unified treatment of higher-order rod models based on hyperbolic and pseudo-hyperbolic equations, with emphasis on model classification and solvability. Further extensions include the analysis of longitudinal vibrations of multi-section

cylindrical rods composed of different materials and diameters using Bishop's theory [42]. Exact solutions for coupled longitudinal and transverse stress-wave propagation in thick isotropic rods were developed by Tenkam et al. [43] within the Mindlin–Herrmann framework, while analytical solutions for rods with variable cross-sections were obtained using the Rayleigh–Love and Bishop models [44]. Time-dependent rod geometries were investigated by Shatalov et al. [45], who examined the dynamics of rods with varying length and cross-section.

In the coming section, detailed derivations and formulations for each of the rod vibration theories are given, using the principles adopted in the original derivation of each theory. Section 3 presents and compares the approximate solutions derived from these theories against Pochhammer's and Chree's exact solution, evaluating their accuracy and practical applicability. Finally, section 4 summarizes the key findings and offers conclusions on the progression, relevance, and limitations of each theory in modeling longitudinal rod vibrations.

Mindlin and Herrmann [11] state: "*The mathematical theory of the longitudinal motions of an elastic bar had its beginnings in the investigations by Daniel Bernoulli, in 1741, on the vibrations of a column of air. Both the bar and air column theories, in their classical form, lead to the one-dimensional wave equation.*" The derivation for classical theory, henceforth, can be found in almost any textbook on vibration of continuous systems. Here, it is adopted from Lord Rayleigh's [3] classical book.

Let us consider a thin element at a distance x from the left end with thickness dx when the rod is unstretched. Due to the longitudinal vibration, the left end of the element is displaced by u and its right end is displaced by $u + du$. The strain in the element is then

$$\epsilon_x = \frac{(u + du) - u}{dx} = \frac{\partial u}{\partial x} \quad (1)$$

According to Hooke's law, stress in the element is

$$\sigma_x = E\epsilon_x = E \frac{\partial u}{\partial x} \quad (2)$$

where E is the Young's modulus of elasticity of the material of the rod. Since the strain is not constant but depends on distance x , the force F_L acting on the left end of the element (towards the left due to the tension force from the element to its left) is

$$F_L = \sigma_x A = E\epsilon_x A = EA \frac{\partial u}{\partial x} \quad (3)$$

where A is the cross-sectional area of the rod. Similarly, the force F_R acting on the right end of the element (towards the right due to the tension force from the element to its right) is

$$F_R = \left(\sigma_x + \frac{\partial \sigma_x}{\partial x} dx \right) A = \left[E \frac{\partial u}{\partial x} + \frac{\partial}{\partial x} \left(E \frac{\partial u}{\partial x} \right) dx \right] A = EA \left(\frac{\partial u}{\partial x} + \frac{\partial^2 u}{\partial x^2} dx \right) \quad (4)$$

This difference in forces on two sides of the element provides inertia to move in the longitudinal direction. If ρ is the density of the material of the rod, then the mass of the element is $\rho A dx$. Using the law of motion in the longitudinal direction,

$$F_R - F_L = \rho A dx \frac{\partial^2 u}{\partial t^2} \quad (5)$$

Substituting F_L and F_R from Eqs. ((3) and (4) in Eq. (5), the equation of motion is obtained

$$EA \frac{\partial^2 u}{\partial x^2} dx = \rho A dx \frac{\partial^2 u}{\partial t^2}$$

$$\frac{\partial^2 u}{\partial t^2} = c_0^2 \frac{\partial^2 u}{\partial x^2} \quad (6)$$

where $c_0 = \sqrt{E/\rho}$ is the velocity with which a disturbance is propagated in space.

2. Sixteen Refined Theories

2.1 The First, Rayleigh-Love Theory

John W. Strutt (Lord Rayleigh) (1842 – 1919) included the lateral effects on rod vibration in his famous book “The Theory of Sound”, first published in 1877 [3]. In section 157, he provides the effect of lateral contraction due to Poisson’s effect. Following his footsteps, let us consider an infinitesimal annular element of thickness dr at a radial distance r from the axis of the rod and the length dx in the longitudinal direction so that the mass dm of the element is $\rho(2\pi r dr)dx$. Due to the lateral contraction of the circular element of radius r inside the annular element, the lateral displacement η of the annular element (towards the axis of the rod) is

$$\eta = (\nu \epsilon_x)r = \nu r \frac{\partial u}{\partial x} \quad (7)$$

where ν is the Poisson’s ratio of the material of the rod. Then the kinetic energy of the rod due to motion in the longitudinal direction is

$$T = \frac{1}{2} \int dm \left(\frac{\partial u}{\partial t} \right)^2 = \frac{1}{2} \int_0^L \int_0^R \rho(2\pi r dr) dx \left(\frac{\partial u}{\partial t} \right)^2 = \frac{\rho A}{2} \int_0^L \left(\frac{\partial u}{\partial t} \right)^2 dx \quad (8)$$

where circular cross-section πR^2 has been replaced by a symbol A in Eq. (8). Similarly, the kinetic energy due to displacement η in the lateral direction is given by

$$\delta T = \frac{1}{2} \int dm \left(\frac{\partial \eta}{\partial t} \right)^2 = \frac{1}{2} \int_0^L \int_0^R \rho(2\pi r dr) dx \left(\nu r \frac{\partial^2 u}{\partial x \partial t} \right)^2 = \frac{\rho A \nu^2 R^2}{4} \int_0^L \left(\frac{\partial^2 u}{\partial x \partial t} \right)^2 dx \quad (9)$$

Next, Lord Rayleigh considered the case of a bar free at both ends with boundary conditions for which

$$u = U(t) \cos\left(\frac{n\pi x}{L}\right)$$

so that

$$\frac{\partial u}{\partial t} = \cos\left(\frac{n\pi x}{L}\right) \frac{\partial U}{\partial t}; \quad \frac{\partial^2 u}{\partial x \partial t} = -\frac{n\pi}{L} \sin\left(\frac{n\pi x}{L}\right) \frac{\partial U}{\partial t} \quad (10)$$

Substituting the first expression of Eq. (10) in Eq. (8) gives

$$T = \frac{\rho A}{2} \int_0^L \left(\frac{\partial U}{\partial t}\right)^2 \cos^2\left(\frac{n\pi x}{L}\right) dx = \frac{\rho A L}{4} \left(\frac{\partial U}{\partial t}\right)^2 \quad (11)$$

Substitution of the second expression of Eq. (10) in Eq. (9) gives

$$\delta T = \frac{\rho A v^2 R^2}{4} \frac{n^2 \pi^2}{L^2} \int_0^L \left(\frac{\partial U}{\partial t}\right)^2 \sin^2\left(\frac{n\pi x}{L}\right) dx = \frac{\rho A v^2 R^2}{8} \frac{n^2 \pi^2}{L} \left(\frac{\partial U}{\partial t}\right)^2 \quad (12)$$

The ratio of kinetic energies with and without lateral inertia is then given by

$$\frac{T + \delta T}{T} = 1 + \frac{n^2 \pi^2 v^2 R^2}{2 L^2}$$

From this, the ratio of time periods with and without the effect of lateral inertia is

$$\frac{t + \delta t}{t} = \sqrt{\frac{T + \delta T}{T}} = \sqrt{1 + \frac{n^2 \pi^2 v^2 R^2}{2 L^2}} = 1 + \frac{n^2 \pi^2 v^2 R^2}{4 L^2} \quad (13)$$

where higher-order terms of the binomial expansion have been neglected for a very thin rod. In the language of Lord Rayleigh [3], “*The effect of the inertia of the lateral motion is therefore to increase the period in the ratio of*”

$$1 : 1 + \frac{n^2 \pi^2 v^2 R^2}{4 L^2}$$

Lord Rayleigh [3] continues, “*This correction will be nearly insensible for the graver modes of bars of ordinary proportions of length to thickness.*” This is the extent to which Lord Rayleigh

included the effect of lateral inertia in his later editions of *The Theory of Sound*. By taking the reciprocal of the ratio of periods in Eq. (13), we can obtain the ratio of frequencies with and without lateral inertia

$$\frac{f_{Rayleigh}}{f_{classical}} = \frac{t}{t + \delta t} = \frac{1}{\sqrt{1 + \frac{n^2 \pi^2 v^2 R^2}{2 L^2}}}$$

Even though Lord Rayleigh included the effect of lateral inertia to derive the period of longitudinal vibration explicitly and natural frequency implicitly, he didn't derive the governing differential equation including the lateral inertia, as he did for the classical equation. This work was done by Augustus E. H. Love (1863 – 1940) in the second edition of his book, published in 1906.

Love used the variational principle to arrive at the classical wave equation in section 263 of his first edition [6]. However, he didn't proceed further to include lateral inertia. He did this work in the second edition of his book [7]. In section 278, he used the same variational principle, including the lateral inertia effect this time, to derive the governing differential equation. Following his derivation, let us consider a small element of thickness dx , dy and dz with x in the longitudinal direction and y and z in lateral directions. If u is the displacement of the left end of the element due to longitudinal vibration of the rod, then the strain in this direction is

$$\epsilon_x = \frac{\partial u}{\partial x}$$

Due to Poisson's effect, the lateral displacements of the elements are

$$v = -(v\epsilon_x)y = -vy \frac{\partial u}{\partial x}; \quad w = -(v\epsilon_x)z = -vz \frac{\partial u}{\partial x}$$

So, the kinetic energy of the rod is given by

$$\begin{aligned} T &= \frac{1}{2} \int_0^L \int \int (\rho \, dx dy dz) \left[\left(\frac{\partial u}{\partial t} \right)^2 + \left(\frac{\partial v}{\partial t} \right)^2 + \left(\frac{\partial w}{\partial t} \right)^2 \right] \\ &= \frac{1}{2} \int_0^L \int \int \rho \left[\left(\frac{\partial u}{\partial t} \right)^2 + v^2 (y^2 + z^2) \left(\frac{\partial^2 u}{\partial x \partial t} \right)^2 \right] dx dy dz \end{aligned}$$

Since

$$\int \int (y^2 + z^2) dy dz = A k_p^2$$

where A is the cross-sectional area of the element and k_p is the polar radius of gyration of the section. Now, the kinetic energy is

$$T = \frac{1}{2} \int_0^L \rho A \left[\left(\frac{\partial u}{\partial t} \right)^2 + v^2 k_p^2 \left(\frac{\partial^2 u}{\partial x \partial t} \right)^2 \right] dx$$

The strain energy of the rod due to motion in the longitudinal direction is

$$V = \frac{1}{2} \int_0^L E \epsilon_x^2 A dx = \frac{1}{2} \int_0^L EA \left(\frac{\partial u}{\partial x} \right)^2 dx$$

Using the variational equation of motion,

$$\delta \int (T - V) dt = 0$$

$$\delta \int dt \int_0^L \left[\frac{1}{2} \rho A \left\{ \left(\frac{\partial u}{\partial t} \right)^2 + v^2 k_p^2 \left(\frac{\partial^2 u}{\partial x \partial t} \right)^2 \right\} - \frac{1}{2} EA \left(\frac{\partial u}{\partial x} \right)^2 \right] dx = 0$$

$$\int dt \int_0^L \left[\rho A \left\{ \left(\frac{\partial u}{\partial t} \right) \delta \left(\frac{\partial u}{\partial t} \right) + v^2 k_p^2 \left(\frac{\partial^2 u}{\partial x \partial t} \right) \delta \left(\frac{\partial^2 u}{\partial x \partial t} \right) \right\} - EA \left(\frac{\partial u}{\partial x} \right) \delta \left(\frac{\partial u}{\partial x} \right) \right] dx = 0$$

Using the commutativity of variational and differential operators,

$$\int dt \int_0^L \left[\rho A \left\{ \left(\frac{\partial u}{\partial t} \right) \frac{\partial}{\partial t} (\delta u) + v^2 k_p^2 \left(\frac{\partial^2 u}{\partial x \partial t} \right) \frac{\partial^2}{\partial x \partial t} (\delta u) \right\} - EA \left(\frac{\partial u}{\partial x} \right) \frac{\partial}{\partial x} (\delta u) \right] dx = 0$$

On integrating by parts and setting the boundary terms to zero, we get

$$\int dt \int_0^L \left[\rho A \left\{ - \frac{\partial}{\partial t} \left(\frac{\partial u}{\partial t} \right) (\delta u) + v^2 k_p^2 \frac{\partial^2}{\partial x \partial t} \left(\frac{\partial^2 u}{\partial x \partial t} \right) (\delta u) \right\} + EA \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) (\delta u) \right] dx = 0$$

Since δu is arbitrary in the domain, which cannot be zero, so the integrand must vanish, providing

$$\rho \left(\frac{\partial^2 u}{\partial t^2} - v^2 k_p^2 \frac{\partial^4 u}{\partial x^2 \partial t^2} \right) = E \frac{\partial^2 u}{\partial x^2} \quad (14)$$

which was the final differential equation obtained by Love for longitudinally vibrating rods. Love [7] goes on to mention, “By retaining the term $\rho v^2 k_p^2 \partial^4 u / \partial x^2 \partial t^2$, we should obtain the correction of the velocity of wave-propagation which was found by Pochhammer (1876) and Chree (1886), of the correction of the frequency of free vibration which was calculated by Lord Rayleigh.”

2.2 The Second, Mindlin-Herrmann Theory

In a 1951 paper, Raymond D. Mindlin (1906-1987) and George Herrmann (1921-2007) used Hamilton's principle to obtain equations of motion governing longitudinal vibration of a rod. These equations were the first time of multi-modal theories in which two displacement equations

of motion coupling two displacement parameters u and w were obtained. Hereinafter, we closely follow the derivations by Mindlin and Herrmann [11].

The increase in energy within a volume V during the time interval from t_0 to t is given by

$$\Delta U = \int_{t_0}^t dt \int_V \frac{\partial}{\partial t} (T + W) d\tau \quad (15)$$

where $d\tau$ represents a differential volume element, and T and W denote the kinetic energy and the elastic strain energy per unit volume, respectively. When expressed in cylindrical coordinates r, θ, z , we have

$$T = \frac{1}{2} \rho (\dot{u}_r^2 + \dot{u}_\theta^2 + \dot{u}_z^2) \quad (16)$$

where u_r, u_θ, u_z represent the displacement components in the radial, angular, and axial directions, respectively, and a dot over a variable signifies its time derivative. Also,

$$\begin{aligned} \frac{\partial W}{\partial t} &= \frac{\partial W}{\partial \epsilon_{rr}} \dot{\epsilon}_{rr} + \frac{\partial W}{\partial \epsilon_{\theta\theta}} \dot{\epsilon}_{\theta\theta} + \frac{\partial W}{\partial \epsilon_{zz}} \dot{\epsilon}_{zz} + \frac{\partial W}{\partial \epsilon_{\theta z}} \dot{\epsilon}_{\theta z} + \frac{\partial W}{\partial \epsilon_{zr}} \dot{\epsilon}_{zr} + \frac{\partial W}{\partial \epsilon_{r\theta}} \dot{\epsilon}_{r\theta} \\ &= \sigma_{rr} \dot{\epsilon}_{rr} + \sigma_{\theta\theta} \dot{\epsilon}_{\theta\theta} + \sigma_{zz} \dot{\epsilon}_{zz} + \sigma_{\theta z} \dot{\epsilon}_{\theta z} + \sigma_{zr} \dot{\epsilon}_{zr} + \sigma_{r\theta} \dot{\epsilon}_{r\theta} \end{aligned} \quad (17)$$

where $\epsilon_{rr}, \epsilon_{\theta\theta}, \epsilon_{zz}, \epsilon_{\theta z}, \epsilon_{zr}, \epsilon_{r\theta}$ and $\sigma_{rr}, \sigma_{\theta\theta}, \sigma_{zz}, \sigma_{\theta z}, \sigma_{zr}, \sigma_{r\theta}$ are the components of strain and stress, respectively.

From the theory of elasticity, strains are related to displacements by relations

$$\begin{aligned} \epsilon_{rr} &= \frac{\partial u_r}{\partial r}; \quad \epsilon_{\theta\theta} = \frac{u_r}{r} + \frac{\partial u_\theta}{r \partial \theta}; \quad \epsilon_{zz} = \frac{\partial u_z}{\partial z}; \\ \epsilon_{\theta z} &= \frac{\partial u_\theta}{\partial z} + \frac{\partial u_z}{r \partial \theta}; \quad \epsilon_{zr} = \frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r}; \quad \epsilon_{r\theta} = \frac{\partial u_r}{r \partial \theta} + \frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r} \end{aligned} \quad (18)$$

Let us examine a bar with a circular cross-section of radius a , and consider, as an approximation, axially symmetric displacements

$$u_r = \frac{r}{a} u(z, t); \quad u_\theta = 0; \quad u_z = w(z, t) \quad (19)$$

Using Eq. (19) in Eq. (18), we get

$$\begin{aligned} \epsilon_{rr} &= \frac{1}{a} u(z, t); \quad \epsilon_{\theta\theta} = \frac{1}{a} u(z, t); \quad \epsilon_{zz} = \frac{\partial w}{\partial z}; \\ \epsilon_{\theta z} &= 0; \quad \epsilon_{zr} = \frac{r}{a} \frac{\partial u}{\partial z}; \quad \epsilon_{r\theta} = 0 \end{aligned} \quad (20)$$

Substituting Eq. (20) in Eq. (17), we arrive at

$$\frac{\partial W}{\partial t} = \sigma_{rr} \frac{\dot{u}}{a} + \sigma_{\theta\theta} \frac{\dot{u}}{a} + \sigma_{zz} \frac{\partial \dot{w}}{\partial z} + \sigma_{zr} \frac{r}{a} \frac{\partial \dot{u}}{\partial z} \quad (21)$$

Differentiating Eq. (16) with respect to time, we get

$$\frac{\partial T}{\partial t} = \rho(\dot{u}_r \ddot{u}_r + \dot{u}_\theta \ddot{u}_\theta + \dot{u}_z \ddot{u}_z) \quad (22)$$

Using Eq. (19) in Eq. (22), we get

$$\frac{\partial T}{\partial t} = \rho \left(\frac{r^2}{a^2} \dot{u} \ddot{u} + \dot{w} \ddot{w} \right) \quad (23)$$

Substituting Eqs. (21) and (23) in Eq. (15), the increment of energy changes to

$$\begin{aligned} \Delta U &= \int_{t_0}^t dt \int_0^{2\pi} \int_0^a \int_b^c \left[\rho \left(\frac{r^2}{a^2} \dot{u} \ddot{u} + \dot{w} \ddot{w} \right) + \sigma_{rr} \frac{\dot{u}}{a} + \sigma_{\theta\theta} \frac{\dot{u}}{a} + \sigma_{zz} \frac{\partial \dot{w}}{\partial z} \right. \\ &\quad \left. + \sigma_{zr} \frac{r}{a} \frac{\partial \dot{u}}{\partial z} \right] r dr dz d\theta \end{aligned} \quad (24)$$

where $c - b$ is the length of the bar and the volume of element $d\tau$ has been replaced by $rdrd\theta dz$.

The integrand in Eq. (24) is independent of θ , so integrating with respect to θ gives the multiplication factor of 2π . Thus,

$$\Delta U = 2\pi \int_{t_0}^t dt \int_0^a \int_b^c \left[\rho \left(\frac{r^2}{a^2} \dot{u}\ddot{u} + \dot{w}\ddot{w} \right) + \sigma_{rr} \frac{\dot{u}}{a} + \sigma_{\theta\theta} \frac{\dot{u}}{a} + \sigma_{zz} \frac{\partial \dot{w}}{\partial z} + \sigma_{zr} \frac{r}{a} \frac{\partial \dot{u}}{\partial z} \right] r dr dz \quad (25)$$

Now, we are led to the definition of four “bar-stress components”, as they are referred to by Mindlin and Herrmann [11]:

$$\begin{aligned} P_r &= \int_0^a \sigma_{rr} r dr; & P_\theta &= \int_0^a \sigma_{\theta\theta} r dr; \\ P_z &= \int_0^a \sigma_{zz} r dr; & Q &= \int_0^a \sigma_{zr} \frac{1}{a} r^2 dr \end{aligned} \quad (26)$$

On integrating the first two terms of the integrand of Eq. (25) and substituting Eq. (26), we get

$$\frac{\Delta U}{2\pi} = \int_{t_0}^t dt \int_b^c \rho \left(\frac{a^2}{4} \dot{u}\ddot{u} + \frac{a^2}{2} \dot{w}\ddot{w} \right) dz + \int_{t_0}^t dt \int_b^c \left[\frac{1}{a} \dot{u} (P_r + P_\theta) + \frac{\partial \dot{w}}{\partial z} P_z + \frac{\partial \dot{u}}{\partial z} Q \right] dz \quad (27)$$

Finally, integrating the last two terms of Eq. (27) and rearranging,

$$\begin{aligned} \frac{\Delta U}{2\pi} &= \int_{t_0}^t dt \int_b^c \dot{u} \left(\frac{\rho a^2}{4} \ddot{u} + \frac{P_r + P_\theta}{a} - \frac{\partial Q}{\partial z} \right) dz + \int_{t_0}^t dt \int_b^c \dot{w} \left(\frac{\rho a^2}{2} \ddot{w} - \frac{\partial P_z}{\partial z} \right) dz \\ &\quad + \int_{t_0}^t [\dot{w} P_z + \dot{u} Q] \Big|_b^c dt \end{aligned} \quad (28)$$

The terms within parentheses in Eq. (28) can be expressed in terms of surface tractions by applying the stress-based equations of motion. For the axially symmetric case, these equations, from the general theory, simplify to

$$\begin{aligned}\frac{\partial \sigma_{rr}}{\partial r} + \frac{\partial \sigma_{zr}}{\partial z} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} &= \rho \ddot{u}_r \\ \frac{\partial \sigma_{zr}}{\partial r} + \frac{\partial \sigma_{zz}}{\partial z} + \frac{\sigma_{zr}}{r} &= \rho \ddot{u}_z\end{aligned}\quad (29)$$

Using Eq. (19) in Eq. (29), we get

$$\begin{aligned}\frac{\partial \sigma_{rr}}{\partial r} + \frac{\partial \sigma_{zr}}{\partial z} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} &= \rho \frac{r}{a} \ddot{u} \\ \frac{\partial \sigma_{zr}}{\partial r} + \frac{\partial \sigma_{zz}}{\partial z} + \frac{\sigma_{zr}}{r} &= \rho \ddot{w}\end{aligned}\quad (30)$$

To transform these into the bar equations of Eq. (26), the first of Eq. (30) is multiplied by r^2 , and the second by r , after which both are integrated across the radius of the cross-section. This procedure is motivated, naturally, by the structure of the expressions in parentheses in Eq. (28).

$$\begin{aligned}[r^2 \sigma_{rr}]_0^a - \int_0^a 2r \sigma_{rr} dr + \int_0^a r^2 \frac{\partial \sigma_{zr}}{\partial z} dr + \int_0^a \sigma_{rr} r dr - \int_0^a \sigma_{\theta\theta} r dr &= \rho \ddot{u} \frac{a^3}{4} \\ [r \sigma_{zr}]_0^a - \int_0^a \sigma_{zr} dr + \int_0^a r \frac{\partial \sigma_{zz}}{\partial z} dr + \int_0^a \sigma_{zr} dr &= \rho \ddot{w} \frac{a^2}{2}\end{aligned}\quad (31)$$

On dividing the first of Eq. (31) throughout by a and making use of Eq. (26) in both of Eq. (31),

$$\begin{aligned}\frac{\partial Q}{\partial z} - \frac{1}{a}(P_r + P_\theta) + aR &= \frac{\rho a^2 \ddot{u}}{4} \\ \frac{\partial P_z}{\partial z} + aZ &= \frac{\rho a^2 \ddot{w}}{4}\end{aligned}\quad (32)$$

where

$$R = \sigma_{rr}|_{r=a} ; \quad Z = \sigma_{zr}|_{r=a} \quad (33)$$

represent the radial and axial components of the traction acting on the cylindrical surface of the bar. Eq. (32) provides the stress equations of motion of the bar, consistent with the assumptions of Eq. (19).

Substituting Eq. (32) in Eq. (28), we get

$$\frac{\Delta U}{2\pi} = a \int_{t_0}^t dt \int_b^c (\dot{u}R + \dot{w}Z) dz + \int_{t_0}^t [\dot{w}P_z + \dot{u}Q] \Big|_b^c dt \quad (34)$$

This is essentially a statement of the work-energy theorem, which asserts that the increase in a system's energy equals the work done on it. Based on the principle of uniqueness, the appropriate boundary and initial conditions must be specified: along the length of the bar, one quantity of each of the products $\dot{u}R$ and $\dot{w}Z$ must be prescribed at every point; at each end of the bar, one component of each of the products $\dot{w}P_z$ and $\dot{u}Q$ must be given. Additionally, the initial displacements and velocities must also be defined.

To derive the equations of motion in terms of displacement, we consider Hooke's Law of the general theory in the isotropic, axisymmetric case

$$\begin{aligned} \sigma_{rr} &= \lambda(\epsilon_{rr} + \epsilon_{\theta\theta} + \epsilon_{zz}) + 2\mu\epsilon_{rr}; & \sigma_{\theta\theta} &= \lambda(\epsilon_{rr} + \epsilon_{\theta\theta} + \epsilon_{zz}) + 2\mu\epsilon_{\theta\theta}; \\ \sigma_{zz} &= \lambda(\epsilon_{rr} + \epsilon_{\theta\theta} + \epsilon_{zz}) + 2\mu\epsilon_{zz}; & \sigma_{zr} &= \mu\epsilon_{zr} \end{aligned} \quad (35)$$

where λ and μ are Lamé's constants. Using Eq. (20) in Eq. (35) and then integrating across the radius of the cross-section in accordance with Eq. (26).

$$\begin{aligned} 2P_r &= 2P_\theta = 2a(\lambda + \mu)u + a^2\lambda \frac{\partial w}{\partial z} \\ 2P_z &= 2a\lambda u + a^2(\lambda + 2\mu) \frac{\partial w}{\partial z}; \end{aligned} \quad (36)$$

$$4Q = a^2 \mu \frac{\partial u}{\partial z}$$

In order to make the wave-velocities of the approximate theory fit those of the Pochhammer–Chree solution more closely, constants κ and κ_1 are introduced into stress-displacement relations (36).

$$\begin{aligned} 2P_r = 2P_\theta &= \kappa_1^2 \left[2a(\lambda + \mu)u + a^2 \lambda \frac{\partial w}{\partial z} \right] \\ 2P_z &= 2a\lambda u + a^2(\lambda + 2\mu) \frac{\partial w}{\partial z}; \\ 4Q &= \kappa^2 a^2 \mu \frac{\partial u}{\partial z} \end{aligned} \quad (37)$$

Combining Eq. (32) and Eq. (37), the displacement equations of motion are obtained as follows:

$$\begin{aligned} a^2 \kappa^2 \mu \frac{\partial^2 u}{\partial z^2} - 8\kappa_1^2(\lambda + \mu)u - 4a\kappa_1^2 \lambda \frac{\partial w}{\partial z} + 4aR &= \rho a^2 \ddot{u} \\ 2a\lambda \frac{\partial u}{\partial z} + a^2(\lambda + 2\mu) \frac{\partial^2 w}{\partial z^2} + 2aZ &= \rho a^2 \ddot{w} \end{aligned} \quad (38)$$

The right side, $\rho a^2 \ddot{u}$, of the first of Eq. (38) is the radial inertial correction and the first term, $a^2 \kappa^2 \mu \frac{\partial^2 u}{\partial z^2}$, of the left side is the radial shear correction.

A single equation on w may be obtained by eliminating u from Eq. (38) with the result

$$\begin{aligned} \frac{E}{\rho} \frac{\partial^2 w}{\partial z^2} - \frac{\partial^2 w}{\partial t^2} + \frac{2Z}{\rho a} \\ + \frac{a^2 \rho}{8\kappa_1^2(\lambda + \mu)} \frac{\partial^2}{\partial t^2} \left[\frac{\lambda + 2\mu}{\rho} \frac{\partial^2 w}{\partial z^2} - \frac{\partial^2 w}{\partial t^2} + \frac{2Z}{\rho a} \right] \\ - \frac{a^2 \kappa^2 \mu}{8\kappa_1^2(\lambda + \mu)} \frac{\partial^2}{\partial z^2} \left[\frac{\lambda + 2\mu}{\rho} \frac{\partial^2 w}{\partial z^2} - \frac{\partial^2 w}{\partial t^2} + \frac{2Z}{\rho a} \right] \\ + \frac{\lambda}{\rho(\lambda + \mu)\kappa_1^2} \frac{\partial R}{\partial z} = 0 \end{aligned} \quad (39)$$

The first line of Eq. (39) contains classical terms, while the second and third lines are the radial inertia and radial shear corrections, respectively.

Considering the assumption $u = -va \frac{\partial w}{\partial z}$ made by Lord Rayleigh [3], and also by Love [7], the second of Eq. (38) converts to

$$\frac{\partial^2 w}{\partial t^2} = \left(\frac{\lambda + 2\mu}{\rho} - \frac{2\lambda\nu}{\rho} \right) \frac{\partial^2 w}{\partial z^2} + \frac{2}{\rho a} \quad (40)$$

With the application of Eq. (40) in Eq. (39), removal of shear correction and surface traction, and assuming $\kappa_1^2 = 1$, Eq. (39) converts to Love's Eq. (14). If the shear correction terms are retained, with dropping of surface traction only, and assuming $\kappa_1^2 = 1 = \kappa^2$, Eq. (39) converts to Bishop's equation, which is the subject of the next theory.

2.3 The Third, Bishop Theory

Richard E. D. Bishop (1925 – 1989) added the effect of radial shear in his 1952 paper [9], “Longitudinal Waves in Beams,” improving the results of longitudinal vibration at higher frequencies. Bishop considered a prismatic beam and applied Newton's laws to a differential volume element $dx dy dz$ in the lateral directions along the Y and Z axes. If the conditions of the stresses at the center of the element are three normal stresses $\sigma_{xx}, \sigma_{yy}, \sigma_{zz}$ and three shear stresses $\tau_{yz}, \tau_{xz}, \tau_{xy}$. Applying force balance in the Y and Z directions gives

$$\begin{aligned} \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} &= \rho \frac{\partial^2 v}{\partial t^2} \\ \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} &= \rho \frac{\partial^2 w}{\partial t^2} \end{aligned} \quad (41)$$

If ν is the Poisson's ratio of the material of the rod, then the lateral displacements v and w in the Y and Z directions are

$$v = -(v\epsilon_x)y = -vy \frac{\partial u}{\partial x}; \quad w = -(v\epsilon_x)z = -vz \frac{\partial u}{\partial x} \quad (42)$$

The shear stresses $\tau_{yz}, \tau_{xz}, \tau_{xy}$ are then given by

$$\begin{aligned} \tau_{xy} &= \mu_{xy} = \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \\ \tau_{xz} &= \mu \gamma_{xz} = \mu \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) \\ \tau_{yz} &= \mu \gamma_{yz} = \mu \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) \end{aligned} \quad (43)$$

where μ is Lamé's constant. On the assumption that plane sections remain plane and their shapes don't change, the longitudinal displacement u should not vary in the Y and Z directions and shear strain γ_{yz} should also vanish.

$$\frac{\partial u}{\partial y} = 0 = \frac{\partial u}{\partial z}; \quad \gamma_{yz} = 0 \quad (44)$$

With the substitution of Eqs. (42) and (44) in Eq. (43),

$$\begin{aligned} \tau_{xy} &= \mu \frac{\partial v}{\partial x} = -v\mu y \frac{\partial^2 u}{\partial x^2} \\ \tau_{xz} &= \mu \frac{\partial w}{\partial x} = -v\mu z \frac{\partial^2 u}{\partial x^2} \\ \tau_{yz} &= 0 \end{aligned} \quad (45)$$

Substituting Eqs. (42) and (45) into the first of Eq. (41),

$$\frac{\partial \sigma_{yy}}{\partial y} = -\rho v y \frac{\partial^3 u}{\partial x \partial t^2} + v\mu y \frac{\partial^3 u}{\partial x^3}$$

On integration with respect to y,

$$\sigma_{yy} = -\frac{\rho\nu}{2}(y^2 - y_e^2)\frac{\partial^3 u}{\partial x \partial t^2} + \frac{\nu\mu}{2}(y^2 - y_e^2)\frac{\partial^3 u}{\partial x^3} \quad (46)$$

where the terms containing y_e^2 represents the constant of integration. Since the stress at the surface of the rod is zero, i.e. $\sigma_{yy} = 0$ at the surface, y_e represents the y-coordinates of the surface.

Substituting Eqs. (42) and (45) into the second of Eq. (41),

$$\frac{\partial \sigma_{zz}}{\partial z} = -\rho\nu z \frac{\partial^3 u}{\partial x \partial t^2} + \nu\mu z \frac{\partial^3 u}{\partial x^3}$$

On integration with respect to z,

$$\sigma_{zz} = -\frac{\rho\nu}{2}(z^2 - z_e^2)\frac{\partial^3 u}{\partial x \partial t^2} + \frac{\nu\mu}{2}(z^2 - z_e^2)\frac{\partial^3 u}{\partial x^3} \quad (47)$$

where the terms containing z_e^2 represents the constant of integration. Since the stress at the surface of the rod is zero i.e. $\sigma_{zz} = 0$ at the surface, z_e represents the z-coordinates of the surface.

Using generalized Hooke's law to the element in the X-direction,

$$E \frac{\partial u}{\partial x} = \sigma_{xx} - \nu(\sigma_{yy} + \sigma_{zz})$$

so that

$$\sigma_{xx} = E \frac{\partial u}{\partial x} - \frac{\rho\nu^2}{2}[(y^2 + z^2) - (y_e^2 + z_e^2)]\frac{\partial^3 u}{\partial x \partial t^2} + \frac{\nu^2\mu}{2}[(y^2 + z^2) - (y_e^2 + z_e^2)]\frac{\partial^3 u}{\partial x^3} \quad (48)$$

Applying the law of motion to the thin slice in the X-direction,

$$\begin{aligned} d \left[\int_A \sigma_{xx} dA \right] &= \rho A dx \frac{\partial^2 u}{\partial t^2} \\ \int_A \frac{\partial \sigma_{xx}}{\partial x} dA &= \rho A \frac{\partial^2 u}{\partial t^2} \end{aligned} \quad (49)$$

Substituting σ_{xx} from Eq. (48) in Eq. (49), carrying out the indicated differentiation and then finally integrating,

$$\begin{aligned} \rho A \frac{\partial^2 u}{\partial t^2} = & EA \frac{\partial^2 u}{\partial x^2} - v^2 \rho \frac{\partial^4 u}{\partial x^2 \partial t^2} \int_A \frac{1}{2} [(y^2 + z^2) - (y_e^2 + z_e^2)] dA \\ & + v^2 \mu \frac{\partial^4 u}{\partial x^4} \int_A \frac{1}{2} [(y^2 + z^2) - (y_e^2 + z_e^2)] dA \end{aligned} \quad (50)$$

To simplify the above Eq. (48), Bishop considers only those sections for which

$$\int (y_0^2 + z_0^2) dA = 3 \int (y^2 + z^2) dA = 3Ak_p^2 \quad (51)$$

This property is satisfied by square, rectangular, circular cross-sections, and probably for I-sections. Substituting Eq. (51) in Eq. (50) gives the equation of motion as follows:

$$\rho A \frac{\partial^2 u}{\partial t^2} = EA \frac{\partial^2 u}{\partial x^2} + v^2 \rho A k_p^2 \frac{\partial^4 u}{\partial x^2 \partial t^2} - v^2 \mu A k_p^2 \frac{\partial^4 u}{\partial x^4}$$

Further simplification results

$$\frac{\partial^2 u}{\partial t^2} = c_0^2 \frac{\partial^2 u}{\partial x^2} + v^2 k_p^2 \frac{\partial^4 u}{\partial x^2 \partial t^2} - v^2 k_p^2 c_2^2 \frac{\partial^4 u}{\partial x^4} \quad (52)$$

where $c_0 [= \sqrt{E/\rho}]$ is the “bar velocity”, k_p is the polar radius of gyration and $c_2 [= \sqrt{\mu/\rho}]$ is the speed of distortion waves in an infinite elastic medium. Eq. (52) is mainly related to Bishop’s theory in the literature. However, he carried out further improvement by improving Eq. (42) with the consideration of

$$\begin{aligned} E \frac{\partial v}{\partial y} &= \sigma_{yy} - v(\sigma_{xx} + \sigma_{zz}) \\ E \frac{\partial w}{\partial z} &= \sigma_{zz} - v(\sigma_{xx} + \sigma_{yy}) \end{aligned} \quad (53)$$

Substituting $\sigma_{xx}, \sigma_{yy}, \sigma_{zz}$ from Eqs. (46) – (48) in Eq. (53), and then integrating,

$$\begin{aligned} v &= -vy \frac{\partial u}{\partial x} + \left(y_e^2 y - \frac{y^3}{3} \right) P + (z_e^2 - z^2) y Q \\ w &= -vz \frac{\partial u}{\partial x} + \left(z_e^2 z - \frac{z^3}{3} \right) P + (y_e^2 - y^2) z Q \end{aligned} \quad (54)$$

where

$$P = \frac{\nu(1-\nu^2)}{2c_0^2} \left[\frac{\partial^3 u}{\partial x \partial t^2} - c_2^2 \frac{\partial^3 u}{\partial x^3} \right]; \quad Q = -\frac{\nu^2(1+\nu)}{2c_0^2} \left[\frac{\partial^3 u}{\partial x \partial t^2} - c_2^2 \frac{\partial^3 u}{\partial x^3} \right]$$

Proceeding in a similar manner henceforth, and considering the particular case of a circular cross-section, one can obtain the equation of motion as

$$\begin{aligned} \frac{\partial^2 u}{\partial t^2} &= c_0^2 \frac{\partial^2 u}{\partial x^2} + \frac{\nu^2 a^2}{2} \frac{\partial^4 u}{\partial x^2 \partial t^2} - \frac{\nu^2 a^2 c_2^2}{2} \frac{\partial^4 u}{\partial x^4} \\ &+ \frac{\nu^2 a^4}{6c_0^2} (2\nu^2 + \nu - 1) \left[\frac{\partial^6 u}{\partial x^2 \partial t^4} - 2c_2^2 \frac{\partial^6 u}{\partial x^4 \partial t^2} + c_2^4 \frac{\partial^6 u}{\partial x^6} \right] \end{aligned} \quad (55)$$

In a similar manner, Bishop obtained the improvement on Love's equation (14) (or merely dropping the radial shear terms in Eq. (55)),

$$\frac{\partial^2 u}{\partial t^2} = c_0^2 \frac{\partial^2 u}{\partial x^2} + \frac{\nu^2 a^2}{2} \frac{\partial^4 u}{\partial x^2 \partial t^2} + \frac{\nu^2 a^4}{6c_0^2} (2\nu^2 + \nu - 1) \frac{\partial^6 u}{\partial x^2 \partial t^4} \quad (56)$$

2.4 The Fourth, Zachmanoglou-Volterra Theory

The theory [12] by Eleftherios C. “Terry” Zachmanoglou (1934 – 2018) and Enrico Volterra (1905 – 1973) is the first time of non-planar cross-sectional theory for the longitudinal vibration of rod, in which displacement components were considered of the form

$$u_x(x, r, \theta; t) = v_x(x; t) + \frac{r^2}{a^2} f_x(x; t)$$

$$u_r(x, r, \theta; t) = r\lambda_r(x; t) + r^3 f_r(x, t) \quad (57)$$

$$u_\theta(x, r, \theta; t) = 0$$

where v_x, f_x, λ_r and f_r are four functions of x and t , whose explicit forms are not known a priori.

The radial stress σ_r is given by

$$\begin{aligned} \sigma_r &= (\lambda + 2G)\epsilon_{rr} + \lambda(\epsilon_{\theta\theta} + \epsilon_{xx}) \\ &= (\lambda + 2G) \frac{\partial u_r}{\partial r} + \lambda \left(\frac{u_r}{r} + \frac{\partial u_\theta}{r \partial \theta} + \frac{\partial u_x}{\partial x} \right) \\ &= (\lambda + 2G)(\lambda_r + 3r^2 f_r) + \lambda \left(\lambda_r + r^2 f_r + \frac{\partial v_x}{\partial x} + \frac{r^2}{a^2} \frac{\partial f_x}{\partial x} \right) \end{aligned} \quad (58)$$

where λ and G are Lamé's constants. Considering the lateral surface of the bar is free from radial stress, i.e. $(\sigma_r)_{r=a} = 0$, one obtains from Eq. (58)

$$f_r = -\frac{1}{a^2} \alpha \left[\lambda_r + v \left(\frac{\partial v_x}{\partial x} + \frac{\partial f_x}{\partial x} \right) \right] \quad (59)$$

where

$$\alpha = \frac{\lambda + G}{2\lambda + 3G} = \frac{1}{3 - 2\nu}$$

Using Eq. (59) in Eq. (57), we get

$$\begin{aligned} u_x(x, r, \theta; t) &= v_x(x; t) + \frac{r^2}{a^2} f_x(x; t) \\ u_r(x, r, \theta; t) &= r\lambda_r(x; t) - \frac{r^3}{a^2} \alpha \left[\lambda_r(x; t) + v \left\{ \frac{\partial v_x}{\partial x}(x; t) + \frac{\partial f_x}{\partial x}(x; t) \right\} \right] \\ u_\theta(x, r, \theta; t) &= 0 \end{aligned} \quad (60)$$

in terms of the three unknowns v_x, f_x and λ_r .

From Eq. (60), the components of strain are derived as

$$\begin{aligned}
\epsilon_{xx} &= \frac{\partial u_x}{\partial x} = \frac{\partial v_x}{\partial x} + \frac{r^2}{a^2} \frac{\partial f_x}{\partial x} \\
\epsilon_{rr} &= \frac{\partial u_r}{\partial r} = \lambda_r - \frac{3r^2}{a^2} \alpha \left[\lambda_r + v \left\{ \frac{\partial v_x}{\partial x} + \frac{\partial f_x}{\partial x} \right\} \right] \\
\epsilon_{\theta\theta} &= \frac{u_r}{r} + \frac{\partial u_\theta}{r \partial \theta} = \lambda_r - \frac{r^2}{a^2} \alpha \left[\lambda_r + v \left\{ \frac{\partial v_x}{\partial x} + \frac{\partial f_x}{\partial x} \right\} \right] \\
\gamma_{xr} &= \frac{\partial u_x}{\partial r} + \frac{\partial u_r}{\partial x} = \frac{2r}{a^2} f_x + r \frac{\partial \lambda_r}{\partial x} - \frac{r^3}{a^2} \alpha \frac{\partial}{\partial x} \left[\lambda_r + v \left\{ \frac{\partial v_x}{\partial x} + \frac{\partial f_x}{\partial x} \right\} \right] \\
\gamma_{r\theta} &= \gamma_{x\theta} = 0
\end{aligned} \tag{61}$$

The components of the particle velocity are given by

$$\begin{aligned}
\frac{\partial u_x}{\partial t} &= \frac{\partial v_x}{\partial t} + \frac{r^2}{a^2} \frac{\partial f_x}{\partial t} \\
\frac{\partial u_r}{\partial t} &= r \frac{\partial \lambda_r}{\partial t} - \frac{r^3}{a^2} \alpha \frac{\partial}{\partial t} \left[\lambda_r + v \left\{ \frac{\partial v_x}{\partial x} + \frac{\partial f_x}{\partial x} \right\} \right] \\
\frac{\partial u_\theta}{\partial t} &= 0
\end{aligned} \tag{62}$$

Let's define a factor K as

$$K = 1 + v \left(\frac{\frac{\partial v_x}{\partial x} + \frac{\partial f_x}{\partial x}}{\lambda_r} \right) \tag{63}$$

Then, the fourth of Eq. (61) and the second of Eq. (62) convert to

$$\begin{aligned}
\gamma_{xr} &= \frac{2r}{a^2} f_x + r \frac{\partial \lambda_r}{\partial x} - \frac{r^3}{a^2} K \alpha \frac{\partial \lambda_r}{\partial x} \\
\frac{\partial u_r}{\partial t} &= r \frac{\partial \lambda_r}{\partial t} - \frac{r^3}{a^2} K \alpha \frac{\partial \lambda_r}{\partial t}
\end{aligned} \tag{64}$$

In deriving the equations of motion, K is taken to be constant. The Potential Energy U , and the Kinetic Energy T are given by

$$U = - \int_V \left[\frac{\lambda + 2G}{2} (\epsilon_x^2 + \epsilon_r^2 + \epsilon_\theta^2) + \lambda (\epsilon_x \epsilon_r + \epsilon_r \epsilon_\theta + \epsilon_\theta \epsilon_x) + \frac{G}{2} (\lambda_{xr}^2 + \lambda_{x\theta}^2 + \lambda_{r\theta}^2) \right] r d\theta dr dx$$

$$T = \frac{\rho}{2} \int_V \left[\left(\frac{\partial u_x}{\partial t} \right)^2 + \left(\frac{\partial u_r}{\partial t} \right)^2 + \left(\frac{\partial u_\theta}{\partial t} \right)^2 \right] r d\theta dr dx$$

The energies in terms of three unknown functions v_x, f_x and λ_r can be written as

$$U = -\pi a^2 \int_0^L \left[\frac{E}{2} \left\{ (1 + \nu_1^2 N_1) \left(\frac{\partial v_x}{\partial x} \right)^2 + \frac{N_2}{3} \left(\frac{\partial f_x}{\partial x} \right)^2 + N_1 (\lambda_r) + N_3 \frac{\partial v_x}{\partial x} \frac{\partial f_x}{\partial x} \right. \right. \quad (65)$$

$$\left. + 2\nu N_1 \lambda_r \frac{\partial v_x}{\partial x} + 2\nu N_4 \lambda_r \frac{\partial f_x}{\partial x} \right\} + \frac{G}{2} \left\{ \frac{2}{a^2} (f_x)^2 + 2 \left(1 - \frac{2}{3} K\alpha \right) f_x \frac{\partial \lambda_r}{\partial x} + \frac{a^2}{2} \left(1 + \frac{K^2 \alpha^2}{2} - \frac{4K\alpha}{3} \right) \left(\frac{\partial \lambda_r}{\partial x} \right)^2 \right\} \right] dx$$

$$T = \frac{\pi a^2 \rho}{2} \int_0^L \left[\left(\frac{\partial v_x}{\partial t} \right)^2 + \frac{1}{3} \left(\frac{\partial f_x}{\partial t} \right)^2 + \frac{\partial v_x}{\partial t} \frac{\partial f_x}{\partial t} + \frac{a^2}{2} \left(1 + \frac{K^2 \alpha^2}{2} - \frac{4K\alpha}{3} \right) \left(\frac{\partial \lambda_r}{\partial x} \right)^2 \right] dx \quad (66)$$

The quantities $N_i (i = 1, 2, 3, 4)$ in Eq. (65) are factors depending only on Poisson's ratio

$$N_1 = \frac{1 - \nu}{(1 + \nu)(1 - 2\nu)} \left[\frac{2}{1 - \nu} (1 - 2\alpha) + \alpha^2 \frac{10 - 4\nu}{3(1 - \nu)} \right]$$

$$N_2 = \frac{1 - \nu}{(1 + \nu)(1 - 2\nu)} \left[1 - \frac{8\alpha\nu^2}{1 - \nu} + \nu^2 \alpha^2 \frac{10 - 4\nu}{1 - \nu} \right]$$

$$N_3 = \frac{1 - \nu}{(1 + \nu)(1 - 2\nu)} \left[1 - \frac{20\alpha\nu^2}{3(1 - \nu)} + 2\nu^2 \alpha^2 \frac{10 - 4\nu}{3(1 - \nu)} \right]$$

$$N_4 = \frac{1 - \nu}{(1 + \nu)(1 - 2\nu)} \left[\frac{3 - 10\alpha}{3(1 - \nu)} + \alpha^2 \frac{10 - 4\nu}{3(1 - \nu)} \right]$$

Under the assumption that the bar is free of body forces and surface tractions, Hamilton's principle is applied in the form

$$\delta \int_{t_1}^{t_2} (U + T) dt = 0$$

And the following equations of motion are obtained

$$\begin{aligned} E(1 + \nu^2 N_1) \frac{\partial^2 v_x}{\partial x^2} + E \frac{N_3}{2} \frac{\partial^2 f_x}{\partial x^2} + E \nu N_1 \frac{\partial \lambda_r}{\partial x} &= \rho \frac{\partial^2 v_x}{\partial t^2} + \frac{\rho}{2} \frac{\partial^2 f_x}{\partial t^2} \\ E \frac{N_2}{3} \frac{\partial^2 f_x}{\partial x^2} + E \frac{N_3}{2} \frac{\partial^2 v_x}{\partial x^2} + E \nu N_4 \frac{\partial \lambda_r}{\partial x} - \frac{2G}{a^2} f_x - G \left(1 - \frac{2}{3} K \alpha\right) \frac{\partial \lambda_r}{\partial x} &= \frac{\rho}{3} \frac{\partial^2 f_x}{\partial t^2} + \frac{\rho}{2} \frac{\partial^2 v_x}{\partial t^2} \\ -E N_1 \lambda_r - E \nu N_1 \frac{\partial v_x}{\partial x} - E \nu N_4 \frac{\partial f_x}{\partial x} + G \left(1 - \frac{2}{3} K \alpha\right) \frac{\partial f_x}{\partial x} & \\ + \frac{G a^2}{2} \left(1 + \frac{K^2 \alpha^2}{2} - \frac{4K \alpha}{3}\right) \frac{\partial^2 \lambda_r}{\partial x^2} &= \frac{\rho a^2}{2} \left(1 + \frac{K^2 \alpha^2}{2} - \frac{4K \alpha}{3}\right) \frac{\partial^2 \lambda_r}{\partial t^2} \end{aligned} \quad (67)$$

along with the following corresponding boundary conditions at $x = 0, x = L$

$$\begin{aligned} \left[(1 + \nu^2 N_1) \frac{\partial v_x}{\partial x} + \frac{N_3}{2} \frac{\partial f_x}{\partial x} + \nu N_1 \lambda_r \right] \delta v_x &= 0 \\ \left[\frac{N_2}{3} \frac{\partial f_x}{\partial x} + \frac{N_3}{2} \frac{\partial v_x}{\partial x} + \nu N_4 \lambda_r \right] \delta f_x &= 0 \\ \left[\left(1 - \frac{2}{3} K \alpha\right) f_x + \frac{a^2}{2} \left(1 + \frac{K^2 \alpha^2}{2} - \frac{4K \alpha}{3}\right) \frac{\partial \lambda_r}{\partial x} \right] \delta \lambda_r &= 0 \end{aligned} \quad (68)$$

The equations of motion given in Eq. (67), in combination with boundary conditions in Eq. (68) form a well-posed boundary value problem for a rod of finite length.

2.5 The Fifth, Mindlin-McNiven Theory

Mindlin-Herrmann theory [11] of rod vibration takes into account the coupling between longitudinal and radial modes of axially symmetric vibration of elastic rods but ignore the axial shear mode which can couple with former two modes. Mindlin-McNiven theory [14] includes the effect of the additional, axial shear mode. The derivation is based on expansions of the displacements in the series of Jacobi polynomials in the radial co-ordinate, retaining only the first

terms representative of axial, radial, and axial shear deformation. To compensate for the errors introduced by omission of higher order terms, four factors are introduced in such a way that the behavior of the first three branches of the Pochhammer and Chree frequency spectrum is reproduced at long wavelengths.

Consider a rod bounded by the cylindrical surface $r = a$ and the planes $z = \pm c$ in a cylindrical co-ordinate system r, θ, z . As there is no motion in circumferential direction, the component of displacement in this direction u_θ is taken as zero. The radial and axial components are expressed in the series of Jacobi polynomials in the radial co-ordinate

$$u_r = \sum_{n=0}^{\infty} \mathcal{U}_n(\alpha) u_n(z, t); \quad u_z = \sum_{n=0}^{\infty} \mathcal{W}_n(\alpha) w_n(z, t) \quad (69)$$

where $\alpha = r/a$ and the Jacobi polynomials are given by

$$\begin{aligned} \mathcal{U}_0(\alpha) = \alpha, \mathcal{U}_1(\alpha) = \alpha - \frac{3}{2}\alpha^3, \dots, \mathcal{U}_n(\alpha) = \alpha + \sum_{k=1}^n (-1)^k \binom{n}{k} \frac{(n+2)_k}{(k+1)!} \alpha^{2k+1}, \\ \mathcal{W}_0(\alpha) = 1, \mathcal{W}_1(\alpha) = 1 - 2\alpha^2, \dots, \mathcal{W}_n(\alpha) = 1 + \sum_{k=1}^n (-1)^k \binom{n}{k} \frac{(n+1)_k}{k!} \alpha^{2k} \end{aligned} \quad (70)$$

in which

$$\begin{aligned} \binom{n}{k} &= \frac{n(n-1)(n-2) \dots (n-k+1)}{k!}; \\ (\beta)_k &= \beta(\beta+1)(\beta+2) \dots (\beta+k-1), \quad (\beta)_0 = 1 \end{aligned} \quad (71)$$

The Jacobi polynomials $\mathcal{U}_n$ and $\mathcal{W}_n$ have convenient orthogonality properties that simplify many of the integrations to be performed.

$$\begin{aligned}
4(n+1)^2 \int_0^1 \alpha \mathcal{U}_m \mathcal{U}_n d\alpha &= \begin{cases} 0, & m \neq n \\ 1, & m = n \end{cases} \\
2(2n+1) \int_0^1 \alpha \mathcal{W}_m \mathcal{W}_n d\alpha &= \begin{cases} 0, & m \neq n \\ 1, & m = n \end{cases}
\end{aligned} \tag{72}$$

The derivative property is

$$\frac{d\mathcal{W}_n}{d\alpha} = \sum_{k=0}^{n-1} C_{nk} \mathcal{U}_k, \quad C_{nk} = 4(k+1)^2 (-1)^{n+k} \tag{73}$$

The equations of motion in variational form are

$$\begin{aligned}
\int_{-c}^c \int_0^a \left(\frac{\partial \sigma_r}{\partial r} + \frac{\partial \tau_{rz}}{\partial z} + \frac{\sigma_r - \sigma_\theta}{r} - \rho \frac{\partial^2 u_r}{\partial t^2} \right) \delta u_r r dr dz &= 0 \\
\int_{-c}^c \int_0^a \left(\frac{\partial \tau_{rz}}{\partial r} + \frac{\partial \sigma_z}{\partial z} + \frac{\tau_{rz}}{r} - \rho \frac{\partial^2 u_z}{\partial t^2} \right) \delta u_z r dr dz &= 0
\end{aligned} \tag{74}$$

where $\sigma_r, \sigma_\theta, \sigma_z, \tau_{rz}$ are components of stress and can be obtained by differentiating strain-energy function U with respect to components of strain $\epsilon_r, \epsilon_\theta, \epsilon_z, \gamma_{rz}$, respectively. With the substitution of Eq. (69) into Eq. (74), performing the integrations with respect to r and setting the coefficients of δu_n and δw_n equal to zero, the stress equations of motion of order n are obtained

$$\begin{aligned}
\frac{dQ_n}{dz} - \frac{P_n}{a} + \frac{(-1)^n R}{n+1} \frac{1}{a} &= \frac{\rho}{4(n+1)^3} \frac{\partial^2 u_n}{\partial t^2} \\
\frac{dF_n}{dz} - \sum_{k=0}^{n-1} C_{nk} \frac{Q_k}{a} + (-1)^n \frac{Z}{a} &= \frac{\rho}{2(2n+1)} \frac{\partial^2 w_n}{\partial t^2}
\end{aligned} \tag{75}$$

where

$$P_n = \int_0^1 \left(\sigma_\theta \mathcal{U}_n + \sigma_r \alpha \frac{d\mathcal{U}_n}{d\alpha} \right) d\alpha, \quad Q_n = \int_0^1 \tau_{rz} \mathcal{U}_n \alpha d\alpha, \quad F_n = \int_0^1 \sigma_z \mathcal{W}_n \alpha d\alpha \quad (76)$$

$$R = [\sigma_r]_{r=a}, \quad Z = [\tau_{rz}]_{r=a} \quad (77)$$

The components of strain after substitution of Eq. (69) in the usual strain-displacement relations are:

$$\begin{aligned} \epsilon_r &= \frac{\partial u_r}{\partial r} = \sum_{n=0}^{\infty} \frac{\partial \mathcal{U}_n}{\partial \alpha} \frac{u_n}{a}, \quad \epsilon_\theta = \frac{u_r}{r} = \sum_{n=0}^{\infty} \frac{\mathcal{U}_n}{\alpha} \frac{u_n}{a} \\ \epsilon_z &= \frac{\partial u_z}{\partial z} = \sum_{n=0}^{\infty} \mathcal{W}_n \frac{\partial w_n}{\partial z}, \quad \gamma_{rz} = \frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} = \sum_{n=0}^{\infty} \left(\mathcal{U}_n \frac{\partial u_n}{\partial z} + \frac{\partial \mathcal{W}_n}{\partial \alpha} \frac{w_n}{a} \right) \end{aligned} \quad (78)$$

The stress-strain relationships are:

$$\begin{aligned} \sigma_r &= \frac{\partial U}{\partial \epsilon_r} = \lambda(\epsilon_r + \epsilon_\theta + \epsilon_z) + 2\mu\epsilon_r, \quad \sigma_\theta = \frac{\partial U}{\partial \epsilon_\theta} = \lambda(\epsilon_r + \epsilon_\theta + \epsilon_z) + 2\mu\epsilon_\theta \\ \sigma_z &= \frac{\partial U}{\partial \epsilon_z} = \lambda(\epsilon_r + \epsilon_\theta + \epsilon_z) + 2\mu\epsilon_z, \quad \tau_{rz} = \frac{\partial U}{\partial \gamma_{rz}} = \mu\gamma_{rz} \end{aligned} \quad (79)$$

where U is the strain-energy-function and λ, μ are Lamé's constants. Substituting Eq. (78) into Eq. (79), then substituting the obtained expressions, in turn, into Eq. (76), and performing the integrations, we obtain

$$\begin{aligned} P_0 &= 2(\lambda + \mu) \frac{u_0}{a} + \lambda \frac{\partial w_0}{\partial z} + \dots, \quad Q_0 = \frac{\mu}{4} \left(\frac{\partial u_0}{\partial z} - \frac{4w_1}{a} \right) + \dots \\ F_0 &= \frac{\lambda + 2\mu}{2} \frac{\partial w_0}{\partial z} + \lambda \frac{u_0}{a} + \dots, \quad F_1 = \frac{\lambda + 2\mu}{6} \frac{\partial w_1}{\partial z} + \dots \end{aligned} \quad (80)$$

where the terms containing amplitudes u_n of order higher than $n = 0$ and w_n of order higher than $n = 1$ have been neglected.

The strain-energy-density at any point of three-dimensional body is given by

$$U = \frac{1}{2} \left[\lambda (\epsilon_r + \epsilon_\theta + \epsilon_z)^2 + 2\mu \left(\epsilon_r^2 + \epsilon_\theta^2 + \epsilon_z^2 + \frac{\gamma_{rz}^2}{2} \right) \right] \quad (81)$$

Then, the energy-density is

$$\bar{U} = \int_0^1 U \alpha d\alpha = \frac{1}{2} \left[P_0 \frac{u_0}{a} + \dots + F_0 \frac{\partial w_0}{\partial z} + F_1 \frac{\partial w_1}{\partial z} + \dots + Q_0 \left(\frac{\partial u_0}{\partial z} - \frac{4w_1}{a} \right) + \dots \right] \quad (82)$$

Similarly, the kinetic-energy-density at any point is given by

$$K = \frac{\rho}{2} \left[\left(\frac{\partial u_r}{\partial t} \right)^2 + \left(\frac{\partial u_z}{\partial t} \right)^2 \right] \quad (83)$$

Then, the energy-density is given by

$$\bar{K} = \int_0^1 K \alpha d\alpha = \frac{\rho}{2} \left[\frac{1}{4} \left(\frac{\partial u_0}{\partial t} \right)^2 + \frac{1}{2} \left(\frac{\partial w_0}{\partial t} \right)^2 + \frac{1}{6} \left(\frac{\partial w_1}{\partial t} \right)^2 + \dots \right] \quad (84)$$

For the second-order approximation, only the terms associated with displacement amplitudes u_0, w_0, w_1 have been retained. The following symbols have been adopted to simplify the notations:

$$u = u_0, \quad w = w_0, \quad \psi = w_1 \quad (85)$$

Then, the three-dimensional displacements are represented in the second-order approximation by

$$u_r \approx \alpha u(z, t), \quad u_z \approx w(z, t) + (1 - 2\alpha^2)\psi(z, t) \quad (86)$$

and the strains by

$$\begin{aligned} \epsilon_r &= \frac{\partial u_r}{\partial r} \approx \frac{u}{a} = \Gamma_r, & \epsilon_\theta &= \frac{u_r}{r} \approx \frac{u}{a} = \Gamma_r \\ \epsilon_z &= \frac{\partial u_z}{\partial z} \approx \frac{\partial w}{\partial z} + (1 - 2\alpha^2) \frac{\partial \psi}{\partial z} = \Gamma_z + (1 - 2\alpha^2)\Gamma_\psi \\ \gamma_{rz} &= \frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} \approx \alpha \frac{\partial u}{\partial z} - 4\alpha \frac{\psi}{a} = \alpha \Gamma_{rz} \end{aligned} \quad (87)$$

Due to these, the stresses are

$$\begin{aligned} P_r = P_0 &= \int_0^1 (\sigma_\theta + \sigma_r) \alpha d\alpha, & P_z = F_0 &= \int_0^1 \sigma_z \alpha d\alpha \\ P_\psi = F_1 &= \int_0^1 \sigma_z (1 - 2\alpha^2) \alpha d\alpha, & P_{rz} = Q_0 &= \int_0^1 \tau_{rz} \alpha^2 d\alpha \end{aligned} \quad (88)$$

and the strains are

$$\Gamma_r = \frac{u}{a}, \quad \Gamma_z = \frac{\partial w}{\partial z}, \quad \Gamma_\psi = \frac{\partial \psi}{\partial z}, \quad \Gamma_{rz} = \frac{\partial u}{\partial z} - 4 \frac{\psi}{a} \quad (89)$$

To compensate for the error introduced by neglecting higher order terms, we replace Γ_r by $\kappa_1 \Gamma_r$ and Γ_{rz} by $\kappa_2 \Gamma_{rz}$ in the strain-energy-density and $\frac{\partial u}{\partial t}$ by $\kappa_3 \frac{\partial u}{\partial t}$ and $\frac{\partial \psi}{\partial t}$ by $\kappa_4 \frac{\partial \psi}{\partial t}$ in the kinetic-energy-density. Here, the κ_i are constants whose values will be determined later to match Pochhammer–Chree solution in the range of longer wavelengths. Then, the energy-densities of the second-order approximation are

$$U_2 = \frac{1}{2} \left[2(\lambda + \mu) \kappa_1^2 \Gamma_r^2 + \frac{1}{2} (\lambda + 2\mu) \Gamma_z^2 + 2\lambda \kappa_1 \Gamma_r \Gamma_z + \frac{1}{4} \mu \kappa_2^2 \Gamma_{rz}^2 + \frac{1}{6} (\lambda + 2\mu) \Gamma_\psi^2 \right] \quad (90)$$

$$K_2 = \frac{\rho}{2} \left[\frac{1}{4} \kappa_3^2 \left(\frac{\partial u}{\partial t} \right)^2 + \frac{1}{2} \left(\frac{\partial w}{\partial t} \right)^2 + \frac{1}{6} \kappa_4^2 \left(\frac{\partial \psi}{\partial t} \right)^2 \right] \quad (91)$$

On differentiating strain-energy-density function (90), the adjusted stress-strain-displacement relations are obtained

$$\begin{aligned} P_r &= \frac{\partial U_2}{\partial \Gamma_r} = 2(\lambda + \mu) \kappa_1^2 \Gamma_r + \lambda \kappa_1 \Gamma_z = [2(\lambda + \mu) \kappa_1^2 u + \lambda \kappa_1 w'] a^{-1} \\ P_z &= \frac{\partial U_2}{\partial \Gamma_z} = \frac{1}{2} (\lambda + 2\mu) \Gamma_z + \lambda \kappa_1 \Gamma_r = \left[\frac{1}{2} (\lambda + 2\mu) w' + \lambda \kappa_1 u \right] a^{-1} \\ P_{rz} &= \frac{\partial U_2}{\partial \Gamma_{rz}} = \frac{1}{4} \mu \kappa_2^2 \Gamma_{rz} = \frac{1}{4} \mu \kappa_2^2 [u' - 4\psi] a^{-1} \end{aligned} \quad (92)$$

$$P_\psi = \frac{\partial U_2}{\partial \Gamma_\psi} = \frac{1}{6}(\lambda + 2\mu)\Gamma_\psi = \frac{1}{6}(\lambda + 2\mu)\psi' a^{-1}$$

where prime over the quantity indicates differentiation with respect to z/a .

Using the variational equations of motion with the constants κ_3 and κ_4 introduced into the kinetic energy, the adjusted stress-equations of motion are

$$\begin{aligned} P'_{rz} - P_r + R &= \frac{1}{4}\rho a \kappa_3^2 \ddot{u} \\ P'_z + Z &= \frac{1}{2}\rho a \ddot{w} \\ P'_\psi + 4P_{rz} - Z &= \frac{1}{6}\rho a \kappa_4^2 \ddot{\psi} \end{aligned} \quad (93)$$

Combining Eqs. (92) and (93), the displacement equations of motion are

$$\begin{aligned} \mu \kappa_2^2 (u'' - 4\psi') - 8(\lambda + \mu) \kappa_1^2 u - 4\lambda \kappa_1 w' + 4aR &= \rho a^2 \kappa_3^2 \ddot{u} \\ (\lambda + 2\mu) w'' + 2\lambda \kappa_1 u' + 2aZ &= \rho a^2 \ddot{w} \\ (\lambda + 2\mu) \psi'' + 6\mu \kappa_2^2 (u' - 4\psi) - 6aZ &= \rho a \kappa_4^2 \ddot{\psi} \end{aligned} \quad (94)$$

The equations given in Eq. (94) can be solved for a rod of infinite length as boundary conditions are not necessary at the ends of the rod.

2.6 The Sixth, Doyle's 3-Mode Theory

Although the Mindlin–Herrmann theory extends the applicability of Love's theory, it loses accuracy in the vicinity of the cut-off frequencies. This breakdown occurs because, in this frequency range, the Lamb spectrum contains a pair of intersecting modes, whereas the Mindlin–Herrmann theory approximates their behavior using only a single mode. To address this limitation, James F. Doyle [15] introduced a third mode by expanding the displacement field to obtain improved approximate solutions.

$$u(x, y) = u_0(x) + \phi_0(x) \cdot h \left(1 - 12 \frac{y^2}{h^2} \right), \quad v(x, y) = \psi_0(x) \cdot y \quad (95)$$

where ϕ_0 is a function which describes a parabolic distribution of the axial displacement along the height of the rod, and h denotes the height of the rod.

The normal and shear strains are given by

$$\begin{aligned} \epsilon_{xx} &= \frac{\partial u}{\partial x} = \frac{\partial u_0}{\partial x} + \frac{\partial \phi_0}{\partial x} h \left(1 - 12 \frac{y^2}{h^2} \right), & \epsilon_{yy} &= \frac{\partial v}{\partial y} = \psi_0 \\ \gamma_{xy} &= \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = \left(-24 \phi_0 \frac{y}{h} + \frac{\partial \psi_0}{\partial x} \right) y \end{aligned} \quad (96)$$

The corresponding stresses are

$$\begin{aligned} \sigma_{xx} &= (2\mu + \lambda) \left[\frac{\partial u_0}{\partial x} + \frac{\partial \phi_0}{\partial x} h \left(1 - 12 \frac{y^2}{h^2} \right) \right] + \lambda \psi_0 \\ \sigma_{yy} &= (2\mu + \lambda) \psi_0 + \lambda \left[\frac{\partial u_0}{\partial x} + \frac{\partial \phi_0}{\partial x} h \left(1 - 12 \frac{y^2}{h^2} \right) \right] \\ \sigma_{xy} &= \mu \left[-24 \phi_0 \frac{y}{h} + \frac{\partial \psi_0}{\partial x} \right] y \end{aligned} \quad (97)$$

Then, the total potential energy is

$$\begin{aligned} U &= \frac{1}{2} \int_0^L \int_S (\sigma_{xx} \epsilon_{xx} + \sigma_{yy} \epsilon_{yy} + \sigma_{xy} \gamma_{xy}) ds dx \\ &= \frac{1}{2} \int_0^L \left[(2\mu + \lambda) A \left\{ \left(\frac{\partial u_0}{\partial x} \right)^2 + \frac{4}{5} h^2 \left(\frac{\partial \phi_0}{\partial x} \right)^2 + \psi_0^2 \right\} + 2\lambda A \frac{\partial u_0}{\partial x} \psi_0 \right. \\ &\quad \left. + \mu I \left\{ \frac{\partial \psi_0}{\partial x} - 24 \frac{\phi_0}{h} \right\}^2 \right] dx \end{aligned} \quad (98)$$

and the total kinetic energy is

$$T = \frac{1}{2} \int_0^L \int_S \rho (\dot{u}^2 + \dot{v}^2) ds dx = \frac{1}{2} \int_0^L \left[\rho A \dot{u}_0^2 + \rho A \frac{4}{5} h^2 \dot{\phi}_0^2 + \rho I \dot{\psi}_0^2 \right] dx \quad (99)$$

The applied surface tractions and loads on the rod have potential given by

$$V = - \int_0^L q(x) v dx - Q_u v|_0^L - Q_\psi \psi_0|_0^L - Q_\phi \phi_0|_0^L \quad (100)$$

where resultants Q_ψ and Q_ϕ have been associated with the degrees of freedom ψ_0 and ϕ_0 respectively. On using Hamilton's principle, we get

$$\begin{aligned} (2\mu + \lambda)A \frac{\partial^2 u_0}{\partial x^2} + \lambda A \frac{\partial \psi_0}{\partial x} &= \rho A \ddot{u}_0 - q \\ \mu I \frac{\partial^2 \psi_0}{\partial x^2} - (2\mu + \lambda)A \psi_0 - \lambda A \frac{\partial u_0}{\partial x} - 2\mu A h \frac{\partial \phi_0}{\partial x} &= \rho I \ddot{\psi}_0 \\ (2\mu + \lambda)I \frac{\partial^2 \phi_0}{\partial x^2} - 5\mu A \phi_0 + \frac{10}{48} \mu A h \frac{\partial \psi_0}{\partial x} &= \rho I \ddot{\phi}_0 \end{aligned} \quad (101)$$

with the associated boundary conditions at each end of the rod given by

$$\begin{aligned} u_0: \quad Q_u &= (2\mu + \lambda)A \frac{\partial u_0}{\partial x} + \lambda A \psi_0 \\ \psi_0: \quad Q_\psi &= \mu I \left(\frac{\partial \psi_0}{\partial x} - 24 \frac{\phi_0}{h} \right) \\ \phi_0: \quad Q_\phi &= \frac{48}{5} (2\mu + \lambda)I \frac{\partial \phi_0}{\partial x} \end{aligned} \quad (102)$$

The equations of motion of the system given in Eq. (101), together with the boundary conditions in Eq. (102), form a well-posed boundary-value problem that can be solved for a rod or bar of finite length.

2.7 The Seventh, Anderson's 4-Mode Theory

Similar to Mindlin-McNiven theory [14], Simon P. Anderson [16] approximated the axial and the radial displacements by the Jacobi polynomials so the displacements are

$$\begin{aligned} u_x(x, r, t) &= u(x, t) + \left(1 - 2 \frac{r^2}{a^2}\right) \phi(x, t) \\ u_r(x, r, t) &= \frac{r}{a} v(x, t) + \left(\frac{r}{a} - \frac{3 r^3}{2 a^3}\right) \psi(x, t) \end{aligned} \quad (103)$$

Then, the strains are given by

$$\begin{aligned} \epsilon_{xx} &= \frac{\partial u_x}{\partial x} = \frac{\partial u}{\partial x} + \left(1 - 2 \frac{r^2}{a^2}\right) \frac{\partial \phi}{\partial x} \\ \epsilon_{rr} &= \frac{\partial u_r}{\partial r} = \frac{v}{a} + \left(\frac{1}{a} - \frac{9 r^2}{2 a^3}\right) \psi \\ \epsilon_{\theta\theta} &= \frac{u_r}{r} + \frac{1}{r} \frac{\partial u_\theta}{\partial \theta} = \frac{v}{a} + \left(\frac{1}{a} - \frac{3 r^2}{2 a^3}\right) \psi \\ \epsilon_{xr} &= \frac{\partial u_x}{\partial r} + \frac{\partial u_r}{\partial x} = -\frac{4r}{a^2} \phi + \frac{r}{a} \frac{\partial v}{\partial x} + \left(\frac{r}{a} - \frac{3 r^3}{2 a^3}\right) \frac{\partial \psi}{\partial x} \end{aligned} \quad (104)$$

The stress-displacement relationships are then

$$\begin{aligned} \sigma_{xx} &= \lambda(\epsilon_{xx} + \epsilon_{rr} + \epsilon_{\theta\theta}) + 2\mu\epsilon_{xx} \\ &= (2\mu + \lambda) \left[\frac{\partial u}{\partial x} + \left(1 - \frac{2r^2}{a^2}\right) \frac{\partial \phi}{\partial x} \right] + \frac{2}{a} \lambda \left[v + \left(1 - \frac{3r^2}{a^2}\right) \psi \right] \\ \sigma_{rr} &= \lambda(\epsilon_{xx} + \epsilon_{rr} + \epsilon_{\theta\theta}) + 2\mu\epsilon_{rr} \\ &= \frac{2}{a} (\mu + \lambda) \left[v + \left(1 - \frac{9r^2}{2a^2}\right) \psi \right] + \lambda \left[\frac{\partial u}{\partial x} + \left(1 - \frac{2r^2}{a^2}\right) \frac{\partial \phi}{\partial x} + \frac{3r^2}{a^3} \psi \right] \\ \sigma_{\theta\theta} &= \lambda(\epsilon_{xx} + \epsilon_{rr} + \epsilon_{\theta\theta}) + 2\mu\epsilon_{\theta\theta} = \sigma_{rr} + \frac{6\mu r^2}{a^3} \psi \end{aligned} \quad (105)$$

$$\sigma_{xr} = \mu \epsilon_{xr} = \frac{\mu}{a} \left[r \frac{\partial v}{\partial x} - \frac{4r}{a} \phi + \left(r - \frac{3r^3}{2a^2} \right) \frac{\partial \psi}{\partial x} \right]$$

The stress equations of motion are given by

$$\begin{aligned} \frac{\partial \sigma_{xr}}{\partial r} + \frac{\partial \sigma_{xx}}{\partial x} + \frac{\sigma_{xr}}{r} &= \rho \frac{\partial^2 u_x}{\partial t^2} \\ \frac{\partial \sigma_{rr}}{\partial r} + \frac{\partial \sigma_{xr}}{\partial x} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} &= \rho \frac{\partial^2 u_r}{\partial t^2} \end{aligned} \quad (106)$$

The acceleration functions on the right-side of the first of Eq. (106) can be written in terms of orthogonal Jacobi polynomials as

$$\sum_{n=0}^{\infty} \rho \mathcal{W}_n \frac{\partial^2 u_n}{\partial t^2}$$

which, when multiplied by $r\mathcal{W}_m$ (given in Eq. (70)) and integrated with respect to r results in

$$\frac{1}{2(2m+1)} \rho a^2 \frac{\partial^2 u_m}{\partial t^2}$$

Similarly, multiplying the second of Eq. (106) by $r\mathcal{U}_m$ (given in Eq. (70)) and integrating with respect to r gives the right-side of this equation in the form of

$$\frac{1}{4(m+1)^3} \rho a^2 \frac{\partial^2 v_m}{\partial t^2}$$

The importance of this result lies in the fact that, regardless of the number of modes retained, multiplying the stress equations of motion in Eq. (106) by the functions $r\mathcal{W}_m$ and $r\mathcal{U}_m$ and integrating over the cross-section yields an exact right-hand side involving only a single variable, namely u_m and v_m respectively. The equations of motion for u and ϕ are obtained by multiplying

the first equation of Eq. (106) by r and $r - 2r^3/a^2$, separately, and then integrating over the cross-section, while making use of the boundary conditions

$$\sigma_{rr} = \sigma_{xx} = 0 \text{ when } r = a.$$

Similarly, the equations of motion for v and ψ are obtained by multiplying the second of Eq. (106) by r^2/a and $\frac{r^2}{a} - \frac{3r^4}{2a^3}$ in turn and integrating over the cross-section. Thus, the four governing equations of motion are

$$\begin{aligned} a^2(2\mu + \lambda) \frac{\partial^2 u}{\partial x^2} + 2a\lambda \frac{\partial v}{\partial x} - a\lambda \frac{\partial \psi}{\partial x} &= \rho a^2 \frac{\partial^2 u}{\partial t^2} \\ -4a\lambda \frac{\partial u}{\partial x} + a^2\mu \frac{\partial^2 v}{\partial x^2} - 8(\mu + \lambda)v - 4a\mu \frac{\partial \phi}{\partial x} + 4(\mu + \lambda)\psi &= \rho a^2 \frac{\partial^2 v}{\partial t^2} \\ 6a\mu \frac{\partial v}{\partial x} + a^2(2\mu + \lambda) \frac{\partial^2 \phi}{\partial x^2} - 24\mu\phi + 3a\lambda \frac{\partial \psi}{\partial x} &= \rho a^2 \frac{\partial^2 \phi}{\partial t^2} \\ 16a\lambda \frac{\partial u}{\partial x} + 32(\mu + \lambda)v - 16a\lambda \frac{\partial \phi}{\partial x} + a^2\mu \frac{\partial^2 \psi}{\partial x^2} - 16(7\mu + 4\lambda)\psi &= \rho a^2 \frac{\partial^2 \psi}{\partial t^2} \end{aligned} \quad (107)$$

The equations given in Eq. (107) can be solved for a rod of infinite length, which does not necessitate the use of boundary conditions at the ends of the rod.

2.8 The Eighth, Żak and Krawczuk Higher Order 2-Mode Theory

Żak and Krawczuk [17] assumed a new form of the longitudinal displacement $u_x(x, r, t)$ and the lateral displacement $u_r(x, r, t)$ of a one-dimensional rod for analysis of propagation of elastic longitudinal waves expressed as:

$$\begin{aligned} u_x(x, r, t) &= u_0(x, t) + \left(1 - \frac{r^2}{a^2}\right) u_2(x, t) \\ u_r(x, r, t) &= ru_1(x, t) + r \left(1 - \frac{r^2}{a^2}\right) u_3(x, t) \end{aligned} \quad (108)$$

where a is the radius of the rod. The six components of strain are given by Eq. (18), where z and θ are used in place of x and ϕ respectively. Taking into account the zero-traction condition

$$\sigma_{rr} = \sigma_{xr} = 0 \quad \text{for} \quad 0 \leq x \leq l, \quad r = a \quad (109)$$

the following conditions are obtained

$$u_2 = \frac{a^2}{2} u_1', \quad u_3 = \frac{\lambda}{2(\lambda + 2\mu)} u_0' + \frac{\lambda + \mu}{(\lambda + 2\mu)} u_1 \quad (110)$$

Substituting Eq. (110) in Eq. (108) converts the degrees of polynomials associated with each mode into higher degrees than those for conventional higher-mode theories. So, Żak and Krawczuk quoted this kind of theory as higher order higher mode theory. Żak and Krawczuk, however, did not provide the equations of motion in their paper explicitly and the authors of this paper have carried out this development in another paper in submission. Deriving the expression for kinetic and potential energies, and then using Hamilton's principle, the equations of motion with necessary boundary conditions can be obtained.

$$\begin{aligned} & \rho a^2 \ddot{u}_0 + \frac{\rho a^4}{4} \ddot{u}_1' - \frac{\rho a^4}{12} C_0 (C_1 + 2) \ddot{u}_1' - \frac{\rho a^4}{12} C_0^2 \ddot{u}_0'' - \frac{a^4}{12} (2C_0 \lambda + 3\lambda + 6\mu) u_1''' \\ & - \frac{a^2}{3} (4C_0^2 + 3)(\lambda + 2\mu) u_0'' - \frac{2a^2}{3} (2C_0 C_1 \lambda + 4C_0 C_1 \mu + 3\lambda) u_1' \\ & + \frac{a^4}{12} C_0^2 \mu u_0^{iv} + \frac{a^4}{12} C_0 C_1 \mu u_1''' = 0 \end{aligned} \quad (111)$$

$$\begin{aligned} & \frac{\rho a^4}{12} (C_1^2 + 4C_1 + 6) \ddot{u}_1 + \frac{\rho a^4}{12} C_0 (C_1 + 2) \ddot{u}_0' - \frac{\rho a^4}{4} \ddot{u}_0' - \frac{\rho a^6}{12} \ddot{u}_1'' + \frac{a^4}{6} (C_1 + 3) \lambda u_1'' \\ & + \frac{2a^2}{3} (2C_0 C_1 \lambda + 4C_0 C_1 \mu + 3\lambda) u_0' + \frac{4a^2}{3} (C_1^2 \lambda + 2C_1^2 \mu + 3\lambda + 3\mu) u_1 \\ & - \frac{a^4}{12} C_0 C_1 \mu u_0''' - \frac{a^4}{12} C_1^2 \mu u_1'' + \frac{a^6}{12} (\lambda + 2\mu) u_1^{iv} + \frac{a^4}{12} (2C_0 \lambda + 3\lambda + 6\mu) u_0''' \end{aligned}$$

$$+\frac{a^4}{6}(C_1+3)\lambda u_1'' = 0$$

The two equations given by Eq. (111) address two displacement modes u_0 and u_1 which can be used to find the longitudinal vibration solution of infinite-length rod.

2.9 The Ninth, Żak and Krawczuk Higher Order 3-Mode Theory

In a similar manner, Żak and Krawczuk [17] assumed the displacements in the form of

$$u_x(x, r, t) = u_0(x, t) + \left(1 - \frac{r^2}{a^2}\right)u_2(x, t) + \left(1 - \frac{r^4}{a^4}\right)u_4(x, t) \quad (112)$$

$$u_r(x, r, t) = ru_1(x, t) + r\left(1 - \frac{r^2}{a^2}\right)u_3(x, t)$$

Using zero-traction conditions given by Eq. (109), the following conditions are obtained

$$u_2 = \frac{a^2}{4}u_1' - \frac{1}{2}u_2, \quad u_3 = \frac{\lambda}{2(\lambda + 2\mu)}u_0' + \frac{\lambda + \mu}{(\lambda + 2\mu)}u_1 \quad (113)$$

Deriving the expression for kinetic and potential energies, and then using Hamilton's principle, the equations of motion with necessary boundary conditions can be obtained.

$$\begin{aligned} &\rho a^2 \ddot{u}_0 + \frac{\rho a^2}{6} \ddot{u}_2 + \frac{\rho a^4}{6} \ddot{u}_1' - \frac{\rho a^4}{12} C_0 (C_1 + 2) \ddot{u}_1' - \frac{\rho a^4}{12} C_0^2 \ddot{u}_0'' \\ &\quad - \frac{a^4}{12} (C_0 \lambda + 2\lambda + 4\mu) u_1''' - \frac{a^2}{3} (4C_0^2 + 3) (\lambda + 2\mu) u_0'' \\ &\quad - \frac{2a^2}{3} (2C_0 C_1 \lambda + 4C_0 C_1 \mu + 3\lambda) u_1' - \frac{a^2}{6} (C_0 \lambda + \lambda + 2\mu) u_2'' \\ &\quad + \frac{a^4}{12} C_0^2 \mu u_0^{iv} + \frac{a^4}{12} C_0 (C_1 + 1) \mu u_1''' - \frac{a^2}{6} C_0 \mu u_2'' = 0 \\ &\frac{\rho a^4}{12} (C_1^2 + 4C_1 + 6) \ddot{u}_1 + \frac{\rho a^4}{12} C_0 (C_1 + 2) \ddot{u}_0' - \frac{\rho a^4}{6} \ddot{u}_0' - \frac{\rho a^6}{30} \ddot{u}_1'' - \frac{3\rho a^4}{80} \ddot{u}_2' \end{aligned} \quad (114)$$

$$\begin{aligned}
& + \frac{a^4}{12} (C_1 + 4) \lambda u_1'' + \frac{2a^2}{3} (2C_0 C_1 \lambda + 4C_0 C_1 \mu + 3\lambda) u_0' \\
& + \frac{4a^2}{3} (C_1^2 \lambda + 2C_1^2 \mu + 3\lambda + 3\mu) u_1 + \frac{a^2}{6} (C_1 + 2) \lambda u_2' \\
& - \frac{a^4}{12} C_0 (C_1 + 1) \mu u_0''' - \frac{a^4}{12} (C_1 + 1)^2 \mu u_1'' + \frac{a^2}{6} (C_1 + 1) \mu u_2' \\
& + \frac{a^6}{30} (\lambda + 2\mu) u_1^{iv} + \frac{a^4}{12} (C_0 \lambda + 2\lambda + 4\mu) u_0''' + \frac{a^4}{12} (C_1 + 4) \lambda u_1'' \\
& + \frac{3a^4}{80} (\lambda + 2\mu) u_2''' = 0 \\
& \frac{\rho a^2}{6} \ddot{u}_0 + \frac{3\rho a^4}{80} \ddot{u}_1' + \frac{\rho a^2}{20} \ddot{u}_2 - \frac{a^2}{6} C_0 \mu u_0'' - \frac{a^2}{6} (C_1 + 1) \mu u_1' + \frac{1}{3} \mu u_2 \\
& - \frac{a^2}{6} (C_0 \lambda + \lambda + 2\mu) u_0'' - \frac{3a^4}{80} (\lambda + 2\mu) u_1''' - \frac{a^2}{6} (C_1 + 2) \lambda u_1' \\
& - \frac{a^2}{20} (\lambda + 2\mu) u_2'' = 0
\end{aligned}$$

The three equations given by Eq. (114) address three displacement modes u_0 , u_1 and u_2 which can be used to find the longitudinal vibration solution of infinite-length rod.

2.10 The Tenth, Żak and Krawczuk Higher Order 4-Mode Theory

For higher order 4-mode theory, the displacements forms are

$$\begin{aligned}
u_x(x, r, t) &= u_0(x, t) + \left(1 - \frac{r^2}{a^2}\right) u_2(x, t) + \left(1 - \frac{r^4}{a^4}\right) u_4(x, t) \\
u_r(x, r, t) &= r u_1(x, t) + r \left(1 - \frac{r^2}{a^2}\right) u_3(x, t) + r \left(1 - \frac{r^4}{a^4}\right) u_5(x, t)
\end{aligned} \tag{115}$$

Using the zero-traction conditions given by Eq. (109), the following conditions are obtained

$$u_4 = \frac{a^2}{4}u_1' - \frac{1}{2}u_2, \quad u_5 = \frac{\lambda}{4(\lambda + 2\mu)}u_0' + \frac{\lambda + \mu}{2(\lambda + 2\mu)}u_1 - \frac{1}{2}u_3 \quad (116)$$

Deriving the expression for kinetic and potential energies, and then using Hamilton's principle, the equations of motion with necessary boundary conditions can be obtained.

$$\begin{aligned} & \rho a^2 \ddot{u}_0 + \frac{\rho a^2}{6} \ddot{u}_2 + \frac{\rho a^4}{6} \ddot{u}_1' - \frac{\rho a^4}{24} C_0 (C_1 + 3) \ddot{u}_1' - \frac{\rho a^4}{24} C_0^2 \ddot{u}_0'' - \frac{\rho a^4}{60} C_0 \ddot{u}_3' \\ & - \frac{a^4}{30} (2C_0 \lambda + 5\lambda + 10\mu) u_1''' - \frac{a^2}{5} (4C_0^2 + 5)(\lambda + 2\mu) u_0'' \\ & - \frac{2a^2}{5} (2C_0 C_1 \lambda + 4C_0 C_1 \mu + 5\lambda) u_1' - \frac{a^2}{60} (7C_0 \lambda + 10\lambda + 20\mu) u_2'' \\ & - \frac{a^2}{5} C_0 (\lambda + 2\mu) u_3' + \frac{a^4}{24} C_0^2 \mu u_0^{iv} + \frac{a^4}{120} C_0 (5C_1 + 7) \mu u_1''' - \frac{7a^2}{60} C_0 \mu u_2'' + \frac{a^4}{60} C_0 \mu u_3''' \\ & = 0 \\ & \frac{\rho a^4}{24} (C_1^2 + 6C_1 + 12) \ddot{u}_1 + \frac{\rho a^4}{24} C_0 (C_1 + 3) \ddot{u}_0' + \frac{\rho a^4}{120} (2C_1 + 5) \ddot{u}_3 - \frac{\rho a^4}{6} \ddot{u}_0' - \frac{\rho a^6}{30} \ddot{u}_1'' \\ & - \frac{3\rho a^4}{80} \ddot{u}_2' + \frac{a^4}{15} (C_1 + 5) \lambda u_1'' + \frac{2a^2}{5} (2C_0 C_1 \lambda + 4C_0 C_1 \mu + 5\lambda) u_0' \\ & + \frac{4a^2}{5} (C_1^2 \lambda + 2C_1^2 \mu + 5\lambda + 5\mu) u_1 + \frac{a^2}{60} (7C_1 + 20) \lambda u_2' \\ & + \frac{a^2}{5} C_1 (\lambda + 2\mu) u_3 - \frac{a^4}{120} C_0 (5C_1 + 7) \mu u_0''' \\ & - \frac{a^4}{120} (5C_1^2 + 14C_1 + 10) \mu u_1'' + \frac{a^2}{60} (7C_1 + 10) \mu u_2' \\ & - \frac{a^4}{120} (2C_1 + 3) \mu u_3'' + \frac{a^6}{30} (\lambda + 2\mu) u_1^{iv} + \frac{a^4}{30} (2C_0 \lambda + 5\lambda + 10\mu) u_0''' \\ & + \frac{a^4}{15} (C_1 + 5) \lambda u_1'' + \frac{3a^4}{80} (\lambda + 2\mu) u_2''' + \frac{a^4}{60} \lambda u_3'' = 0 \end{aligned} \quad (117)$$

$$\begin{aligned}
& \frac{\rho a^2}{6} \ddot{u}_0 + \frac{3\rho a^4}{80} \ddot{u}'_1 + \frac{\rho a^2}{20} \ddot{u}_2 - \frac{7a^2}{60} C_0 \mu u''_0 - \frac{a^2}{60} (7C_1 + 10) \mu u'_1 + \frac{1}{3} \mu u_2 - \frac{a^2}{20} \mu u'_3 \\
& - \frac{a^2}{60} (7C_0 \lambda + 10\lambda + 20\mu) u''_0 - \frac{3a^4}{80} (\lambda + 2\mu) u'''_1 - \frac{a^2}{60} (7C_1 + 10) \lambda u'_1 \\
& - \frac{a^2}{20} (\lambda + 2\mu) u''_2 - \frac{a^2}{20} \lambda u'_3 = 0 \\
& \frac{\rho a^4}{60} C_0 \ddot{u}'_0 + \frac{\rho a^4}{120} (2C_1 + 5) \ddot{u}_1 + \frac{\rho a^4}{120} \ddot{u}_3 + \frac{a^2}{5} C_0 (\lambda + 2\mu) u'_0 + \frac{a^2}{5} C_1 (\lambda + 2\mu) u_1 \\
& + \frac{a^4}{60} \lambda u''_1 + \frac{a^2}{20} \lambda u'_2 + \frac{2a^2}{15} (\lambda + 2\mu) u_3 - \frac{a^4}{60} C_0 \mu u'''_0 \\
& - \frac{a^4}{120} (2C_1 + 3) \mu u''_1 + \frac{a^2}{20} \mu u'_2 - \frac{a^4}{120} \mu u''_3 = 0
\end{aligned}$$

The four equations given by Eq. (117) address four displacement modes u_0, u_1, u_2 and u_3 which can be used to find the longitudinal vibration solution of infinite-length rod.

2.11 The Eleventh, Shatalov's et al. Three Term Rayleigh-Bishop Type Theory

Shatalov et. al. [19] considered the longitudinal displacement $u(x, r, t)$ and the lateral displacement $w(x, r, t)$ in the form

$$u(x, r, t) = u_0(x, t) + r^2 u_2(x, t), \quad w(x, r, t) = r u_1(x, t) \quad (118)$$

The displacements in the tangential direction are considered to be negligible i.e. $v(x, r, \phi, t) = 0$.

The strains are given by

$$\begin{aligned}
\epsilon_{xx} &= \frac{\partial u}{\partial x}, \quad \epsilon_{rr} = \frac{\partial w}{\partial r}, \quad \epsilon_{\phi\phi} = \frac{\partial v}{r \partial \phi} + \frac{w}{r} = \frac{w}{r} \\
\epsilon_{xr} &= \frac{\partial u}{\partial r} + \frac{\partial w}{\partial x}, \quad \epsilon_{\phi x} = \frac{\partial u}{r \partial \phi} + \frac{\partial v}{\partial x} = 0, \quad \epsilon_{\phi r} = \frac{\partial v}{\partial r} - \frac{v}{r} + \frac{\partial w}{r \partial \phi} = 0
\end{aligned} \quad (119)$$

From Hooke's law, the stresses are

$$\begin{aligned}
\sigma_{xx} &= (\lambda + 2\mu)\epsilon_{xx} + \lambda(\epsilon_{rr} + \epsilon_{\phi\phi}), & \sigma_{rr} &= (\lambda + 2\mu)\epsilon_{rr} + \lambda(\epsilon_{xx} + \epsilon_{\phi\phi}), \\
\sigma_{\phi\phi} &= (\lambda + 2\mu)\epsilon_{\phi\phi} + \lambda(\epsilon_{xx} + \epsilon_{rr}), \\
\sigma_{\phi r} &= \mu\epsilon_{\phi r} = 0, & \sigma_{xr} &= \mu\epsilon_{xr}, & \sigma_{\phi x} &= \mu\epsilon_{\phi x} = 0
\end{aligned} \tag{120}$$

Substituting Eq. (118) into Eq. (119), and then the resulting strains into radial stress σ_{rr} and axial shear stress σ_{xr} given by Eq. (120), we get

$$\sigma_{rr} = (\lambda + 2\mu) u_1 + \lambda(u'_0 + r^2 u'_2 + u_1), \quad \sigma_{xr} = \mu(2r u_2 + r u'_1) \tag{121}$$

Equating the coefficients of r^0 and r^1 to zero yields

$$u_1 = -\frac{\lambda}{2(\lambda + \mu)} u'_0 = -\eta u'_0, \quad u_2 = -\frac{1}{2} u'_1 = \frac{\eta}{2} u''_0 \tag{122}$$

where η is the Poisson's ratio of the material of the rod. Using Eq. (122), Eq. (121) converts to

$$\sigma_{rr} = \lambda r^2 u'_2 = \lambda r^2 \frac{\eta}{2} u'''_0, \quad \sigma_{xr} = 0 \tag{123}$$

and Eq. (118) converts to

$$u(x, r, t) = u_0(x, t) + r^2 \frac{\eta}{2} u''_0, \quad w(x, r, t) = -r \eta u'_0 \tag{124}$$

The conventional Rayleigh–Bishop theory, which has displacement given by Eq. (42), is single-degree in the lateral coordinate. However, this theory has a displacement function given by Eq. (124), which is second-degree in the lateral coordinate r . So, it is a higher order unimodal theory.

The kinetic and potential energies of the system are given by

$$T = \frac{\rho}{2} \int_0^l \int_S (\dot{u}^2 + \dot{w}^2) ds dx \tag{125}$$

$$P = \frac{1}{2} \int_0^l \int_S (\sigma_{xx} \epsilon_{xx} + \sigma_{rr} \epsilon_{rr} + \sigma_{\phi\phi} \epsilon_{\phi\phi} + \sigma_{xr} \epsilon_{xr})$$

Using Eq. (124) in the first of Eq. (125), the kinetic energy is

$$T = \frac{1}{2} \int_0^l \rho \left[S \dot{u}_0^2 + \eta I_2 \dot{u}_0 \dot{u}_0'' + \frac{\eta^2}{4} I_4 \dot{u}_0''^2 + \eta^2 I_2 \dot{u}_0'^2 \right] dx \quad (126)$$

Using Eq. (124) into Eq. (119), the strains are

$$\epsilon_{xx} = u_0' + r^2 \frac{\eta}{2} u_0''', \quad \epsilon_{rr} = -\eta u_0', \quad \epsilon_{\phi\phi} = -\eta u_0', \quad \epsilon_{xr} = 0, \quad (127)$$

Using strains given by Eq. (127) into stresses σ_{xx} and $\sigma_{\phi\phi}$ of Eq. (120) and simplifying, we get

$$\sigma_{xx} = E u_0' + r^2 (\lambda + 2\mu) \frac{\eta}{2} u_0''', \quad \sigma_{\phi\phi} = \lambda r^2 \frac{\eta}{2} u_0''' \quad (128)$$

where E is the Young's modulus of elasticity of the material of the rod.

Substituting Eqs. (123), (127) and (128) in the second of Eq. (125), the potential energy is

$$P = \frac{1}{2} \int_0^l \left[E S u_0'^2 + I_2 \eta E u_0' u_0''' + I_4 (\lambda + 2\mu) \frac{\eta^2}{4} u_0'''^2 \right] dx \quad (129)$$

The Lagrangian density Λ is defined as

$$T - P = \int_0^l \Lambda dx \quad (130)$$

Using Eqs. (126) and (129) in Eq. (130), the Lagrangian density is

$$\begin{aligned} \Lambda(\dot{u}_0, u_0', \dot{u}_0'', u_0''') \\ = \frac{1}{2} \left[\rho \left(S \dot{u}_0^2 + \eta I_2 \dot{u}_0 \dot{u}_0'' + \frac{\eta^2}{4} I_4 \dot{u}_0''^2 + \eta^2 I_2 \dot{u}_0'^2 \right) \right. \\ \left. - \left(E S u_0'^2 + I_2 \eta E u_0' u_0''' + I_4 (\lambda + 2\mu) \frac{\eta^2}{4} u_0'''^2 \right) \right] \end{aligned} \quad (131)$$

Using Euler-Lagrange differential equation

$$\frac{\partial}{\partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_0} \right) + \frac{\partial}{\partial x} \left(\frac{\partial \Lambda}{\partial u_0'} \right) - \frac{\partial^2}{\partial x \partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_0''} \right) + \frac{\partial^3}{\partial x^2 \partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_0'''} \right) + \frac{\partial^3}{\partial x^3} \left(\frac{\partial \Lambda}{\partial u_0'''} \right) = 0$$

we get the equation of motion

$$S(\rho\ddot{u}_0 - Eu_0'') + \eta I_2[\rho(1 - \eta)\ddot{u}_0'' - Eu_0^{(iv)}] + \frac{\eta^2}{4}I_4[\rho\ddot{u}_0^{(iv)} - (\lambda + 2\mu)u_0^{(vi)}] = 0 \quad (132)$$

with the boundary conditions at $x = 0$ and l

$$\begin{aligned} \left[-SEu_0' - \eta I_2Eu_0''' - \rho\eta^2 I_2 \ddot{u}_0' + \rho\frac{\eta^2}{4}I_4 \ddot{u}_0''' - \frac{\eta^2}{4}I_4(\lambda + 2\mu)u_0^{(v)} \right] \delta u_0 &= 0 \\ \left[-\rho\frac{\eta}{2}I_2\ddot{u}_0 - \rho\frac{\eta^2}{4}I_4\ddot{u}_0'' + \frac{\eta}{2}I_2Eu_0'' + \frac{\eta^2}{4}I_4(\lambda + 2\mu)u_0^{(iv)} \right] \delta u_0' &= 0 \\ \left[-\frac{\eta}{2}I_2Eu_0' - \frac{\eta^2}{4}I_4(\lambda + 2\mu)u_0''' \right] \delta u_0'' &= 0 \end{aligned} \quad (133)$$

The equations of motion given by Eq. (132) can be combined with boundary conditions provided by Eq. (133) to obtain the solutions of a finite rod.

2.12 The Twelfth, Shatalov's et al. Three Term Rayleigh-Love Type Theory

In contrast to three term Rayleigh-Bishop type theory, an additional Rayleigh-Love type assumption is made to neglect the radial stress σ_{rr} when calculating the potential energy of the system. Under this assumption, kinetic energy remains the same as before given by Eq. (126) while the potential energy takes the form

$$P = \frac{1}{2} \int_0^l \left[ESu_0'^2 + I_2\eta Eu_0'u_0''' + I_2\frac{\eta^2}{2}\lambda u_0'u_0''' + I_4(\lambda + 2\mu)\frac{\eta^2}{4}u_0'''^2 \right] dx \quad (134)$$

Using Eqs. (126) and (134) into Eq. (130), the Lagrangian density is

$$\Lambda(\dot{u}_0, u_0', \dot{u}_0', u_0'', u_0''') \quad (135)$$

$$= \frac{1}{2} \left[\rho \left(S \dot{u}_0^2 + \eta I_2 \dot{u}_0 \dot{u}_0'' + \frac{\eta^2}{4} I_4 \dot{u}_0''^2 + \eta^2 I_2 \dot{u}_0'^2 \right) - \left(E S u_0'^2 + I_2 \eta \left(E + \frac{\eta \lambda}{2} \right) u_0' u_0''' + I_4 (\lambda + 2\mu) \frac{\eta^2}{4} u_0'''^2 \right) \right]$$

Using Euler-Lagrange differential equation

$$\frac{\partial}{\partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_0} \right) + \frac{\partial}{\partial x} \left(\frac{\partial \Lambda}{\partial u_0'} \right) - \frac{\partial^2}{\partial x \partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_0'} \right) + \frac{\partial^3}{\partial x^2 \partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_0'} \right) + \frac{\partial^3}{\partial x^3} \left(\frac{\partial \Lambda}{\partial u_0'''} \right) = 0$$

we get the equation of motion

$$S(\rho \ddot{u}_0 - E u_0'') + \eta I_2 \left[\rho(1 - \eta) \ddot{u}_0'' - \left(E + \frac{\eta \lambda}{2} \right) u_0^{(iv)} \right] + \frac{\eta^2}{4} I_4 \left[\rho \ddot{u}_0^{(iv)} - (\lambda + 2\mu) u_0^{(vi)} \right] = 0 \quad (136)$$

with the boundary conditions at $x = 0$ and l

$$\begin{aligned} \left[-S E u_0' - \eta I_2 E u_0''' - \frac{\eta^2}{2} \lambda I_2 u_0''' - \rho \eta^2 I_2 \ddot{u}_0' + \rho \frac{\eta}{2} I_2 \ddot{u}_0' + \rho \frac{\eta^2}{4} I_4 \ddot{u}_0''' - \frac{\eta^2}{4} I_4 (\lambda + 2\mu) u_0^{(v)} \right] \delta u_0 &= 0 \\ \left[-\rho \frac{\eta}{2} I_2 \ddot{u}_0 - \rho \frac{\eta^2}{4} I_4 \ddot{u}_0'' + \frac{\eta}{2} I_2 E u_0'' + \frac{\eta^2}{4} I_4 (\lambda + 2\mu) u_0^{(iv)} + \frac{\eta^2}{4} \lambda I_2 u_0'' \right] \delta u_0' &= 0 \\ \left[-\frac{\eta}{2} I_2 E u_0' - \frac{\eta^2}{4} I_4 (\lambda + 2\mu) u_0''' - \frac{\eta^2}{4} \lambda I_2 u_0' \right] \delta u_0'' &= 0 \end{aligned} \quad (137)$$

The equations of motion given by Eq. (136) can be combined with boundary conditions given by Eq. (137) to obtain the solutions of a rod of finite length.

2.13 The Thirteenth, Shatalov's et al. Four Term Rayleigh-Bishop Type Theory

Shatalov et. al. [19] considered the longitudinal displacement $u(x, r, t)$ and the lateral displacement $w(x, r, t)$ in the form

$$u(x, r, t) = u_0(x, t) + r^2 u_2(x, t), \quad w(x, r, t) = r u_1(x, t) + r^3 u_3(x, t) \quad (138)$$

Substituting Eq. (138) into Eq. (119), and then the resulting strains into radial stress σ_{rr} and axial shear stress σ_{xr} given by Eq. (120), we get

$$\sigma_{rr} = (\lambda + 2\mu) (u_1 + 3r^2 u_3) + \lambda(u'_0 + r^2 u'_2 + u_1 + r^2 u_3) \quad (139)$$

$$\sigma_{xr} = \mu(2r u_2 + r u'_1 + r^3 u'_3)$$

Equating the coefficients of r^0 , r^1 and r^2 to zero, and simplifying yields

$$\begin{aligned} u_1 &= -\frac{\lambda}{2(\lambda + \mu)} u'_0 = -\eta u'_0, & u_2 &= -\frac{1}{2} u'_1 = \frac{\eta}{2} u''_0, \\ u_3 &= -\frac{\lambda}{2(2\lambda + 3\mu)} u'_2 = -\frac{\eta^2}{2(3 - 2\eta)} u'''_0 \end{aligned} \quad (140)$$

By use of Eq. (140), Eq. (139) converts to

$$\sigma_{rr} = 0, \quad \sigma_{xr} = \mu r^3 u'_3 = -\frac{\eta^2}{2(3 - 2\eta)} \mu r^3 u_0^{(iv)} \quad (141)$$

and Eq. (138) converts to

$$u(x, r, t) = u_0 + r^2 \frac{\eta}{2} u''_0, \quad w(x, r, t) = -r\eta u'_0 - \frac{\eta^2}{2(3 - 2\eta)} r^3 u'''_0 \quad (142)$$

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Using Eq. (142) in the first of Eq. (125), the kinetic energy is

$$\begin{aligned} T &= \frac{1}{2} \int_0^l \rho \left[S \dot{u}_0^2 + \eta I_2 \dot{u}_0 \dot{u}''_0 + \frac{\eta^2}{4} I_4 \dot{u}''_0^2 + \eta^2 I_2 \dot{u}_0'^2 + \frac{\eta^3}{(3 - 2\eta)} I_4 \dot{u}'_0 \dot{u}'''_0 \right. \\ &\quad \left. + \frac{\eta^4}{4(3 - 2\eta)^2} I_6 \dot{u}'''_0^2 \right] dx \end{aligned} \quad (143)$$

Using Eq. (134) into Eq. (119), the strains are

$$\epsilon_{xx} = u'_0 + r^2 \frac{\eta}{2} u_0''', \quad \epsilon_{rr} = -\eta u'_0 - \frac{3\eta^2}{2(3-2\eta)} r^2 u_0''', \quad (144)$$

$$\epsilon_{\phi\phi} = -\eta u'_0 - \frac{\eta^2}{2(3-2\eta)} r^2 u_0''', \quad \epsilon_{xr} = -\frac{\eta^2}{2(3-2\eta)} r^3 u_0^{(iv)}$$

Using Eq. (144) into stresses σ_{xx} and $\sigma_{\phi\phi}$ of Eq. (120), and simplifying,

$$\sigma_{xx} = E u'_0 + r^2 \left(\frac{3+\eta}{3-2\eta} \right) \eta \mu u_0''', \quad \sigma_{\phi\phi} = \frac{2\eta^2}{3-2\eta} \mu r^2 u_0''' \quad (145)$$

Substituting Eqs. (141), (144) and (145) in the second of Eq. (125), the potential energy is

$$P = \frac{1}{2} \int_0^l \left[E S u_0'^2 + I_2 \eta E u_0' u_0''' + I_4 \left(\frac{9-3\eta-4\eta^2}{(3-2\eta)^2} \right) \frac{\eta^2}{2} \mu u_0'''^2 \right. \\ \left. + \frac{\eta^4}{4(3-2\eta)^2} \mu I_6 u_0^{(v)2} \right] dx \quad (146)$$

Using Eqs. (143) and (146) in Eq. (130), the Lagrangian density is

$$\Lambda(\dot{u}_0, u'_0, \dot{u}_0', \dot{u}_0'', u_0''', \dot{u}_0''', u_0''''') \\ = \frac{1}{2} \left[\rho \left(S \dot{u}_0^2 + \eta I_2 \dot{u}_0 \dot{u}_0'' + \frac{\eta^2}{4} I_4 \dot{u}_0''^2 + \eta^2 I_2 \dot{u}_0'^2 + \frac{\eta^3}{(3-2\eta)} I_4 \dot{u}_0' \dot{u}_0''' \right. \right. \\ \left. \left. + \frac{\eta^4}{4(3-2\eta)^2} I_6 \dot{u}_0'''^2 \right) \right. \\ \left. - \left(E S u_0'^2 + I_2 \eta E u_0' u_0''' + I_4 \left(\frac{9-3\eta-4\eta^2}{(3-2\eta)^2} \right) \frac{\eta^2}{2} \mu u_0'''^2 \right. \right. \\ \left. \left. + \frac{\eta^4}{4(3-2\eta)^2} \mu I_6 u_0''''^2 \right) \right] \quad (147)$$

Using Euler-Lagrange differential equation

$$\begin{aligned} & \frac{\partial}{\partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_0} \right) + \frac{\partial}{\partial x} \left(\frac{\partial \Lambda}{\partial u'_0} \right) - \frac{\partial^2}{\partial x \partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}'_0} \right) + \frac{\partial^3}{\partial x^2 \partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}''_0} \right) + \frac{\partial^3}{\partial x^3} \left(\frac{\partial \Lambda}{\partial u'''_0} \right) - \frac{\partial^4}{\partial x^3 \partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}'''_0} \right) \\ & - \frac{\partial^4}{\partial x^4} \left(\frac{\partial \Lambda}{\partial u^{(iv)}_0} \right) = 0, \end{aligned}$$

we get the equation of motion

$$\begin{aligned} & S(\rho \ddot{u}_0 - E u''_0) + \eta I_2 [\rho(1 - \eta) \ddot{u}''_0 - E u'''_0] \\ & + \frac{\eta^2}{2} I_4 \left[\frac{3(1 - 2\eta)}{2(3 - 2\eta)} \rho \ddot{u}''''_0 - \left(\frac{9 - 3\eta - 4\eta^2}{9 - 12\eta + 4\eta^2} \right) \mu u^{(vi)}_0 \right] \\ & - \frac{\eta^4}{4(3 - 2\eta)^2} I_6 (\rho \ddot{u}^{(vi)}_0 - \mu u^{(viii)}_0) = 0 \end{aligned} \quad (148)$$

with the boundary conditions at $x = 0$ and l

$$\begin{aligned} & \left[\frac{\partial \Lambda}{\partial u'_0} - \frac{\partial}{\partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}'_0} \right) + \frac{\partial^2}{\partial x \partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}''_0} \right) + \frac{\partial^2}{\partial x^2} \left(\frac{\partial \Lambda}{\partial u'''_0} \right) - \frac{\partial^3}{\partial x^2 \partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}'''_0} \right) \right. \\ & \left. - \frac{\partial^2}{\partial x^3} \left(\frac{\partial \Lambda}{\partial u^{(iv)}_0} \right) \right] \delta u_0 = 0 \end{aligned} \quad (149)$$

$$\left[\frac{\partial \Lambda}{\partial u_0^{(iv)}} \right] \delta u_0''' = 0$$

The equations of motion given by Eq. (148) can be combined with boundary conditions provided by Eq. (149) to obtain the solutions of a rod of finite length

2.14 The Fourteenth, Shatalov's et al. Four Term Rayleigh-Love Type Theory

In contrast to four term Rayleigh-Bishop type theory, an additional Rayleigh-Love type assumption is made to neglect the axial shear stress σ_{xr} when calculating the potential energy of the system. Under this assumption, kinetic energy remains the same as before given by Eq. (143) while the potential energy takes the form

$$P = \frac{1}{2} \int_0^l \left[ESu_0'^2 + I_2 \eta E u_0' u_0''' + I_4 \left(\frac{9 - 3\eta - 4\eta^2}{(3 - 2\eta)^2} \right) \frac{\eta^2}{2} \mu u_0'''^2 \right] dx \quad (150)$$

Using Eqs. (143) and (150) in Eq. (130), the Lagrangian density is

$$\begin{aligned} & \Lambda(\dot{u}_0, u_0', \dot{u}_0', u_0'', u_0''', \dot{u}_0''') \\ &= \frac{1}{2} \left[\rho \left(S \dot{u}_0^2 + \eta I_2 \dot{u}_0 u_0'' + \frac{\eta^2}{4} I_4 \dot{u}_0'^2 + \eta^2 I_2 \dot{u}_0'^2 + \frac{\eta^3}{(3 - 2\eta)} I_4 \dot{u}_0' u_0''' \right. \right. \\ & \quad \left. \left. + \frac{\eta^4}{4(3 - 2\eta)^2} I_6 \dot{u}_0'''^2 \right) \right. \\ & \quad \left. - \left(ESu_0'^2 + I_2 \eta E u_0' u_0''' + I_4 \left(\frac{9 - 3\eta - 4\eta^2}{(3 - 2\eta)^2} \right) \frac{\eta^2}{2} \mu u_0'''^2 \right) \right] \end{aligned} \quad (151)$$

Using Euler-Lagrange differential equation

$$\frac{\partial}{\partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_0} \right) + \frac{\partial}{\partial x} \left(\frac{\partial \Lambda}{\partial u_0'} \right) - \frac{\partial^2}{\partial x \partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_0'} \right) + \frac{\partial^3}{\partial x^2 \partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_0''} \right) + \frac{\partial^3}{\partial x^3} \left(\frac{\partial \Lambda}{\partial u_0'''} \right) - \frac{\partial^4}{\partial x^3 \partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_0'''} \right) = 0$$

we get the equation of motion

$$\begin{aligned}
& S(\rho\ddot{u}_0 - Eu_0'') + \eta I_2 [\rho(1 - \eta)\ddot{u}_0'' - Eu_0^{(iv)}] \\
& + \frac{\eta^2}{2} I_4 \left[\frac{3(1 - 2\eta)}{2(3 - 2\eta)} \rho\ddot{u}_0^{(iv)} - \left(\frac{9 - 3\eta - 4\eta^2}{9 - 12\eta + 4\eta^2} \right) \mu u_0^{(vi)} \right] - \frac{\eta^4}{4(3 - 2\eta)^2} I_6 \rho\ddot{u}_0^{(vi)} = 0
\end{aligned} \tag{152}$$

with the boundary conditions at $x = 0$ and l

$$\begin{aligned}
& \left[\frac{\partial \Lambda}{\partial u_0'} - \frac{\partial}{\partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_0'} \right) + \frac{\partial^2}{\partial x \partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_0''} \right) + \frac{\partial^2}{\partial x^2} \left(\frac{\partial \Lambda}{\partial u_0'''} \right) - \frac{\partial^3}{\partial x^2 \partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_0'''} \right) \right] \delta u_0 = 0 \\
& \left[-\frac{\partial}{\partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_0''} \right) - \frac{\partial}{\partial x} \left(\frac{\partial \Lambda}{\partial u_0'''} \right) + \frac{\partial^2}{\partial x \partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_0'''} \right) \right] \delta u_0' = 0 \\
& \left[\frac{\partial \Lambda}{\partial u_0'''} - \frac{\partial^2}{\partial x \partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_0'''} \right) \right] \delta u_0'' = 0
\end{aligned} \tag{153}$$

The equations of motion given by Eq. (152) can be combined with boundary conditions given by Eq. (153) to obtain the solutions of a rod of finite length.

2.15 The Fifteenth, Shatalov's et al. Three Mode Theory

Shatalov et. al. [19] represented the longitudinal and lateral displacements using a power series. In their three-mode theory, they retained two terms for the longitudinal displacement $u(x, r, t)$ and a single term for the lateral displacement $w(x, r, t)$, given by

$$u(x, r, t) = u_0(x, t) + r^2 u_2(x, t), \quad w(x, r, t) = r u_1(x, t) \tag{154}$$

Substituting Eq. (154) into Eq. (119), the strains are

$$\epsilon_{xx} = u_0' + r^2 u_2', \quad \epsilon_{rr} = u_1, \quad \epsilon_{\phi\phi} = u_1, \quad \epsilon_{xr} = 2r u_2 + r u_1' \tag{155}$$

Using Eq. (155) in Eq. (120), the stresses are

$$\begin{aligned}
\sigma_{xx} &= (\lambda + 2\mu)(u_0' + r^2 u_2') + \lambda(2u_1) \\
\sigma_{rr} &= \sigma_{\phi\phi} = (\lambda + 2\mu) u_1 + \lambda(u_0' + r^2 u_2' + u_1)
\end{aligned} \tag{156}$$

$$\sigma_{xr} = \mu(2r u_2 + r u_1')$$

Substituting Eq. (154) in the first of Eq. (125), the kinetic energy is

$$T = \frac{1}{2} \int_0^l \rho [S \dot{u}_0^2 + 2I_2 \dot{u}_0 \dot{u}_2 + I_4 \dot{u}_2^2 + I_2 \dot{u}_1^2] dx \quad (157)$$

Using Eq. (155) and (156) in the second of Eq. (125) and simplifying, the potential energy is

$$P = \frac{1}{2} \int_0^l [(\lambda + 2\mu)(S u_0'^2 + 2I_2 u_0' u_2' + I_4 u_2'^2) + 4\lambda(S u_0' u_1 + I_2 u_1 u_2') + 4(\lambda + \mu)S u_1^2 + \mu I_2(u_1'^2 + 4u_1' u_2 + 4u_2^2)] dx \quad (158)$$

Using Eqs. (157) and (158) in Eq. (130), the Lagrangian density is

$$\begin{aligned} \Lambda(\dot{u}_0, \dot{u}_1, \dot{u}_2, u_0', u_1', u_2', u_1, u_2) \\ = \frac{1}{2} [\rho \{S \dot{u}_0^2 + 2I_2 \dot{u}_0 \dot{u}_2 + I_4 \dot{u}_2^2 + I_2 \dot{u}_1^2\} \\ - \{(\lambda + 2\mu)(S u_0'^2 + 2I_2 u_0' u_2' + I_4 u_2'^2) + 4\lambda(S u_0' u_1 + I_2 u_1 u_2') \\ + 4(\lambda + \mu)S u_1^2 + \mu I_2(u_1'^2 + 4u_1' u_2 + 4u_2^2)\}] \end{aligned} \quad (159)$$

Using Euler-Lagrange differential equations given by

$$\begin{aligned} \frac{\partial}{\partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_0} \right) + \frac{\partial}{\partial x} \left(\frac{\partial \Lambda}{\partial u_0'} \right) &= 0, \quad \frac{\partial}{\partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_1} \right) + \frac{\partial}{\partial x} \left(\frac{\partial \Lambda}{\partial u_1'} \right) - \frac{\partial \Lambda}{\partial u_1} = 0, \\ \frac{\partial}{\partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_2} \right) + \frac{\partial}{\partial x} \left(\frac{\partial \Lambda}{\partial u_2'} \right) - \frac{\partial \Lambda}{\partial u_2} &= 0 \end{aligned}$$

and simplifying, we get the equations of motion

$$\begin{aligned} S[\rho \ddot{u}_0 - (\lambda + 2\mu)u_0''] - 2\lambda S u_1' + I_2[\rho \ddot{u}_2 - (\lambda + 2\mu)u_2''] &= 0 \\ 2\lambda S u_0' + I_2[\rho \ddot{u}_1 - \mu u_1''] + 4(\lambda + \mu)S u_1 + 2(\lambda - \mu)I_2 u_2' &= 0 \end{aligned} \quad (160)$$

$$I_2[\rho\ddot{u}_0 - (\lambda + 2\mu)u_0''] - 2(\lambda - \mu)I_2u_1' + I_4[\rho\ddot{u}_2 - (\lambda + 2\mu)u_2''] + 4\mu I_2u_2 = 0$$

subjected to the boundary conditions at $x = 0$ and l

$$\begin{aligned} [(\lambda + 2\mu)Su_0' + 2\lambda Su_1 + (\lambda + 2\mu)I_2u_2']\delta u_0 &= 0 \\ [\mu I_2u_1' + 2\mu I_2u_2]\delta u_1 &= 0 \\ [(\lambda + 2\mu)I_2u_0' + 2\lambda I_2u_1 + (\lambda + 2\mu)I_4u_2']\delta u_2 &= 0 \end{aligned} \quad (161)$$

The equations of motion given by Eq. (160) can be combined with boundary conditions given by Eq. (161) to obtain the solutions of a rod of finite length

2.16 The Sixteenth, Boström -Shatalov's Four Mode Theory

In their four-mode theory, Boström [23] and Shatalov et. al. [19] retained two terms for the longitudinal displacement $u(x, r, t)$ and two terms for the lateral displacement $w(x, r, t)$, given by

$$u(x, r, t) = u_0(x, t) + r^2u_2(x, t), \quad w(x, r, t) = ru_1(x, t) + r^3u_3(x, t) \quad (162)$$

Substituting Eq. (162) into Eq. (119), the strains are

$$\begin{aligned} \epsilon_{xx} &= u_0' + r^2u_2', & \epsilon_{rr} &= u_1 + 3r^2u_3, & \epsilon_{\phi\phi} &= u_1 + r^2u_3, \\ \epsilon_{xr} &= 2ru_2 + ru_1' + r^3u_3' \end{aligned} \quad (163)$$

Using Eq. (163) in Eq. (120), the stresses are

$$\begin{aligned} \sigma_{xx} &= (\lambda + 2\mu)(u_0' + r^2u_2') + \lambda(2u_1 + 4r^2u_3) \\ \sigma_{rr} &= (\lambda + 2\mu)(u_1 + 3r^2u_3) + \lambda(u_0' + r^2u_2' + u_1 + r^2u_3) \\ \sigma_{\phi\phi} &= (\lambda + 2\mu)(u_1 + r^2u_3) + \lambda(u_0' + r^2u_2' + u_1 + 3r^2u_3) \\ \sigma_{xr} &= \mu(2ru_2 + ru_1' + r^3u_3') \end{aligned} \quad (164)$$

Using Eq. (162) in the first of Eq. (125), the kinetic energy is

$$T = \frac{1}{2} \int_0^l \rho [S\dot{u}_0^2 + 2I_2\dot{u}_0\dot{u}_2 + I_4\dot{u}_2^2 + I_2\dot{u}_1^2 + 2I_4\dot{u}_1\dot{u}_3 + I_6\dot{u}_3^2] dx \quad (165)$$

Using Eqs. (163) and (164) in the second of Eq. (125) and simplifying, the potential energy is

$$\begin{aligned} P = \frac{1}{2} \int_0^l [& (\lambda + 2\mu)(Su_0'^2 + 2I_2u_0'u_2' + I_4u_2'^2) \\ & + 2\lambda(2Su_0'u_1 + 4I_2u_0'u_3 + Su_1^2 + 2I_2u_1u_2' + 4I_2u_1u_3 + 4I_4u_2'u_3 \\ & + 3I_4u_3^2) \\ & + \mu(I_2u_1'^2 + 4I_2u_2^2 + I_6u_3'^2 + 4I_2u_1'u_2 + 2I_4u_1'u_3 + 4I_4u_2u_3')] dx \end{aligned} \quad (166)$$

Using Eqs. (165) and (166) in Eq. (130), the Lagrangian density is

$$\begin{aligned} \Lambda(\dot{u}_0, \dot{u}_1, \dot{u}_2, \dot{u}_3, u_0', u_1', u_2', u_3', u_1, u_2, u_3) \\ = \frac{1}{2} [\rho \{ S\dot{u}_0^2 + 2I_2\dot{u}_0\dot{u}_2 + I_4\dot{u}_2^2 + I_2\dot{u}_1^2 + 2I_4\dot{u}_1\dot{u}_3 + I_6\dot{u}_3^2 \} \\ - \{ (\lambda + 2\mu)(Su_0'^2 + 2I_2u_0'u_2' + I_4u_2'^2) \\ + 2\lambda(2Su_0'u_1 + 4I_2u_0'u_3 + Su_1^2 + 2I_2u_1u_2' + 4I_2u_1u_3 + 4I_4u_2'u_3 \\ + 3I_4u_3^2) \\ + \mu(I_2u_1'^2 + 4I_2u_2^2 + I_6u_3'^2 + 4I_2u_1'u_2 + 2I_4u_1'u_3 + 4I_4u_2u_3') \}] \end{aligned} \quad (167)$$

Using Euler-Lagrange differential equations given by

$$\begin{aligned} \frac{\partial}{\partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_0} \right) + \frac{\partial}{\partial x} \left(\frac{\partial \Lambda}{\partial u_0'} \right) &= 0, & \frac{\partial}{\partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_1} \right) + \frac{\partial}{\partial x} \left(\frac{\partial \Lambda}{\partial u_1'} \right) - \frac{\partial \Lambda}{\partial u_1} &= 0, \\ \frac{\partial}{\partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_2} \right) + \frac{\partial}{\partial x} \left(\frac{\partial \Lambda}{\partial u_2'} \right) - \frac{\partial \Lambda}{\partial u_2} &= 0, & \frac{\partial}{\partial t} \left(\frac{\partial \Lambda}{\partial \dot{u}_3} \right) + \frac{\partial}{\partial x} \left(\frac{\partial \Lambda}{\partial u_3'} \right) - \frac{\partial \Lambda}{\partial u_3} &= 0 \end{aligned}$$

and simplifying, we get the equations of motion

$$\begin{aligned}
& S[\rho\ddot{u}_0 - (\lambda + 2\mu)u_0''] - 2\lambda Su_1' + I_2[\rho\ddot{u}_2 - (\lambda + 2\mu)u_2''] - 4\lambda I_2 u_3' = 0 \\
& 2\lambda Su_0' + I_2[\rho\ddot{u}_1 - \mu u_1''] + 2\lambda Su_1 + 2(\lambda - \mu)I_2 u_2' + I_4[\rho\ddot{u}_3 - \mu u_3''] + 4\lambda I_2 u_3 = 0 \quad (168) \\
& I_2[\rho\ddot{u}_0 - (\lambda + 2\mu)u_0''] - 2(\lambda - \mu)I_2 u_1' + I_4[\rho\ddot{u}_2 - (\lambda + 2\mu)u_2''] + 4\mu I_2 u_2 \\
& \quad - 2I_4(2\lambda - \mu)u_3' = 0 \\
& 4\lambda I_2 u_0' + I_4[\rho\ddot{u}_1 - \mu u_1''] + 4\lambda I_2 u_1 + 2I_4(2\lambda - \mu)u_2' + I_6[\rho\ddot{u}_3 - \mu u_3''] + 6\lambda I_4 u_3 = 0
\end{aligned}$$

subjected to the boundary conditions at $x = 0$ and l

$$\begin{aligned}
& [(\lambda + 2\mu)Su_0' + 2\lambda Su_1 + (\lambda + 2\mu)I_2 u_2' + 4\lambda I_2 u_3]\delta u_0 = 0 \\
& [\mu I_2 u_1' + 2\mu I_2 u_2 + \mu I_4 u_3']\delta u_1 = 0 \quad (169) \\
& [(\lambda + 2\mu)I_2 u_0' + 2\lambda I_2 u_1 + (\lambda + 2\mu)I_4 u_2' + 4\lambda I_4 u_3]\delta u_2 = 0 \\
& [\mu I_4 u_1' + 2\mu I_2 u_2 + \mu I_6 u_3']\delta u_3 = 0
\end{aligned}$$

The equations of motion given by Eq. (168) can be combined with boundary conditions given by Eq. (169) to obtain the solutions of a finite length rod.

3. Comparison with Pochhammer-Chree's Exact Solution

In his 1876 paper [46], Leo A. Pochhammer (1841–1920) solved the three-dimensional equations of elasticity to obtain exact solutions that encompassed all forms of vibration: torsional, longitudinal, transverse, and combinations thereof. Remarkably, this was achieved at a time when only classical wave theory was available, and Lord Rayleigh's, refined theory was apparently still in development, eventually published a year later, in his 1877 treatise *The Theory of Sound*. Possibly, independently, in 1886 [47], Charles Chree (1860–1928) also derived exact solutions to the same equations of elasticity. Indeed, Chree [48] notes, “*That Pochhammer's work was wholly unknown to me until the appearance of Love's 'Elasticity'.*”

According to H. D. McNiven and J. J. McCoy [49], “*Even though the exact solution was established by Pochhammer in 1876 and received prominence by being included and discussed at*

length in Love's 1892 monograph, no systematic exploration of its roots was undertaken until 1941." Although early attempts were made by Ruedy [50] and Field [51] to analyze the curves mathematically, significant progress came with the contributions of Bancroft [52] and Hudson [53], which laid the groundwork for subsequent researchers.

Bancroft [52] made a pivotal breakthrough by reducing the complex frequency equation, originally involving six variables (wavelength, frequency, radius of the bar, density, and the two elastic moduli of the isotropic material), to a more tractable form involving just three (wave velocity to bar velocity ratio (v/v_0), diameter-to-wavelength ratio, and Poisson's ratio. Owing to the transcendental nature of the functions involved, analytical solutions were not feasible. Bancroft instead numerically solved the equation and plotted v/v_0 against the diameter-to-wavelength ratio for various values of Poisson's ratio. These plots became known as dispersion curves.

Building on this approach, Hudson [53] plotted the ratio of longitudinal to shear wave velocities against the $\tau_0/(1 + \tau_0)$ for various Poisson's ratios, where τ_0 is the shear wave number. He discovered that all longitudinal wave curves were tangent at a single point, which he identified as the universal point, a significant insight into wave behavior.

Further contributions included Davies's 1948 paper [54], in which he numerically solved the frequency equation, making it more accessible to applied researchers. In 1951, Holden [55] made another important observation: real values of non-dimensional velocity could exist even for imaginary values of the propagation constant, expanding the understanding of wave propagation characteristics.

A major advancement came in 1962 [56] when Onoe and co-authors plotted multiple branches of dimensionless frequency versus dimensionless propagation constant for Poisson's ratio of 0.31.

This achievement produced what are now referred to as the frequency spectrum curves, marking a milestone in the visualization and practical interpretation of wave propagation in elastic media.

Due to the complexity of Pochhammer-Chree's exact solution, several simplified, refined theories have been developed over time. However, the validity and accuracy of these approximations are typically assessed by comparing them against Pochhammer-Chree's original solution. Therefore, it is worthwhile to first examine Pochhammer-Chree's exact solution, for which we will reference Love's 1892 [57] monograph.

3.1 Pochhammer-Chree's Exact Theory

The three equations of vibration referred to cylindrical coordinates r, θ, z are

$$\begin{aligned}\rho \frac{\partial^2 u_r}{\partial t^2} &= (\lambda + 2\mu) \frac{\partial \Delta}{\partial r} - \frac{2\mu}{r} \frac{\partial \varpi_z}{\partial \theta} + 2\mu \frac{\partial \varpi_\theta}{\partial z} \\ \rho \frac{\partial^2 u_\theta}{\partial t^2} &= (\lambda + 2\mu) \frac{1}{r} \frac{\partial \Delta}{\partial \theta} - 2\mu \frac{\partial \varpi_r}{\partial z} + 2\mu \frac{\partial \varpi_z}{\partial r} \\ \rho \frac{\partial^2 u_z}{\partial t^2} &= (\lambda + 2\mu) \frac{\partial \Delta}{\partial z} - \frac{2\mu}{r} \frac{\partial (r \varpi_\theta)}{\partial r} + \frac{2\mu}{r} \frac{\partial \varpi_r}{\partial \theta}\end{aligned}\quad (170)$$

in which the dilatation Δ is related to displacement components by

$$\Delta = \frac{1}{r} \frac{\partial (r u_r)}{\partial r} + \frac{1}{r} \frac{\partial u_\theta}{\partial \theta} + \frac{\partial u_z}{\partial z} \quad (171)$$

and satisfies the differential equation

$$\frac{\rho}{(\lambda + 2\mu)} \frac{\partial^2 \Delta}{\partial t^2} = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \Delta}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \Delta}{\partial \theta^2} + \frac{\partial^2 \Delta}{\partial z^2} \quad (172)$$

Also, the rotational quantities $\varpi_r, \varpi_\theta, \varpi_z$ are given by

$$2\varpi_r = \frac{1}{r} \frac{\partial u_z}{\partial \theta} - \frac{\partial u_\theta}{\partial z}, \quad 2\varpi_\theta = \frac{\partial u_r}{\partial z} - \frac{\partial u_z}{\partial r}, \quad 2\varpi_z = \frac{1}{r} \left(\frac{\partial (r u_\theta)}{\partial r} - \frac{\partial u_r}{\partial \theta} \right) \quad (173)$$

The stress-components $\sigma_{rr}, \sigma_{r\theta}, \sigma_{rz}$ vanish at the surface of the cylinder $r = a$. These stress-components are expressed by the formulae

$$\sigma_{rr} = \lambda\Delta + 2\mu \frac{\partial u_r}{\partial r}, \quad \sigma_{r\theta} = \mu \left\{ \frac{1}{r} \frac{\partial u_r}{\partial \theta} + r \frac{\partial}{\partial r} \left(\frac{u_\theta}{r} \right) \right\}, \quad \sigma_{rz} = \mu \left(\frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} \right) \quad (174)$$

Now, we shall take u_r, u_θ, u_z to be of the forms

$$u_r = Ue^{i(\gamma z + pt)}, \quad u_\theta = Ve^{i(\gamma z + pt)}, \quad u_z = We^{i(\gamma z + pt)}, \quad (175)$$

in which U, V, W are functions of r , but independent of θ , for the case of longitudinal vibration.

Then, the expressions for $\varpi_r, \varpi_\theta, \varpi_z$ given by Eq. (173) reduce to

$$2\varpi_r = -\frac{\partial u_\theta}{\partial z}, \quad 2\varpi_\theta = \frac{\partial u_r}{\partial z} - \frac{\partial u_z}{\partial r}, \quad 2\varpi_z = \frac{1}{r} \frac{\partial(ru_\theta)}{\partial r} \quad (176)$$

Substituting Eq. (176) in the first and the third of Eq. (170), we get

$$\begin{aligned} \rho \frac{\partial^2 u_r}{\partial t^2} &= (\lambda + 2\mu) \frac{\partial \Delta}{\partial r} + 2\mu \frac{\partial \varpi_\theta}{\partial z} \\ \rho \frac{\partial^2 u_z}{\partial t^2} &= (\lambda + 2\mu) \frac{\partial \Delta}{\partial z} - \frac{2\mu}{r} \frac{\partial(r\varpi_\theta)}{\partial r} \end{aligned} \quad (177)$$

Differentiating the first and the second of Eq. (177) with z and r respectively, and then subtracting the latter from the first, we get

$$\frac{\rho}{\mu} \frac{\partial^2 \varpi_\theta}{\partial t^2} = \frac{\partial^2 \varpi_\theta}{\partial z^2} + \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial(r\varpi_\theta)}{\partial r} \right) \quad (178)$$

Substituting u_r, u_θ, u_z given by Eq. (175) in Eq. (171) and the second of Eq. (176), we get

$$\begin{aligned} \Delta &= \left(\frac{1}{r} \frac{\partial(rU)}{\partial r} + i\gamma W \right) e^{i(\gamma z + pt)} = \Pi e^{i(\gamma z + pt)} \\ 2\varpi_\theta &= \left(i\gamma U - \frac{\partial W}{\partial r} \right) e^{i(\gamma z + pt)} = \Psi e^{i(\gamma z + pt)} \end{aligned} \quad (179)$$

where Π, Ψ are functions of r only.

Substituting the first and the second of Eq. (179) in Eqs. (172) and (178) respectively, we get

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \Pi}{\partial r} \right) + h'^2 \Pi = 0, \quad \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial (r \Psi)}{\partial r} \right) + \kappa'^2 \Psi = 0 \quad (180)$$

where

$$h'^2 = \frac{p^2 \rho}{\lambda + 2\mu} - \gamma^2, \quad \kappa'^2 = \frac{p^2 \rho}{\mu} - \gamma^2 \quad (181)$$

The solutions to the differential equations given by Eq. (180) are Bessel's functions $J_0(h'r)$ and $J_1(\kappa'r)$ respectively.

Now, substituting Eqs. (175) and (179) in Eq. (177), we get

$$\begin{aligned} -\rho \gamma^2 U &= (\lambda + 2\mu) \frac{\partial \Pi}{\partial r} + i\mu \gamma \Psi = (\lambda + 2\mu) \frac{\partial J_0(h'r)}{\partial r} + i\mu \gamma J_1(\kappa'r) \\ -\rho \gamma^2 W &= (\lambda + 2\mu) i\gamma \Pi - \frac{\mu}{r} \frac{\partial (r \Psi)}{\partial r} = (\lambda + 2\mu) i\gamma J_0(h'r) - \frac{\mu}{r} \frac{\partial (r J_1(\kappa'r))}{\partial r} \end{aligned}$$

which can be written as

$$U = A \frac{\partial J_0(h'r)}{\partial r} + C \gamma J_1(\kappa'r), \quad W = A i\gamma J_0(h'r) + \frac{iC}{r} \frac{\partial (r J_1(\kappa'r))}{\partial r} \quad (182)$$

where A, C are constants.

Since the traction across the cylindrical surface $r = a$ vanishes, substituting Eq. (182) in the first and the third of Eq. (174), and setting each to zero gives two homogeneous equations in A and C .

$$\begin{aligned} A \left[2\mu \frac{\partial^2 J_0(h'r)}{\partial r^2} - \frac{p^2 \rho \lambda}{\lambda + 2\mu} J_0(h'r) \right]_{r=a} + C \left[2\mu \gamma \frac{\partial J_1(\kappa'r)}{\partial r} \right]_{r=a} &= 0 \\ 2A \gamma \frac{\partial J_0(h'r)}{\partial r} \Big|_{r=a} + C \left(2\gamma^2 - \frac{p^2 \rho}{\mu} \right) J_1(\kappa'r) \Big|_{r=a} &= 0 \end{aligned} \quad (183)$$

In the words of Love [57], “On eliminating the ratio $A : C$, we obtain the frequency equation”, but he did not mention the frequency equation. However, he derived the first and the second approximation to the velocity of a longitudinal wave.

If we eliminate the ratio $A:C$, we obtain

$$\begin{aligned} & \left[2\mu \frac{\partial^2 J_0(h'r)}{\partial r^2} - \lambda(h'^2 + \gamma^2)J_0(h'r) \right] (\gamma^2 - \kappa'^2)J_1(\kappa'r) \Big|_{r=a} \\ & - \left[2\mu\gamma \frac{\partial J_1(\kappa'r)}{\partial r} \right] 2\gamma \frac{\partial J_0(h'r)}{\partial r} \Big|_{r=a} = 0 \end{aligned} \quad (184)$$

Now, let's use the properties, $J'_0(x) = -J_1(x)$ and $J'_1(x) = J_0(x) - J_1(x)/x$, of Bessel's function to convert derivatives of Bessel's functions to that without derivatives and use non-dimensional analysis.

$$\begin{aligned} \frac{\partial J_0(h'r)}{\partial r} \Big|_{r=a} &= \frac{\partial J_0(h'r)}{\partial(h'r)} \frac{\partial(h'r)}{\partial r} \Big|_{r=a} = -h'J_1(h'a) \\ \frac{\partial^2 J_0(h'r)}{\partial r^2} \Big|_{r=a} &= \frac{\partial}{\partial r} [-h'J_1(h'r)] \Big|_{r=a} = -h'^2 \left[J_0(h'a) - \frac{J_1(h'a)}{h'a} \right] \\ \frac{\partial J_1(\kappa'r)}{\partial r} \Big|_{r=a} &= \kappa' \left[J_0(\kappa'a) - \frac{J_1(\kappa'a)}{\kappa'a} \right] \end{aligned} \quad (185)$$

Substituting Eq. (185) in Eq. (184) and simplifying, we get

$$\begin{aligned} & -2h'J_1(h'a)J_1(\kappa'a)[\kappa'^2 + \gamma^2] \\ & - a \left[\frac{(\lambda + 2\mu)}{\mu} h'^2 + \frac{\lambda}{\mu} \gamma^2 \right] (\gamma^2 - \kappa'^2)J_0(h'a)J_1(\kappa'a) \\ & + 4\gamma^2 \kappa' h'a J_0(\kappa'a)J_1(h'a) = 0 \end{aligned} \quad (186)$$

Since Bessel's function can take only non-dimensional arguments, $h'a, \kappa'a$ are already dimensionless. However, h', κ', a separately have dimensions. To address the variability of $h'a, \kappa'a$, we introduce dimensionless variables α, β for $h'a/\delta, \kappa'a/\delta$ where δ is the lowest nonzero root of $J_1(\delta_n) = 0$ and is a scaling factor for our problem. Then, with the use of Eq. (181),

$$\alpha^2 = \left(\frac{h'a}{\delta}\right)^2 = \frac{p^2 \frac{\rho}{\mu} \left(\frac{a}{\delta}\right)^2}{\frac{\lambda + 2\mu}{\mu}} - \left(\frac{\gamma a}{\delta}\right)^2 = \frac{p^2}{k^2} - \zeta^2 = \frac{\Omega^2}{k^2} - \zeta^2$$

$$\beta^2 = \left(\frac{\kappa'a}{\delta}\right)^2 = \frac{p^2 \rho}{\mu} \left(\frac{a}{\delta}\right)^2 - \left(\frac{\gamma a}{\delta}\right)^2 = \frac{p^2}{p_s^2} - \zeta^2 = \Omega^2 - \zeta^2$$
(187)

where

$\Omega = p/p_s =$ dimensionless frequency

$\zeta = \gamma a/\delta =$ dimensionless propagation constant

$p =$ circular frequency as introduced in Eq. (175)

$p_s = (\delta/a)(\mu/\rho)^{1/2} =$ circular frequency of first axial shear mode

$k = (\lambda + 2\mu)/\mu =$ ratio of dilatational velocity to equivoluminal velocity in an infinite, isotropic medium $= [2(1 - \nu)/(1 - 2\nu)]^{1/2}$, where ν is Poisson's ratio

Now, multiplying throughout of Eq. (186) by a^3/δ^3 , we get

$$\begin{aligned} & -2\alpha J_1(\delta\alpha)J_1(\delta\beta)[\beta^2 + \zeta^2] - \delta[k^2\alpha^2 + (k^2 - 2)\zeta^2][\zeta^2 - \beta^2]J_0(\delta\alpha)J_1(\delta\beta) \\ & + 4\zeta^2\beta\alpha\delta J_0(\delta\beta)J_1(\delta\alpha) = 0 \end{aligned}$$
(188)

Multiplying Eq. (188) by $\alpha/J_1(\delta\alpha)J_1(\delta\beta)$, and introducing new function $\mathcal{J}_1(x)$ for $xJ_0(x)/J_1(x)$, we get

$$-2\alpha^2\Omega^2 + (2\zeta^2 - \Omega^2)^2\mathcal{J}_1(\delta\alpha) + 4\zeta^2\alpha^2\mathcal{J}_1(\delta\beta) = 0$$
(189)

which coincides with the frequency equation of Onoe et al. [56], and apparently obtained for the first time, in similar form, by Bancroft [52].

As already mentioned above, there are two ways to graphically represent the frequency equation: dispersion curves and frequency spectrum. To generate the dispersion curves as obtained by Bancroft [52], let's convert Eq. (189) to appropriate form by dividing throughout by $4\zeta^4$,

introducing x for $\Omega^2/2\zeta^2$, and replacing $\delta\alpha$ and $\delta\beta$ by their previous form $h'a$ and $\kappa'a$ respectively.

$$(x-1)^2 \mathcal{J}_1(h'a) - \frac{\alpha^2}{\zeta^2} [x - \mathcal{J}_1(\kappa'a)] = 0 \quad (190)$$

Now, dividing the first of Eq. (187) by ζ^2 , we get

$$\frac{\alpha^2}{\zeta^2} = \frac{\Omega^2}{k^2 \zeta^2} - 1 = 2x \frac{1-2\nu}{2(1-\nu)} - 1 = \varepsilon x - 1 \quad (191)$$

where $\varepsilon = (1-2\nu)/(1-\nu)$. Furthermore,

$$\begin{aligned} x &= \frac{\Omega^2}{2\zeta^2} = \frac{\frac{p^2}{p_s^2}}{2\left(\frac{\gamma a}{\delta}\right)^2} = \frac{\frac{p^2}{\left(\frac{\delta p_s}{a}\right)^2}}{2\gamma^2} = \frac{\frac{p^2 \rho}{\mu}}{2\gamma^2} = \frac{(v\gamma)^2 2(1+\nu)}{\frac{E}{\rho} 2\gamma^2} = \left(\frac{v}{v_0}\right)^2 (1+\nu) \\ h'^2 &= \frac{p^2 \rho}{\lambda + 2\mu} - \gamma^2 = \frac{(v\gamma)^2}{\frac{(1-\nu)}{(1-2\nu)(1+\nu)} v_0^2} - \gamma^2 = \gamma^2 (\beta x - 1) \\ \kappa'^2 &= \frac{p^2 \rho}{\mu} - \gamma^2 = \frac{(v\gamma)^2}{\frac{1}{2(1+\nu)} v_0^2} - \gamma^2 = \gamma^2 (2x - 1) \end{aligned} \quad (192)$$

where

$v = p/\gamma$ = Velocity of the wave

$v_0 = (E/\rho)^{1/2}$ = “Bar velocity”

Using Eq. (191) in Eq. (190), we obtain the form of Dancroft [52]

$$(x-1)^2 \mathcal{J}_1(h'a) - (\varepsilon x - 1)[x - \mathcal{J}_1(\kappa'a)] = 0 \quad (193)$$

where φ, h, k, β and σ have been denoted by $\mathcal{J}_1, h', \kappa', \varepsilon$ and ν respectively. In the words of Bancroft, “Referring to the definitions, we see that the latter equation is of the form $F(x, \beta, \gamma a) =$

0, and that therefore the transformation to a function of three variables has been accomplished.”

Now, Eq. (193) is solved numerically by considering Bessel’s functions J_0 and J_1 up to 20th order to generate **Table 1**, where more conventional variables v/v_0 , v and d/L have been used. Here, d and L denotes rod diameter and wavelength respectively.

Two particular cases arise corresponding to $\gamma a = 0$ and $\gamma a \rightarrow \infty$. When $\gamma a = 0$, Eq. (193) yields at once

$$x(\varepsilon, 0) = \frac{3 - 2\varepsilon}{2 - \varepsilon} = 1 + v \quad (194)$$

where the property $J_1(0) = 2$, of function J_1 has been used.

For some large value of γa , the arguments of J_1 function become and remain pure imaginaries, and also that the arguments, $h'a$ and $\kappa'a$, increase without limit. Under these conditions, $J_1(h'a) = -ih'a$ and $J_1(\kappa'a) = -i\kappa'a$, and Eq. (193) reduces to

$$x^3 - 4x^2 + (6 - 2\beta)x - (2 - \beta) = 0 \quad (195)$$

Table 1: v/v_0 as a function of d/L and v (adapted from Bancroft [52] except for the cases of Poisson’s ratio of 0.00 and 0.50, and independently verified by authors)

d/L	v								
	0.00	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.50
0.00	1	1	1	1	1	1	1	1	1
0.05	1	0.99994	0.99986	0.99975	0.99961	0.99944	0.99924	0.99901	0.99846
0.10	1	0.99975	0.99943	0.99899	0.99843	0.99774	0.99694	0.99602	0.99385
0.15	1	0.99941	0.99868	0.99767	0.99638	0.99482	0.99302	0.99097	0.98620
0.20	1	0.99890	0.99754	0.99568	0.99333	0.99054	0.98732	0.98373	0.97557
0.25	1	0.99816	0.99591	0.99287	0.98909	0.98467	0.97967	0.97418	0.96205
0.30	1	0.99710	0.99362	0.98899	0.98337	0.97691	0.96979	0.96214	0.94579

0.35	1	0.99556	0.99038	0.98366	0.97573	0.96688	0.95739	0.94747	0.92698
0.40	1	0.99323	0.98569	0.97627	0.96559	0.95411	0.94218	0.93007	0.90594
0.45	1	0.98951	0.97866	0.96592	0.95220	0.93810	0.92397	0.91001	0.88310
0.50	1	0.98296	0.96771	0.95133	0.93479	0.91854	0.90277	0.88758	0.85901
0.55	1	0.97014	0.95037	0.93119	0.91288	0.89549	0.87899	0.86333	0.83431
0.60	0.98015	0.94487	0.92436	0.90502	0.88681	0.86964	0.85341	0.83806	0.80968
0.65	0.91768	0.90658	0.89086	0.87433	0.85800	0.84222	0.82709	0.81265	0.78572
0.70	0.86664	0.86493	0.85470	0.84201	0.82841	0.81466	0.80111	0.78793	0.76294
0.75	0.82466	0.82653	0.82009	0.81074	0.79982	0.78818	0.77632	0.76452	0.74167
0.80	0.78995	0.79306	0.78893	0.78202	0.77332	0.76358	0.75331	0.74285	0.72211
0.85	0.76111	0.76445	0.76167	0.75644	0.74943	0.74125	0.73236	0.72310	0.70432
0.90	0.73706	0.74013	0.73812	0.73402	0.72826	0.72130	0.71355	0.70532	0.68829
0.95	0.71692	0.71949	0.71791	0.71454	0.70967	0.70365	0.69682	0.68946	0.67393
1.00	0.70001	0.70196	0.70058	0.69768	0.69344	0.68814	0.68203	0.67537	0.66112
1.20	0.65493	0.65419	0.65266	0.65030	0.64712	0.64321	0.63869	0.63369	0.62270
1.40	0.63142	0.62836	0.62623	0.62361	0.62048	0.61687	0.61284	0.60844	0.59885
1.60	0.61884	0.61393	0.61118	0.60815	0.60479	0.60111	0.59713	0.59289	0.58381
1.80	0.61201	0.60565	0.60237	0.59892	0.59526	0.59139	0.58731	0.58305	0.57407
2.00	0.60830	0.60080	0.59707	0.59326	0.58932	0.58524	0.58101	0.57664	0.56760
∞	0.61803	0.60213	0.59491	0.58805	0.58148	0.57516	0.56903	0.56307	0.55155

With the help of **Table 1**, the dispersion curves for various Poisson's ratios ν have been plotted, as shown in **Figure 1** and will be used to compare the accuracy of refined theories in later sections. However, these curves correspond to the first mode only, as the frequency equation is a transcendental equation and has infinitely many solutions corresponding to higher modes. Higher

mode dispersion curves can also be obtained by considering other roots and will be obtained for Mindlin-Hermann theory.

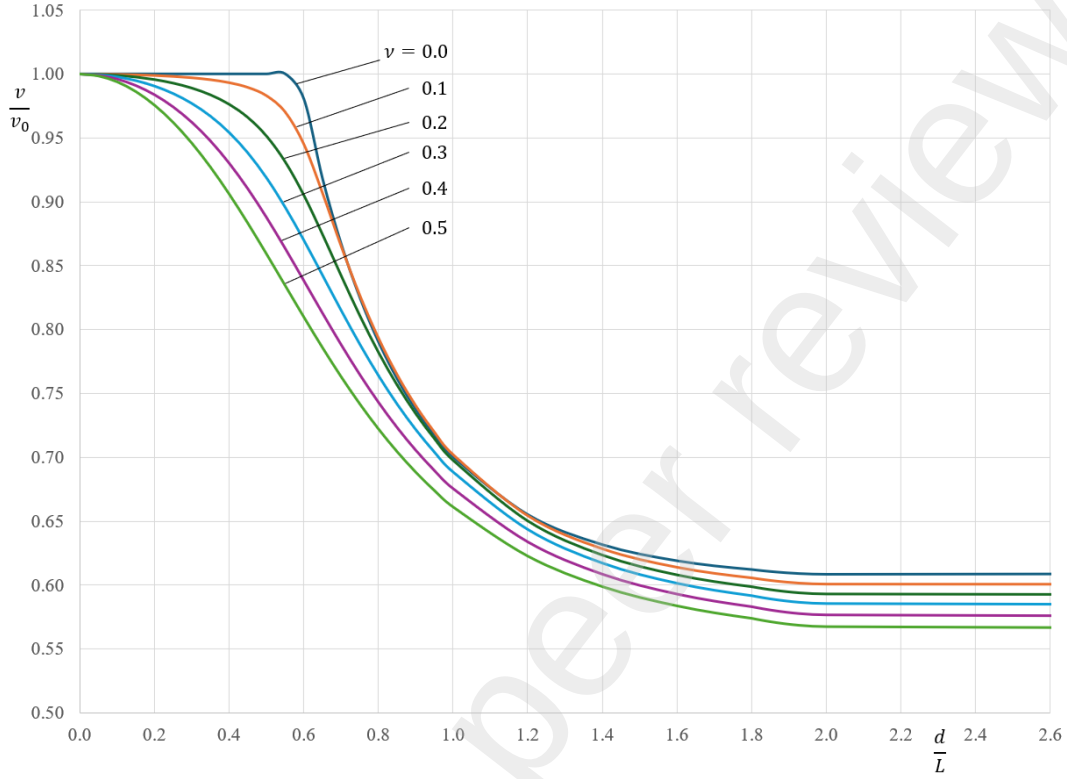


Figure 1: v/v_0 as a function of d/L for various values of ν (adapted from Bancroft [52], and independently verified by authors)

To obtain the frequency spectrum, the frequency equation given by Eq. (189), and the expressions of auxiliary parameters α, β given by Eq. (187) have been used. The main idea is to generate families of simple curves corresponding to α and β , and obtain their points of intersection, which satisfy Eq. (189).

The first kind is ϕ – families of curves given by

$$\Omega^2 = \zeta^2 + \phi_m^2, \quad \Omega^2 = k^2(\zeta^2 + \phi_n^2) \quad (196)$$

where ϕ_m and ϕ_n satisfy the equations

$$J_1(\delta\phi_m) = 0, \quad J_1(\delta\phi_n) = 0 \quad (197)$$

respectively. Since Eq. (197) represents transcendental equations, there are infinitely many roots corresponding to each of ϕ_m and ϕ_n . Using these values of ϕ_m and ϕ_n , Eq. (196) can be plotted in $\zeta - \Omega$ plane. When a pair of Ω and ζ satisfies both of Eq. (196) and both of Eq. (197), only then is the frequency equation (189) satisfied. So, each intersection points of ϕ_m and ϕ_n curves give the locus of the branches of the frequency spectrum.

In the same way, the second kind is $\sigma -$ families of curves given by

$$\Omega^2 = \zeta^2 + \sigma_m^2, \quad \Omega^2 = k^2(\zeta^2 + \sigma_n^2) \quad (198)$$

where σ_m and σ_n satisfy the equations

$$J_1(\delta\sigma_m) = 1, \quad J_1(\delta\sigma_n) = \frac{2\sigma_n^2}{\zeta^2(k^2 - 2) + k^2\sigma_n^2} \quad (199)$$

respectively. Since the σ_n given by the second of Eq. (199) are themselves dependent on the coordinates (ζ, Ω) , so the values of the σ_n or coordinates must be arrived at by an iterative procedure. Due to this, other kinds of families are expected to be more complex if tried for more intersection points.

The $\phi -$ and $\sigma -$ families of curves can be plotted for both real and pure imaginary ζ and have been shown in **Figure 2**. These are represented by full lines and dashed lines, respectively. All the curves are quarter-circles, quarter ellipses, and hyperbolas except for two straight full lines passing through the origin, which correspond to solutions, $\phi_m = 0$ and $\phi_n = 0$, of the first and the second, respectively, of Eq. (197). Each branch of the frequency spectrum must pass alternately through the intersection of full lines and the intersection of dashed lines. The other two straight dashed-dot lines passing through the origin and represented by R and L correspond to the Rayleigh

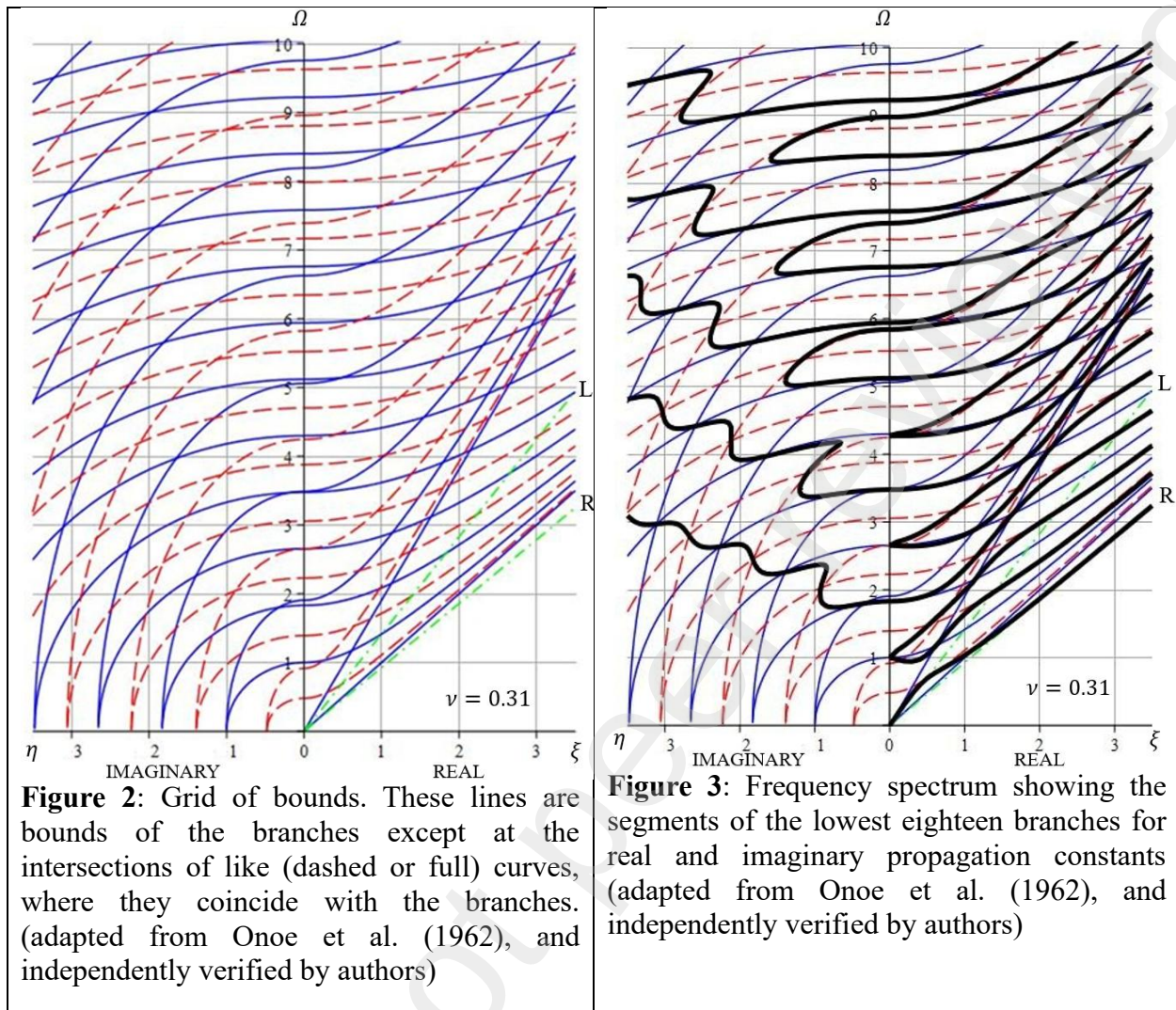
surface frequency and Lamb's equivoluminal modes of rods, respectively. These lines are given by

$$\Omega = s_1 \xi, \quad \Omega = \sqrt{2} \xi \quad (200)$$

respectively where s_1 is the real root, in the range of $0 < s < 1$, of

$$s^6 - 8s^4 + 8(3 - 2k^{-2})s^2 - 16(1 - k^{-2}) = 0 \quad (201)$$

After obtaining the slopes of the branches at the intersection points, and their curvatures at zero propagation constant, the sketching of the branches is accomplished, which is shown in **Figure 3**.



The branches can also be illustrated by means of a three-dimensional plot, as shown in Fig. 4. In this representation, the solid curves in the $\xi - \Omega$ plane correspond to the real segments of the branches, whereas the solid curves in the $\eta - \Omega$ plane represent the imaginary segments. The dashed curves indicate the complex segments of the branches.

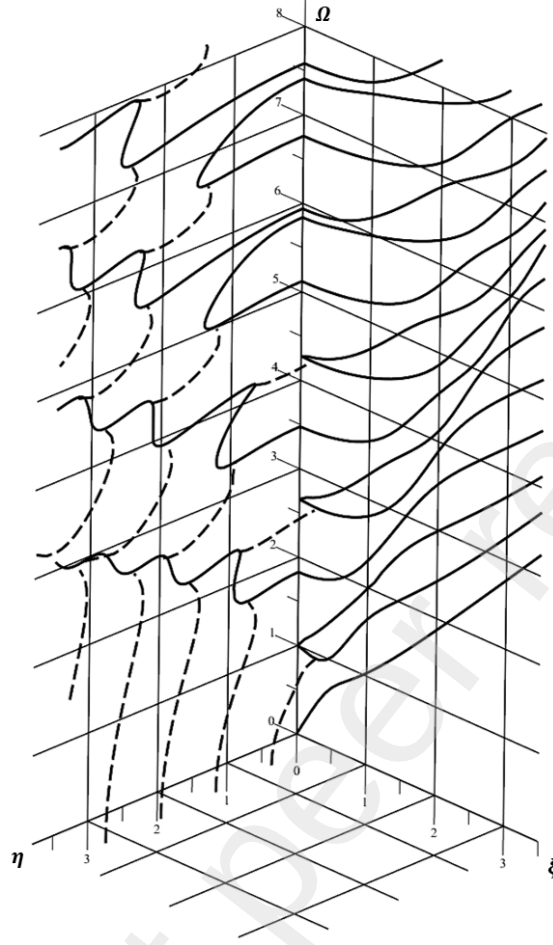


Figure 4: Illustrating linkage of real, imaginary, and complex segments to form branches at higher frequencies (adapted from Onoe et al. [56], and independently verified by authors)

3.2 Solutions Furnished by Classical and Refined Theories

(a) Classical Theory

For the dispersion curve, consider the displacement u of the form

$$u = Ae^{i\gamma(x-ct)} \quad (202)$$

Then, Eq. (6) converts to

$$-\gamma^2 c^2 = -c_0^2 \gamma^2$$

giving $c/c_0 = 1$, a horizontal line.

For the frequency spectrum, consider the displacement u of the form

$$u = Ae^{i(\gamma x - \omega t)} \quad (203)$$

Then, Eq. (6) converts to

$$-\omega^2 = -c_0^2 \gamma^2$$

To convert to the form of Eq. (189), applying some manipulation

$$\omega = c_0 \gamma = \left(\frac{E}{\rho}\right)^{1/2} \gamma = \left(\frac{E}{\mu} \frac{\mu}{\rho}\right)^{1/2} \frac{\gamma a}{\delta} \frac{\delta}{a} = \frac{\delta}{a} \left(\frac{\mu}{\rho}\right)^{1/2} \frac{\gamma a}{\delta} \left(\frac{E}{\mu}\right)^{1/2} = \omega_s \zeta \left(\frac{E}{\mu}\right)^{1/2}$$

finally results

$$\Omega = \zeta \sqrt{2(1 + \nu)} \quad (204)$$

So, the frequency spectrum for classical theory at a constant Poisson's ratio (ν) is a straight line passing through the origin.

(b) Rayleigh-Love Theory

For the dispersion curve, consider the displacement u of the form $Ae^{i\gamma(x-ct)}$

Then, Eq. (14) converts to

$$\rho(-\gamma^2 c^2 - \nu^2 k_p^2 \gamma^4 c^2) = -E \gamma^2$$

$$\frac{c}{c_0} = \frac{1}{\sqrt{1 + \nu^2 k_p^2 \gamma^2}} = \frac{1}{\sqrt{1 + \nu^2 \frac{a^2}{2} \frac{4\pi^2}{L^2}}} = \frac{1}{\sqrt{1 + 2\pi^2 \nu^2 \left(\frac{a}{L}\right)^2}} \quad (205)$$

For $a/L \rightarrow 0, c/c_0 = 1$ giving the result of classical theory.

For $a/L \rightarrow \infty, c/c_0 = 0$, a result contradicted by experimental values for higher a/L ratios, and therefore the need for better approximation theories arose.

For the frequency spectrum, consider the displacement u of the form $Ae^{i(\gamma x - \omega t)}$

Then, Eq. (14) converts to

$$\rho(-\omega^2 - \nu^2 k_p^2 \gamma^2 \omega^2) = -E\gamma^2$$

$$\omega = \left(\frac{E}{\rho}\right)^{\frac{1}{2}} \frac{\gamma}{\sqrt{1 + \nu^2 k_p^2 \gamma^2}} = \left(\frac{E}{\mu} \frac{\mu}{\rho}\right)^{\frac{1}{2}} \frac{\gamma a \delta}{\delta a} \frac{1}{\sqrt{1 + \nu^2 \frac{a^2}{2} \gamma^2}} = \omega_s \zeta \frac{\sqrt{2(1 + \nu)}}{\sqrt{1 + \frac{\nu^2}{2} \zeta^2 \delta^2}}$$

finally providing

$$\Omega = \zeta \frac{\sqrt{2(1 + \nu)}}{\sqrt{1 + \frac{\nu^2}{2} \zeta^2 \delta^2}} \quad (206)$$

where δ is a constant and for a particular value of Poisson's ratio ν , frequency spectrum can be plotted with Ω against ζ .

(c) Mindlin-Herrmann Theory

For the dispersion curve, consider the displacements u, w of the form

$$u = Ae^{i\gamma(z-ct)}, \quad w = Be^{i\gamma(z-ct)}, \quad R = Z = 0$$

Then, the first and the second of Eq. (38) convert to

$$[a^2 \kappa^2 \mu \gamma^2 + 8\kappa_1^2 (\lambda + \mu) - \rho a^2 \gamma^2 c^2]A + 4a\kappa_1^2 \lambda (i\gamma)B = 0$$

and

$$2a\lambda (i\gamma)A + [a^2 \gamma^2 (\lambda + 2\mu) - \rho a^2 \gamma^2 c^2]B = 0$$

respectively. The determinant of these equations set equal to zero, yields the frequency, or velocity, equation

$$[a^2 \kappa^2 \mu \gamma^2 + 8\kappa_1^2 (\lambda + \mu) - \rho a^2 \gamma^2 c^2][a^2 \gamma^2 (\lambda + 2\mu) - \rho a^2 \gamma^2 c^2] - 8a^2 \kappa_1^2 \lambda^2 \gamma^2 = 0$$

On dividing throughout by $(\rho a^2 \gamma^2)^2$, we get

$$\left[\frac{\kappa^2 \mu}{\rho} + \frac{8\kappa_1^2(\lambda + \mu)}{\rho a^2 \gamma^2} - c^2 \right] \left[\frac{\lambda + 2\mu}{\rho} - c^2 \right] - \frac{8\kappa_1^2 \lambda^2}{\rho^2 a^2 \gamma^2} = 0 \quad (207)$$

For long waves ($a/L \rightarrow 0$), Eq. (207) gives

$$c^2 = \frac{(3\lambda + 2\mu)\mu}{(\lambda + \mu)\rho} = \frac{E}{\rho} = c_0^2, \quad c^2 = \infty$$

which are identical with the corresponding limiting velocities of the first two modes in the exact (Pochhammer–Chree) solution.

For short waves ($a/L \rightarrow \infty$), frequency equation gives

$$c^2 = \frac{\kappa^2 \mu}{\rho} = \kappa^2 c_s^2, \quad c^2 = \frac{\lambda + 2\mu}{\rho} = c_c^2$$

In the exact theory, the first of these is the velocity of Rayleigh surface waves. Hence, to make the present approximation match in this respect, κ^2 should be taken as the root of [58]

$$16(1 - \alpha\kappa^2)(1 - \kappa^2) - (2 - \kappa^2)^4 = 0, \quad 0 < \kappa < 1 \quad (208)$$

where $\alpha = (1 - 2\nu)/2(1 - \nu)$.

In order to further adjust the curves so as to be tangent at the correct common point of Hudson [53] coordinates, κ_1 must satisfy

$$\kappa_1^2 = 0.422(2 - \kappa^2) \quad (209)$$

On the basis of Eqs. (207) – (209), the dispersion curves of c/c_e versus a/L have been plotted for several values of Poisson's ratio. These curves exhibit a typical coupling phenomenon between two modes of motion.

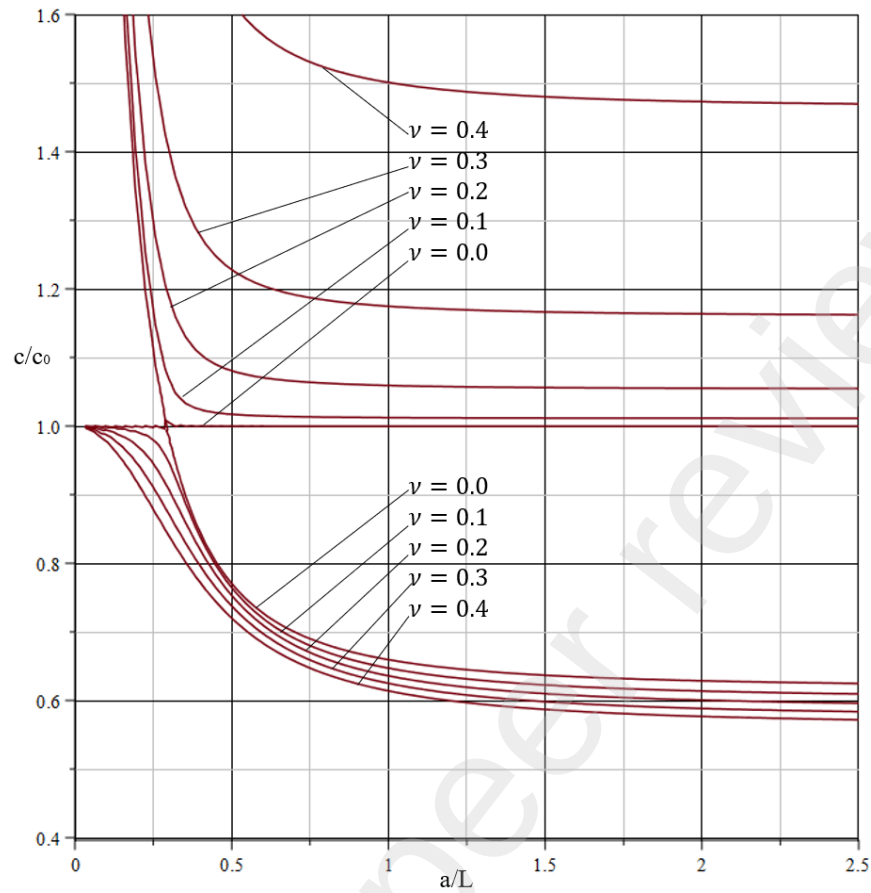


Figure 5: Effect of coupling, through Poisson's ratio, on wave velocities of axial and radial modes according to Mindlin-Herrmann theory (adapted from Mindlin and Herrmann [11], and independently verified by authors)

In **Figure 5**, the dispersion curves from Mindlin-Herrmann theory for the first mode and Poisson's ratio of 0.29 have been compared with those of Pochhammer's and Chree's exact solution (after Davies), Classical theory and Rayleigh-Love theory and found to be in better approximation with the exact theory than other refined theories. However, the Mindlin-Herrmann theory without the radial shear term provides a better approximation only for lower values of a/L less than and near to the value of 0.5.

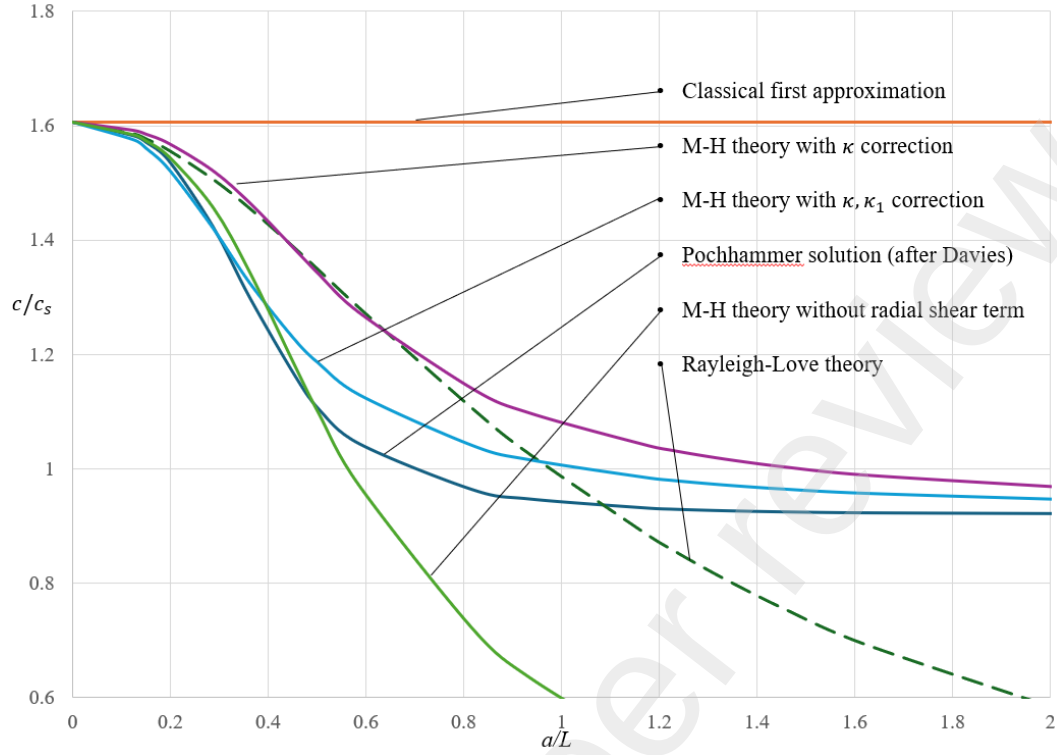


Figure 6: Longitudinal wave velocities of first mode of motion (adapted from Mindlin and Herrmann [11], and independently verified by authors)

(d) Bishop Theory

For the dispersion curve, consider the displacement u of the form $e^{i\gamma(x-ct)}$

Then, Eq. (52) converts to

$$-\gamma^2 c^2 = c_0^2(-\gamma^2) + v^2 k_p^2(\gamma^4 c^2) - v^2 k_p^2 c_2^2(\gamma^4)$$

providing

$$\frac{c^2}{c_0^2} = \frac{1 + v^2 k_p^2 \gamma^2 \left(\frac{c_2}{c_0}\right)^2}{1 + v^2 k_p^2 \gamma^2} = \frac{1 + v^2 \frac{a^2}{2} \gamma^2 \frac{1}{2(1+v)}}{1 + v^2 \frac{a^2}{2} \gamma^2} = \frac{1 + \frac{v^2}{4(1+v)} (a\gamma)^2}{1 + \frac{v^2}{2} (a\gamma)^2} \quad (210)$$

Similarly, Eq. (55) converts to

$$\begin{aligned}
-\gamma^2 c^2 &= c_0^2(-\gamma^2) + \frac{\nu^2 a^2}{2}(\gamma^4 c^2) - \frac{\nu^2 a^2 c_2^2}{2}(\gamma^4) \\
&+ \frac{\nu^2 a^4}{6c_0^2}(2\nu^2 + \nu - 1)[- \gamma^6 c^4 + 2c_2^2(\gamma^6 c^2) - c_2^4(\gamma^6)]
\end{aligned}$$

giving

$$\begin{aligned}
\frac{c^4}{c_0^4}(a\gamma)^4 \frac{\nu^2}{6}(2\nu^2 + \nu - 1) - \frac{c^2}{c_0^2} \left[1 + \frac{\nu^2}{2}(a\gamma)^2 + \frac{\nu^2}{3}(a\gamma)^4 \frac{(2\nu^2 + \nu - 1)}{2(1 + \nu)} \right] \\
+ \left[1 + \frac{\nu^2}{2}(a\gamma)^2 \frac{1}{2(1 + \nu)} + \frac{\nu^2}{6}(a\gamma)^4 \frac{(2\nu^2 + \nu - 1)}{4(1 + \nu)^2} \right] = 0
\end{aligned} \tag{211}$$

Finally, the improved Love's equation (56) changes to

$$\frac{c^4}{c_0^4}(a\gamma)^4 \frac{\nu^2}{6}(2\nu^2 + \nu - 1) - \frac{c^2}{c_0^2} \left[1 + \frac{\nu^2}{2}(a\gamma)^2 \right] + 1 = 0 \tag{212}$$

In **Figure 6**, for a Poisson's ratio of 0.29, the dispersion curves from Bishop theory (Eq. (210)), Bishop theory with improvement (Eq. (211)) and Rayleigh-Love theory with improvement (Eq. (212)) are compared with Classical theory ($c/c_0 = 1$), Pochhammer-Chree's exact solution (Eq. (193)) and Rayleigh-Love theory (Eq. (205)). For shorter waves ($a\gamma > 10$), Bishop theory with and without improvement provide better results while other refined theories are better for longer waves.

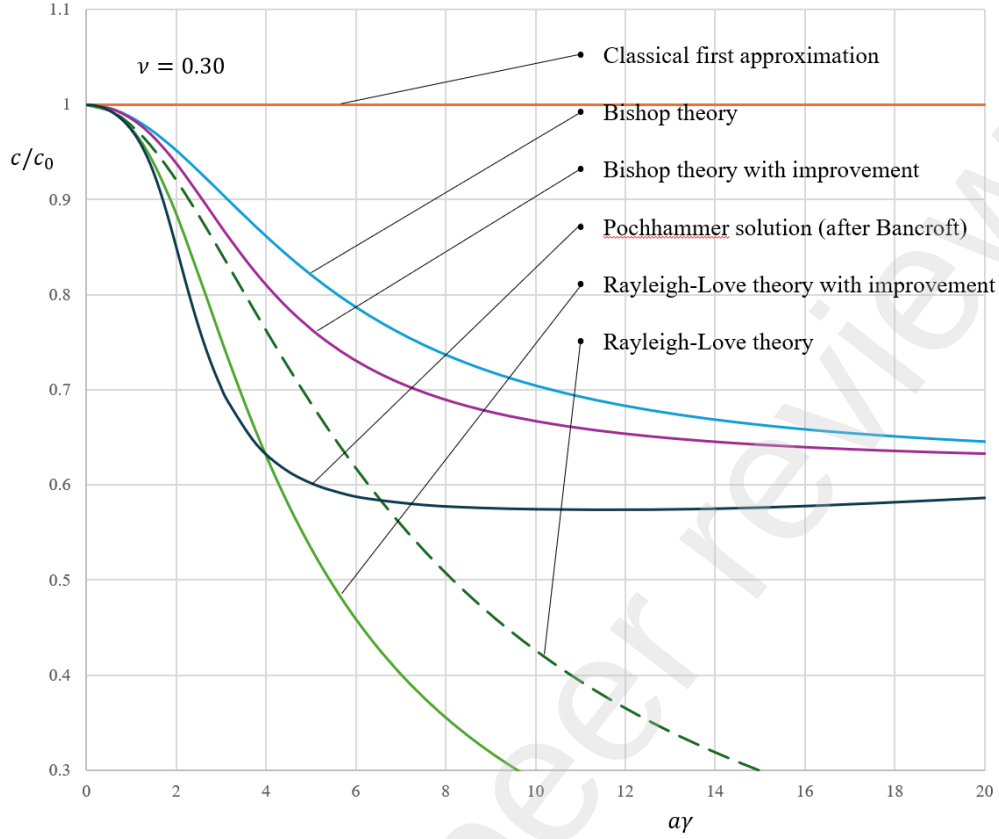


Figure 7: c/c_0 as a function of $a\gamma$ (adapted from Bishop [9], and independently verified by authors)

(e) Zachmanoglou–Volterra Theory

For the dispersion curve, consider the displacements v_x, f_x, λ_r of the form

$$v_x = A_1 e^{i\gamma(x-ct)}, \quad f_x = A_2 e^{i\gamma(x-ct)}, \quad \lambda_r = A_3 e^{i\gamma(x-ct)} \quad (213)$$

Then, Eq. (67) converts to

$$\begin{aligned} Z_1 A_1 + Z_2 A_2 + \nu N_1 \frac{i}{\gamma} A_3 &= 0 \\ 2\pi^2 \left(\frac{a}{\Lambda}\right)^2 Z_2 A_1 + \left[2\pi^2 \left(\frac{a}{\Lambda}\right)^2 Z_3 - \beta\right] A_2 + 2\pi^2 \left(\frac{a}{\Lambda}\right)^2 \epsilon \frac{i}{\gamma} A_3 &= 0 \\ \nu N_1 A_1 + \epsilon A_2 + \frac{i}{\gamma} A_3 \left[2\pi^2 \left(\frac{a}{\Lambda}\right)^2 Z_4 - N_1\right] &= 0 \end{aligned} \quad (214)$$

where

$$Z_1 = \left(\frac{c}{c_0}\right)^2 - (1 + v^2 N_1), \quad Z_2 = \frac{1}{2} \left[\left(\frac{c}{c_0}\right)^2 - N_3 \right], \quad Z_3 = \frac{1}{3} \left[\left(\frac{c}{c_0}\right)^2 - N_2 \right],$$

$$Z_4 = \left(1 + \frac{K^2 \alpha^2}{2} - \frac{4K\alpha}{3} \right) \left[\left(\frac{c}{c_0}\right)^2 - \beta \right], \quad \beta = \frac{1}{2(1+v)}, \quad \epsilon = vN_4 - \beta \left(1 - \frac{2}{3} K\alpha \right)$$

Setting the determinant of the above system of equations given by Eq. (214) to zero, gives the frequency equation

$$\left(\frac{a}{\Lambda}\right)^4 A + \left(\frac{a}{\Lambda}\right)^2 B + C = 0 \quad (215)$$

where

$$A = 4\pi^4 Z_4 [Z_1 Z_3 - Z_2^2],$$

$$B = 2\pi^2 [-N_1 (Z_1 Z_3 - Z_2^2) + 2vN_1 \epsilon Z_2 - \epsilon^2 Z_1 - \beta Z_1 Z_4 - v^2 N_1^2 Z_3]$$

$$C = v^2 N_1^2 \beta + \beta N_1 Z_1$$

The ratios

$$\frac{A_2}{A_1} = \frac{vN_1 Z_2 - \epsilon Z_1}{\epsilon Z_2 - vN_1 Z_3 + \frac{vN_1 \beta}{2\pi^2 \left(\frac{a}{\Lambda}\right)^2}}, \quad (216)$$

$$\frac{iA_3}{\gamma A_1} = \frac{(Z_1 Z_3 - Z_2^2) - Z_1 \beta / 2\pi^2 (a/\Lambda)^2}{\epsilon Z_2 - vN_1 Z_3 + vN_1 \beta / 2\pi^2 (a/\Lambda)^2}$$

can also be directly obtained from Eq. (214). Substituting Eq. (213) in Eq. (63) provides

$$K = 1 + v \left(\frac{i\gamma A_1 + i\gamma A_2}{A_3} \right) = 1 - v \left(\frac{1 + A_2/A_1}{iA_3/\gamma A_1} \right) \quad (217)$$

For Poisson's ratio of 0.29, Eq. (215) can be solved numerically by taking $K = 1$ and plotted for dispersion curves as shown in **Figure 7**. The value of K can be improved upon by using Eqs. (216) and (217), and Eq. (215) can again be solved to obtain better dispersion results which are shown by square boxes in **Figure 7**.

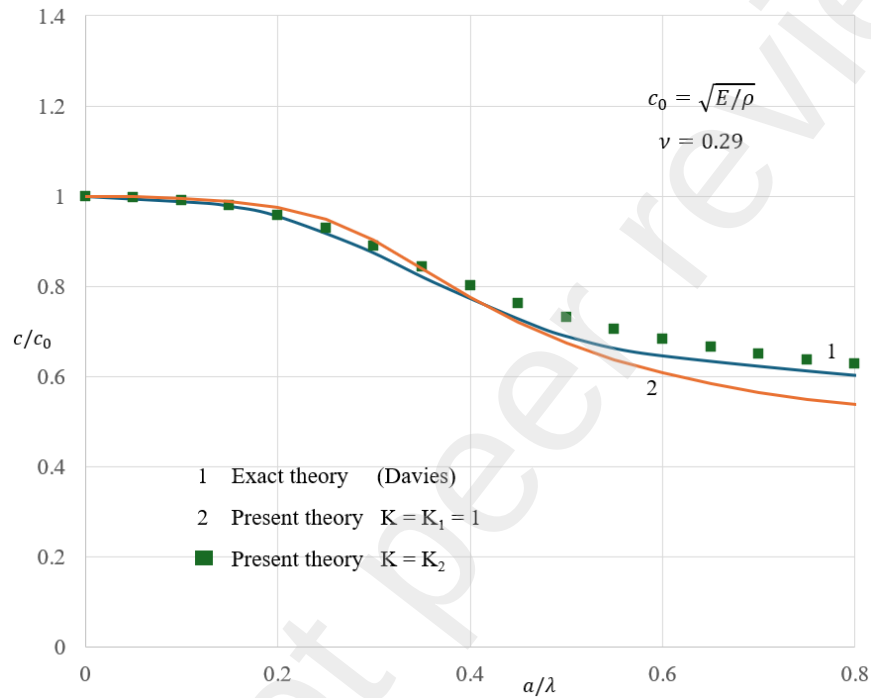


Figure 8: First mode of Longitudinal waves of wavelength λ in cylindrical bars of radius a (adapted from Zachmanoglou and Volterra [12] and independently verified by authors)

(f) Mindlin-McNiven's Theory

Following Mindlin and McNiven [14], consider the displacement u, w, ψ of the form given below along with zero tractions at the cylindrical surface.

$$u = A \cos \gamma z e^{i\omega t}, \quad w = B \sin \gamma z e^{i\omega t}, \quad \psi = C \sin \gamma z e^{i\omega t}, \quad R = 0 = Z \quad (218)$$

Then, substituting Eq. (218) in Eq. (94) and eliminating the constants A, B , and C , we find, for the cubic equation governing the branches of the approximate equations,

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & 0 \\ a_{13} & 0 & a_{33} \end{vmatrix} = 0 \quad (219)$$

where

$$\begin{aligned} a_{11} &= \kappa_2^2 \delta^2 \zeta^2 + 8(k^2 - 1)\kappa_1^2 - \kappa_3^2 \delta^2 \Omega^2, & a_{22} &= 2\delta^2(k^2 \zeta^2 - \Omega^2) \\ a_{33} &= 6(k^2 \delta^2 \zeta^2 + 24\kappa_2^2 - \kappa_4^2 \delta^2 \Omega^2), & a_{12} &= 4(k^2 - 2)\kappa_1 \delta \zeta, & a_{13} &= 12\kappa_2^2 \delta \zeta \end{aligned} \quad (220)$$

When $\zeta = 0$, Eq. (219) has the three roots

$$\Omega^2 = 0, \quad \frac{24\kappa_2^2}{\delta^2 \kappa_4^2}, \quad \frac{8(k^2 - 1)\kappa_1^2}{\delta^2 \kappa_3^2} \quad (221)$$

for the longitudinal (w), axial shear (ψ), and radial (u) modes, respectively. To make these frequencies the same as the cutoff frequencies of the first three branches of Pochhammer's and Chree's exact solution [14], we set

$$\frac{\kappa_2^2}{\kappa_4^2} = \frac{\delta^2}{24}, \quad \frac{\kappa_1^2}{\kappa_3^2} = \frac{k^2 \chi^2}{8(k^2 - 1)} \quad (222)$$

where δ and χ are the lowest (nonzero) roots of

$$J_1(\delta) = 0, \quad k^2 J_1(\chi) = 2 \quad (223)$$

respectively.

For $\zeta \ll 1$, Eq. (219) becomes

$$\Omega = \zeta [2(1 + \nu)]^{1/2}, \quad \Omega = 1 + \frac{1}{2} \zeta^2 \bar{C}_s, \quad \Omega = \frac{k\chi}{\delta} + \frac{1}{2} \zeta^2 \bar{C}_r \quad (224)$$

where

$$\bar{C}_s = \frac{k^2 \delta^2}{24} \left[\frac{1}{\kappa_2^2} - \frac{3\chi^2 \kappa_2^2}{\kappa_1^2 (k^2 - 1)(k^2 \chi^2 - \delta^2)} \right] \quad (225)$$

$$\bar{C}_r = \frac{k\chi\delta}{k^2 - 1} \left[\frac{(k^2 - 2)^2}{k^2\chi^2} + \frac{k^2\chi^2\kappa_2^2}{8\kappa_1^2(k^2\chi^2 - \delta^2)} \right]$$

For small ζ , the first three branches of the Pochhammer and Chree equation correspond to

$$\Omega = \zeta[2(1 + \nu)]^{1/2}, \quad \Omega = 1 + \frac{1}{2}\zeta^2 C_s, \quad \Omega = \frac{k\chi}{\delta} + \frac{1}{2}\zeta^2 C_r \quad (226)$$

where

$$C_s = 1 - 8/[k^2 J(\delta/k) - 2]$$

$$C_r = \frac{k\delta}{\chi} \left[1 + \frac{4(k^2 + 2J_1(k\chi) - 4)}{k^4\chi^2 - 4(k^2 - 1)} \right] \quad (227)$$

Comparing Eq. (224) with Eq. (226), we see that the longitudinal mode needs no adjustment, while the others will match if

$$\bar{C}_r = C_r, \quad \bar{C}_s = C_s \quad (228)$$

from which

$$\frac{\kappa_2^2}{\kappa_1^2} = \frac{8(k^2\chi^2 - \delta^2)}{k^2\chi^2} \left[\frac{(k^2 - 1)C_r}{k\chi\delta} - \frac{(k^2 - 2)^2}{k^2\chi^2} \right]$$

$$\frac{1}{\kappa_2^2} = \frac{24}{k^2\delta^2} \left[\frac{k^2\chi^2\kappa_2^2\delta^2}{8\kappa_1^2(k^2 - 1)(k^2\chi^2 - \delta^2)} + C_s \right] \quad (229)$$

Eqs. (222) and (229) fix the four coefficients κ_i as functions of Poisson's ratio. With these values, the ordinates, slopes, and curvatures of the three branches from the approximate and exact equations are identical at $\zeta = 0$. Eq. (219) can then be used to plot the frequency spectrum shown in **Figure 9**, for Poisson's ratio of 0.31. In this depiction, solid curves in the $\xi - \Omega$ plane denote the real portions of the branches, while solid curves in the $\eta - \Omega$ plane correspond to their imaginary portions. The dashed curves represent the complex segments of the branches.

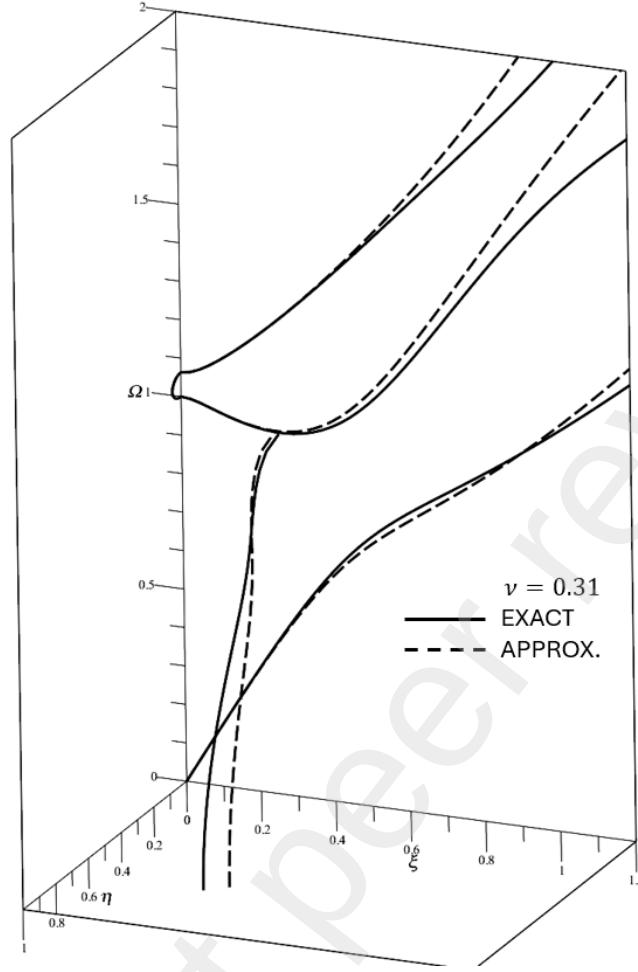


Figure 9: Frequency spectra of Mindlin-McNiven theory (dashed) compared with Pochhammer-Chree exact theory (solid) for Poisson's ratio of 0.31 (adapted from Mindlin and McNiven [14] and independently verified by authors)

(g) Doyle's Theory

If the applied load q is zero and the rod is circular in cross-section, then Eq. (101) converts to

$$(2\mu + \lambda) \frac{\partial^2 u_0}{\partial x^2} + \lambda \frac{\partial \psi_0}{\partial x} = \rho \ddot{u}_0$$

$$\mu \frac{h^2}{8} \frac{\partial^2 \psi_0}{\partial x^2} - (2\mu + \lambda) \psi_0 - \lambda \frac{\partial u_0}{\partial x} - 2\mu h \frac{\partial \phi_0}{\partial x} = \rho \frac{h^2}{8} \ddot{\psi}_0 \quad (230)$$

$$(2\mu + \lambda) \frac{h^2}{8} \frac{\partial^2 \phi_0}{\partial x^2} - 5\mu \phi_0 + \frac{10}{48} \mu h \frac{\partial \psi_0}{\partial x} = \rho \frac{h^2}{8} \ddot{\phi}_0$$

Let the displacements u_0 , ψ_0 , ϕ_0 be of the form

$$u_0 = A_1 e^{i(kx - \omega t)}, \quad \psi_0 = A_2 e^{i(kx - \omega t)}, \quad \phi_0 = A_3 e^{i(kx - \omega t)}$$

Then, Eq. (230) convert into three homogeneous equations in A_1, A_2 and A_3 . The determinant of the coefficient matrix yields the frequency equation

$$\begin{aligned} & [-\mu I^2 (2\mu + \lambda)^2 A] k^6 + (2\mu + \lambda) I [\rho I \omega^2 (4\mu + \lambda) A - 4\mu (\mu + \lambda) A^2] k^4 \\ & + [-(2\mu + \lambda) A \{\rho I \omega^2 - (7\mu + \lambda) A\} \rho I \omega^2 - 20\mu^2 (\mu + \lambda) A^3] k^2 \\ & + \rho A \omega^2 I \{-\rho I \omega^2 (3\mu + \lambda) 4\mu (\mu + \lambda) A\} k^2 \\ & + [\rho I \omega^2 - (2\mu + \lambda) A] (\rho I \omega^2 - 5\mu A) \rho A \omega^2 = 0 \end{aligned} \quad (231)$$

For a given material characterized by density ρ , Young's modulus E , and Poisson's ratio ν , the Lamé constants λ and μ can be determined. The wave speed $c_0 = \sqrt{E/\rho}$ then follows directly. Upon selecting the geometric parameters, namely the cross-sectional area A and the moment of inertia I , Eq. (231) contains only two unknowns, the wavenumber k and the angular frequency ω . By prescribing values of ω , the corresponding values of k can be obtained. Differentiation of Eq. (231) with respect to ω yields the group velocity, $c_g = d\omega/dk$, which can be evaluated for each (k, ω) pair. The resulting dispersion curve, shown in **Figure 10**, is then plotted as velocity ratio (c_g/c_0) versus frequency $(\omega/2\pi)$. Using the same procedure, dispersion curves corresponding to the other theories presented in **Figure 10** are obtained.

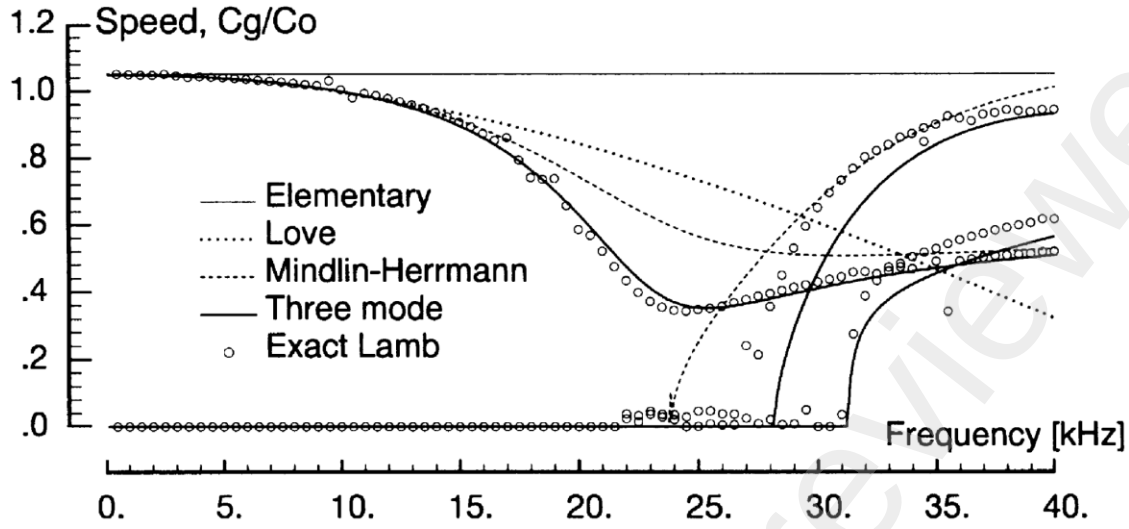


Figure 10: Velocity ratio c_g/c_0 dispersion curves for different rod theories (adapted from Doyle [15])

(h) Anderson's Theory

Let the displacements u, v, ϕ, ψ be of the form

$$u = A_1 e^{i\gamma(x-ct)}, \quad v = A_2 e^{i\gamma(x-ct)}, \quad \phi = A_3 e^{i\gamma(x-ct)}, \quad \psi = A_4 e^{i\gamma(x-ct)}$$

Then, Eq. (107) converts to

$$\begin{bmatrix} a_{11} & 2a\lambda i\gamma & 0 & -a\lambda i\gamma \\ -4a\lambda i\gamma & a_{22} & -4a\mu i\gamma & 4(\mu + \lambda) \\ 0 & 6a\mu i\gamma & a_{33} & 3a\lambda i\gamma \\ 16a\lambda i\gamma & 32(\mu + \lambda) & -16a\lambda i\gamma & a_{44} \end{bmatrix} \begin{bmatrix} u \\ v \\ \phi \\ \psi \end{bmatrix} = 0 \quad (232)$$

where

$$a_{11} = a^2 \gamma^2 (\rho c^2 - (2\mu + \lambda)), \quad a_{22} = \rho a^2 \gamma^2 c^2 - a^2 \mu \gamma^2 - 8(\mu + \lambda),$$

$$a_{33} = \rho a^2 \gamma^2 c^2 - a^2 (2\mu + \lambda) \gamma^2 - 24\mu, \quad a_{44} = \rho a^2 \gamma^2 c^2 - a^2 \mu \gamma^2 - 16(7\mu + 4\lambda)$$

Equating the determinant of the coefficient matrix of Eq. (232) to zero gives the frequency equation

$$C_1 \gamma^8 + C_2 \gamma^6 + C_3 \gamma^4 + C_4 \gamma^2 + C_5 = 0 \quad (233)$$

where C_1, C_2, C_3, C_4 and C_5 are the known functions of ρ, a, c, λ and μ . For a proper choice of material properties, the wave velocity $c_0 = \sqrt{E/\rho}$ and Lamé's constant λ and μ can be determined. Eq. (233) then has unknowns $(a\gamma)$ and c which for a prescribed values of $(a\gamma)$ can be solved for c . The velocity ratio c/c_0 can then be plotted against nondimensional propagation constant $a\gamma$. Figure 11 compares the dispersion curves for a typical 100mm diameter aluminum rod ($E = 70 \text{ GPa}$, $\nu = 0.3$, $\rho = 2700 \text{ kg/m}^3$) using the Pochhammer–Chree equation and four rod approximations: Love, Mindlin-Herrmann, Mindlin-McNiven and Anderson theories.

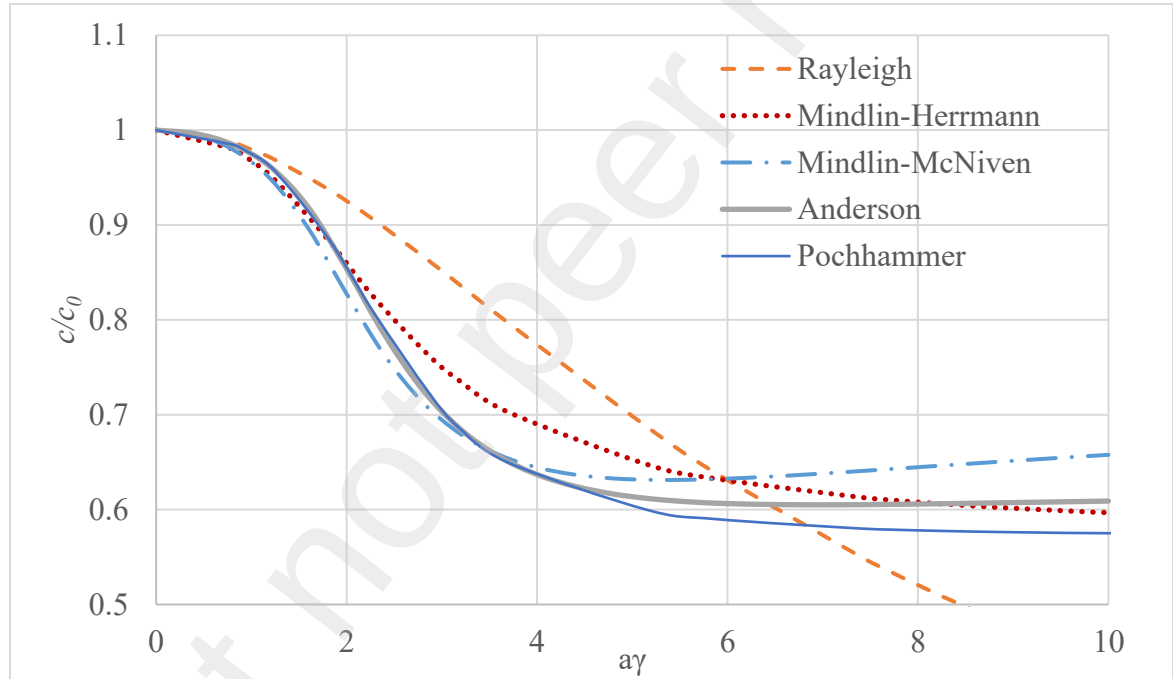


Figure 11: Dispersion curves for an aluminum bar (adapted from Anderson [16], and independently verified by authors)

(i) Żak -Krawczuk Higher Order Two-, Three- and Four – Mode Theories

By following the procedure used to obtain the solution of Doyle's theory, the velocity ratio (c_g/c_p) dispersion curves are obtained for higher order two-, three- and four-mode theories as shown in Figure 12 to Figure 14. Eqs. (111), (114) and (117) are used for constructing these plots.

A rod made from aluminum with Young's modulus $E = 72.7 \text{ GPa}$, Poisson's ratio $\nu = 0.33$ and material density $\rho = 2700 \text{ kg/m}^3$ and of the diameter $d = 0.01 \text{ m}$, is considered.

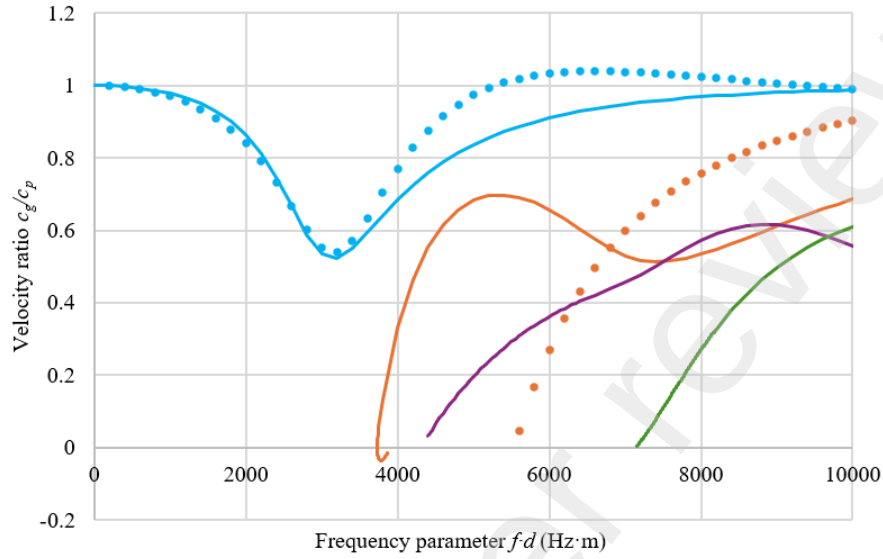


Figure 12: Velocity ratio c_g/c_p dispersion curves for higher higher-order two-mode theory of rods ($c_l = 6300 \text{ m/s}$, $c_t = 3200 \text{ m/s}$) (blue, orange, purple, and green continuous lines represent first, second, third, and fourth branches of exact theory, blue and orange dotted lines represent first and second branches of associated refined theory)

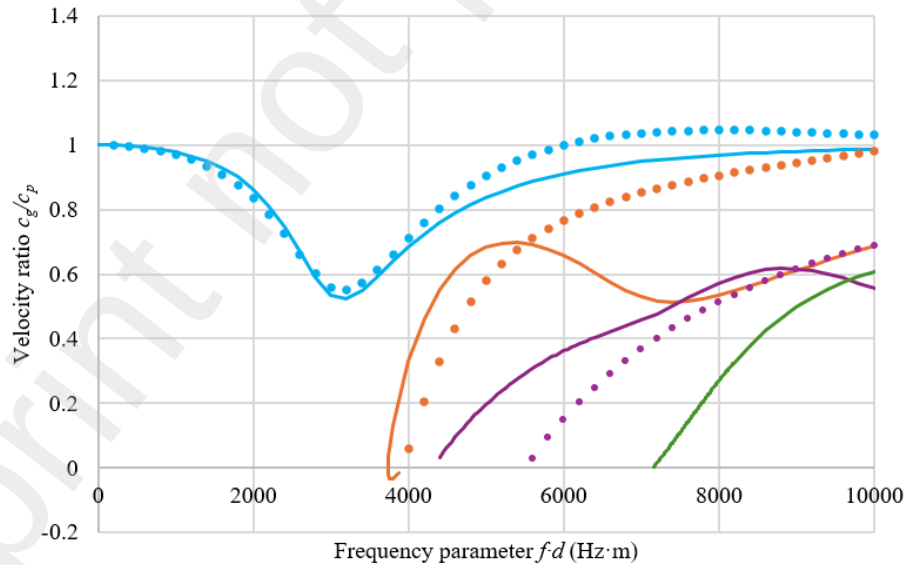


Figure 13: Velocity ratio c_g/c_p dispersion curves for higher higher-order three-mode theory of rods ($c_l = 6300 \text{ m/s}$, $c_t = 3200 \text{ m/s}$) (blue, orange, purple and green continuous lines represent first, second, third, and fourth branches of exact theory, blue, orange, and purple dotted lines represent first, second, and third branches of associated refined theory)

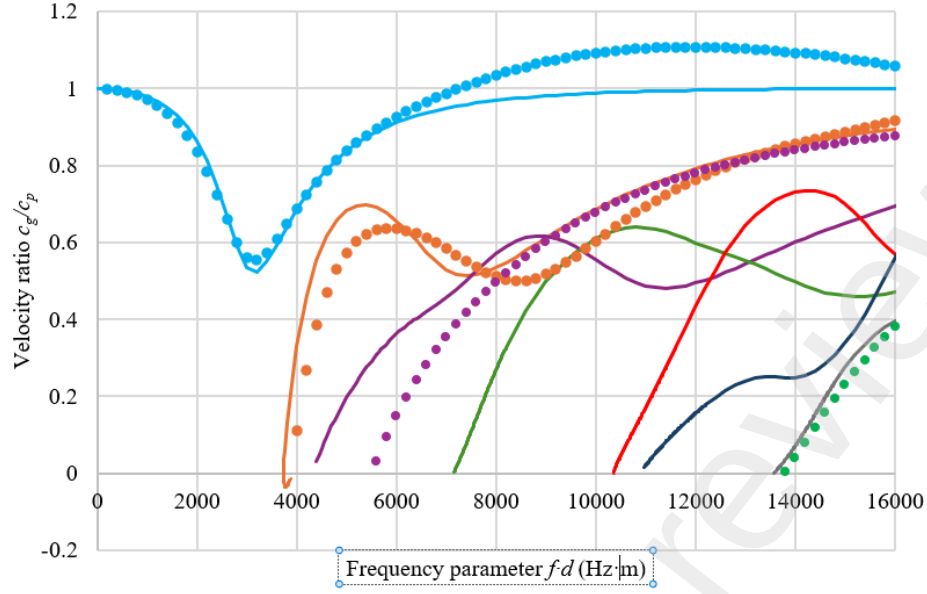


Figure 14: Velocity ratio c_g/c_p dispersion curves for higher-order four-mode theory of rods ($c_l = 6300 \text{ m/s}$, $c_t = 3200 \text{ m/s}$) (blue, orange, purple, green, red, dark blue, and gray continuous lines represent first, second, third, fourth, fifth, sixth, and seventh branches of exact theory; blue, orange, purple, and green dotted lines represent first, second, third, and fourth branches of associated refined theory)

(j) Shatalov et al. Theories

By following a procedure similar to that employed for the unimodal and multimodal theories discussed above, the frequency spectra for Shatalov et al.'s theories can be obtained. For an infinitely long cylindrical rod composed of an aluminum alloy with Young's modulus $E = 70 \text{ GPa}$, density $\rho = 2700 \text{ kg/m}^3$, and Poisson's ratio $\eta = 0.33$, the resulting frequency spectra are presented in **Figure 15** to **Figure 17**. These plots are generated using Eqs. (132), (136), (148), (152), (160), and (168), in which harmonic wave solutions of the form $u_i = U_i e^{i(kx - \omega t)}$ are substituted to derive the corresponding frequency equations.

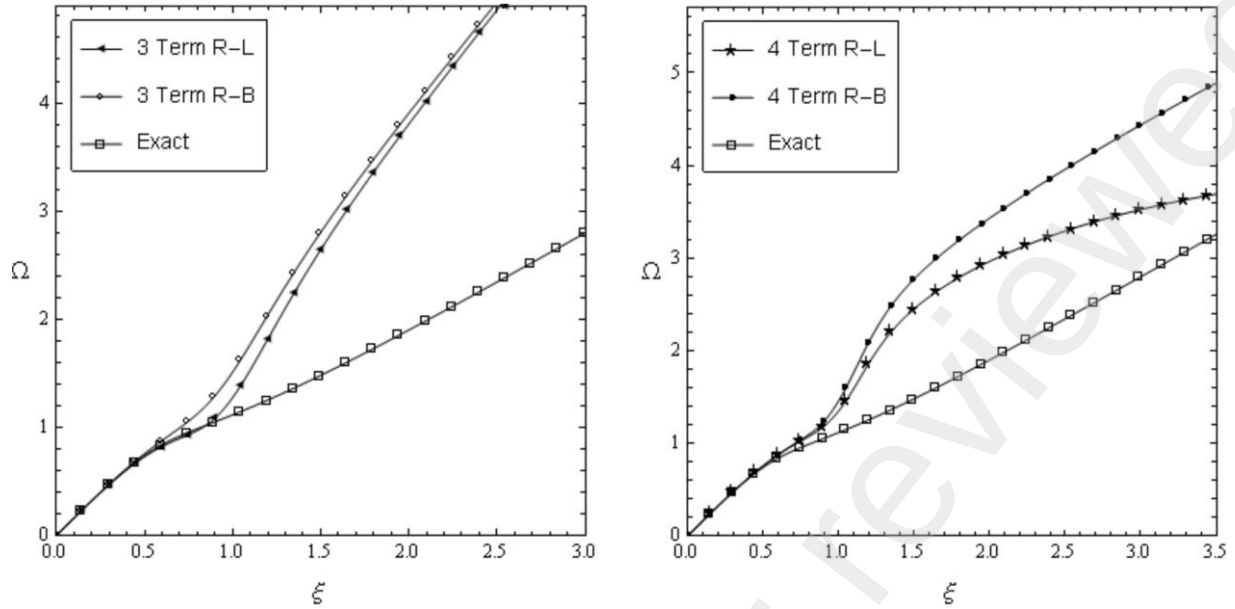


Figure 15: Frequency spectra of three-term and four-term Rayleigh-Love and Rayleigh-Bishop type theories (adapted from Shatalov et. al. [19])

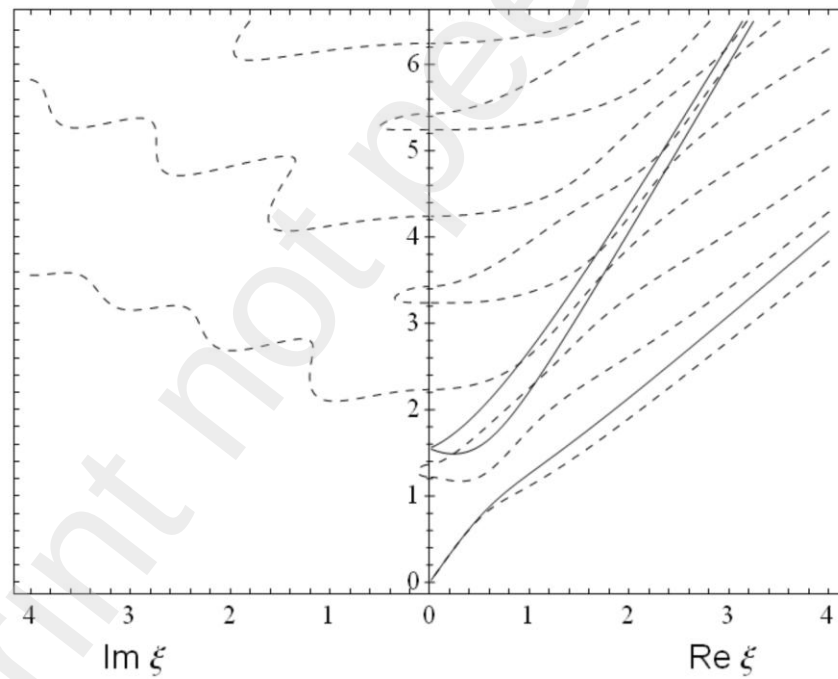


Figure 16: Frequency spectrum of Shatalov's et al. three mode theory (solid) compared with Pochhammer-Chree exact theory (dashed) (adapted from Shatalov et. al. [19])

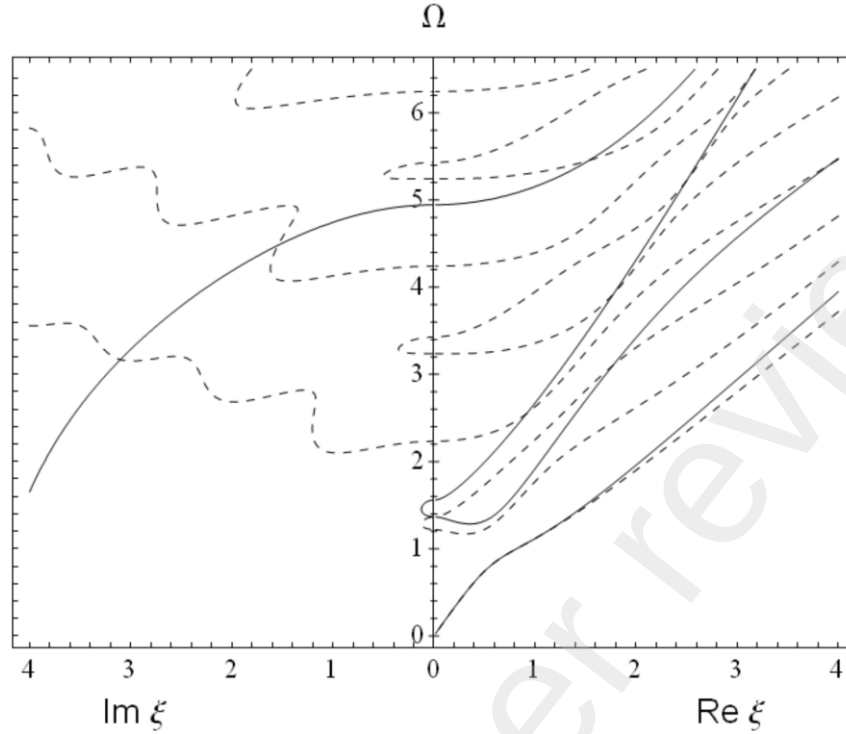


Figure 17: Frequency spectrum of Shatalov et al.'s four-mode theory (solid) compared with Pochhammer-Chree exact theory (dashed) (adapted from Shatalov et. al. [19])

4. Conclusion

A comprehensive review of the available refined theories suggests that the theoretical development of longitudinal rod vibration models can proceed along three principal directions: unimodal theories, multimodal theories, and higher-order multimodal theories.

Unimodal theories assume that all displacement components depend linearly on the first longitudinal displacement mode and its derivatives. This class of theories has been thoroughly investigated by Shatalov and co-authors [19], who demonstrated that refined unimodal formulations can achieve improved accuracy while retaining analytical simplicity.

Multimodal theories allow for the inclusion of an arbitrary number of axial and radial displacement modes. Representative examples include the Mindlin–Herrmann, Mindlin–

McNiven, Doyle, and Anderson theories, all of which demonstrate the effectiveness of incorporating multiple kinematic modes to more accurately approximate the exact elasticity solution.

Higher-order multimodal theories begin with an expanded set of displacement modes, some of which are subsequently eliminated by enforcing physical boundary conditions, such as the zero-traction condition at the rod surface. It should be emphasized that higher-order multimodal theories constitute only a subset of higher-order theories; the latter represents a more general class, defined by the order of the displacement field approximation rather than solely by the number of retained modes. A prominent example is the Zachmanoglou–Volterra theory. This class of theories has been studied in detail by Żak and Krawczuk [17], [18]. Notably, the displacement assumptions employed in the Żak–Krawczuk formulations are relatively simple, resulting in significantly reduced computational complexity when compared with other higher-order theories that rely on power-series expansions or more elaborate displacement representations.

From a classification standpoint, since the degree of the polynomial used in the displacement fields defines the theoretical order, the classical, Rayleigh–Love, Bishop, and Mindlin–Herrmann theories may be regarded as lower-order theories, while the remaining formulations constitute higher-order theories. Furthermore, because polynomial degrees greater than one permit nonplanar deformation of the cross-section, lower-order theories may be classified as planar theories, whereas higher-order theories correspond to nonplanar theories. In terms of modal content, the classical, Rayleigh–Love, Rayleigh–Bishop, and Shatalov et al. three- and four-term Rayleigh–Love and Rayleigh–Bishop type models can be categorized as unimodal theories, while all remaining formulations fall within the multimodal class. Since the Mindlin–Herrmann and Mindlin–McNiven theories introduce correction factors to match the Pochhammer–Chree’s exact

solution at zero propagation constant, the theories considered in this paper may also be categorized into two groups: theories that include the correction factors and those that do not.

The results reported in this review demonstrate that the predictions of refined longitudinal rod theories can be successfully reproduced in close agreement with those originally reported by their respective authors. All figures presented in this paper compare either dispersion curves or natural frequency spectra obtained from the sixteen refined theories with the exact Pochhammer–Chree solution, which serves as a rigorous benchmark. When considering the trade-off between model complexity and accuracy, Bishop’s theory emerges as the simplest refined theory that predicts the lowest order dispersion curve more accurately than the Rayleigh–Love theory. When all predicted branches are considered without regard to the complexity, the Mindlin–McNiven theory provides the highest overall prediction accuracy among the theories examined. Because the Mindlin–Herrmann and Mindlin–McNiven theories incorporate a correction factor, all branches within this family of theories exhibit comparatively higher accuracy in both dispersion curves and frequency spectra than those of other formulations.

For theories that do not introduce correction factors, a closer examination of the comparative results reveals that systematically increasing the number of retained modes improves the accuracy of the lower-order dispersion branches. Consequently, for obtaining refined and reliable predictions at a given theoretical order, it appears advantageous to employ models that incorporate two or three additional higher-order modes rather than relying solely on minimal modal representations. The Żak–Krawczuk higher-order four-mode theory considers displacement fields represented by polynomials up to the fifth degree, which is the highest degree considered to date. As a result, it provides the most accurate predictions among theories that do not introduce correction factors.

In summary, the unimodal, multimodal, and higher-order multimodal approaches each provide a systematic pathway for constructing an infinite number of refined theories. The choice among these approaches depends on the desired balance between predictive accuracy and computational complexity, as well as the specific frequency range and physical effects of interest. This review highlights the underlying connections among these formulations and provides a unified perspective on their relative capabilities and limitations.

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